FriendMate
Books

DROPBOX

A Book on Computer Education with Artificial Intelligence

Name ....

Class .... Section ....

Witty

6

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(M.Tech)

PREPARATORY CURRICULUM

Inculcating 21st-Century Skills

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Preface

In the context of the National Education Policy 2020, Computer Science education must undergo a paradigm shift where our children transition from being 'users' to 'developer's at a young age through mastery in coding. Newer technologies such as AI, Natural Language Processing, Data Science, Computer Vision and Neural Networks must be Introduced.

As per CBSE directives, the series stresses upon sustainable development goals while also ensuring that the right attitudes and values are inculated through of cyber Ethios.

Special Features :

- Carefully selected and graded content
- Stepwise explanation of programs with clear screen shots.
- Inculcation of in depth problem solving and programming.
- Sufficient Lab Activities to ensure practical leaning.
- A section on computers and Health including tips to minimize health issues while working on computers.
- A section on Olympiad Bot containing practice questions for Natural Cyber olympiad.

—Author and Publisher

INCORPORATES NEP 2020

Created with new dimension of joyful learning along with completely mapped parameters of National Education Policy, 2020

Critical Thinking

Strengthening decision making and innovation

Creative Learning

Thinking out of the box

Art Integration

Promoting art based learning

Computational Thinking

Enables to help solve problems and make thinking more engaging.

Inter Disciplinary

Connects knowledge across the curriculum for holistic development

Subject Enrichment

Aimed at understanding and skill developing in specific subject

Communication

Emphasis on soft skills and power of language

Experimental Learning

Encourages learning by doing through hands-on activities

Life Skills and Values

Inclined towards empowering moral values in learner

INSIDE THE BOOK

Outline includes the topics to be learnt in the chapter.

Outlines

- Computer : An Electronic Machine
- Features of a Computer
- Types of a Computer

Info Zone include some facts related to chapter to enhance the knowledge of the students.

Info Zone

The first version of MS Paint was introduced with the first version of windows in 1985.

Smart Tip provide keyboard shortcuts for menu commands, to help users save time while performing routine operations.

Smart Tip

You can also open Paint by typing Paint in the Search Box next to the Start button. Then, click on the Paint button.

Computer Etiquette presents computer ethics and manners to be followed while working on a computer.

COMPUTER ETIQUETTES

While working on a computer, keep your neck relaxed. Practise this exercise to avoid any strain on your neck.

Let's Implement contains carefully graded exercises to test the knowledge of concepts learnt in a chapter.

Let's Implement

A Tick (✓) the correct option.

1. A monitor is also called :
   (a) LCU ◯ (b) VDU ◯
2. Which part of a computer is called the brain of a computer?
   (a) CPU ◯ (b) VDU ◯
3. It is used to give the output of our work on paper.
   (a) Speakers ◯ (b) Printer ◯

Refreshment Corner includes interesting exercises for enhancing the skills with fun.

Refreshment Corner

A Colour the alphabet keys that spell your name with yellow.

Practical Session for practical hands-on exercises to strong then the concepts learned by doing.

Practical Session

- Visit your computer lab and open Paint. Draw and colour the following.

Racking Your Brain includes the life skills and values questions.

RACKING YOUR BRAIN

Smart devices are used to make our tasks easier. Should we always use them at small tasks? Justify your answer.

Just Have a Discussion stimulates a small discussion for the topics being covered in the chapter which in turn improves the communication and logical skills.

Just Have a Discussion

Discuss the different components of Scratchjr in class in detail.

Teacher's Note provides important information and suggestions on creative approaches to a chapter or a topic.

TEACHER'S NOTE

- Give enough exercises to students so that they get practice of using the tools of Paint as well as using the mouse.

CONTENTS

1

COMPUTER FUNDAMENTALS

Scan Me

Outlines

- What is a Computer?
- Why is Computer so Powerful?
- How Does a Computer Know What to Do?
- What are the Components of a Computer?
- What Does a Computer Do?
- The IPO Cycle
- Categories of Computers

Today, there is no one who is not familiar with the computer. So, it is not necessary to give its introduction. However, you must know the definition of the computer.

In simple words, Computer is a machine which has mechanical, electrical or electronic parts and performs work to help the human beings.

There are unnumerable machines which help us in our physical or mental works. Machines save our labour and time and give accurate results.

Computer is also a machine that helps in doing the mental works.

For example : If you have a computer and want to make a marksheet, you need not have to do all the processing by using your mind repeatedly for all the students, instead you can just set the logic in the computer program which will automatically handle the task.

What is a Computer?

The most obvious question related to understanding of the computer and its impact on our lives is "What is a computer?". A computer is an electronic device, operating under the control of instructions stored in its own memory unit that can accept data input, process data arithmetically and logically, produce output from the processing and store the results for future use.

What Does a Computer Do?

Whether small or large, computers can perform four general operations. These operations comprise the Information Processing Cycle. They are Input, Process, Output and Storage. Collectively, these operations describe the procedures that a computer performs to process data into information and store it for future use.

All computer processing require data. Data refers to the raw facts, including numbers and words, given to a computer during the input operation. In the processing phase, the computer manipulates the data to create information.

> **Info Zone**
> The information can also be stored electronically for future use.

Information refers to data that has been processed into a form that has meaning and is useful. The production of information by processing data on a computer is called Information processing or sometimes Data processing (DP). During the output operation, the information that has been created is put into some form, such as printed report, that people can use.

> **Info Zone**
> The use of personal computer illustrates the four operations of the information cycle : input, process, output and storage.

The people who either use the computer directly or use the information it provides are called computer users, end users or sometimes just simply users. The figure shows a computer user and demonstrates how the four operations of the information processing cycle can occur on a personal computer.

1. The computer user inputs data by pressing the keys on the keyboard.
2. The data is then processed by the unit called the processor.
3. The output or results from the processing, are displayed on the screen or printed on the paper with the help of printer providing information to the user.
4. Finally, the output may be stored on a disk for future reference.

Why is Computer so Powerful?

The input, process, output and storage operations that a computer performs may seem very basic and simple. However, the computer's power derives from its capability to perform these operations very quickly, accurately and reliably. In a computer, operations occur through the use of electronic circuits contained on small chips. As in the above figure, when data flows along these circuits, it travels close to the speed of light. This allows processing to be accomplished in billionths of a second. The electronic circuits in modern computers are very reliable and seldom fail. Storage capability is another reason why computers are so powerful. They can store enormous amount of data and keep that data readily available for processing. This capability combined with the factors of speed, accuracy and reliability consider computers to be such a powerful tool for information processing.

Info Zone

Inside a computer are chips and other electronic components that process data in billionths of a second.

How Does a Computer Know What to Do?

For a computer to perform the operations in the information processing cycle, it must be given a detailed set of instructions that tell it exactly what to do. These instructions are called a computer program, program instructions or software.

Before the information processing cycle for a specific job begins, the computer program corresponding to that job is stored in the computer. Once the program is stored, the computer can begin to process data by executing the program's first instruction. The computer executes one program instruction after another until the job is complete.

The IPO Cycle

To review, the information processing cycle consists of four operations. They are input, process, output and storage.

INUPUT → CPU → OUTPUT

Main memory

↕

Auxiliary storage

The first three of these operations, input, process and output, describe the procedures that a computer performs to process data into information. The fourth operation, storage describes a computer's electronic storage capability. As you learn more about the computers, you will see that these four operations apply to both the computer equipments and the computer software. The equipments or 0devices of a computer, are classified according to the operations that they perform. Computer software is made up of instructions that describe how the operations are to be performed.

What are the Components of a Computer?

A computer is composed of input devices through which data is entered into the computer, the processor that processes data stored in the main memory, output devices on which the results of the processing are made available and auxiliary storage units that store data for future processing.

Info Zone

Main memory, also called primary storage is a part of the processor unit. Main memory electronically stores data and program instructions when they are being processed.

Data is processed by a specific equipment that is often called computer hardware. This equipment consists of input devices, a processing unit, output devices and auxiliary storage units.

Input Devices : Input devices are used to enter data into a computer. A common input device is the keyboard as shown in the figure. As the data is entered, or keyed, it is stored in the computer and displayed on a screen.

Processor Unit : Above figure shows the processor unit of a computer, which contains the electronic circuits that actually causes the processing of data to occur. The processor unit is divided into two parts, the Central Processing Unit and Main Memory.

The Central Processing Unit (CPU) contains a Control Unit that executes the program instructions and an Arithmetic/Logic Unit (ALU) that performs numeric and logical operations. Arithmetic operations include numeric calculations, such as addition, subtraction, multiplication and division.

Output Devices

Output from a computer can be presented in many forms. The two most commonly used output devices are the printers and the computer screen. Other frequently used name for the screen is the monitor, or the CRT, which stands for Cathode Ray Tube.

Auxiliary Storage Units : Auxiliary storage units store instructions and data when they are not being used by the processor unit. A common auxiliary storage device on personal computers is a diskette drive or CD-drive, which stores data as magnetic spots on a small plastic disk or fibre disk. Another important auxiliary storage device is called a hard disk drive. Hard disk drives contain non-removable metal disks and provide larger storage capacities than diskettes or CDs.

Categories of Computers

Computers are generally classified according to their size, speed, processing capabilities and price. However, rapid changes in the technology make firm definitions of these categories difficult. This year's speed, performance and price classification of a mainframe might fit next year's classification of a minicomputer. Even though they are not firmly defined, the categories are frequently used and should be generally understood.

1. Microcomputers

Microcomputers, also called Personal Computers or Micros, are the small desktop sized systems that have become so widely used in recent years. These machines are generally priced under ₹ 40,000. This category also includes hand held notebook, laptop, portable and super microcomputer.

2. Minicomputers

Minicomputers are more powerful than microcomputers and can support a number of users performing different tasks. Originally developed to perform specific tasks such as engineering calculations, their use grew rapidly as their

performance and capabilities increased. These systems can cost from approximately ₹ 60,000 up to several hundred thousand rupees. The most powerful minis are called super mini computers.

3. Mainframe Computers

Mainframe computers are large systems that can handle hundreds of users, store large amounts of data and process transactions at a very high rate. Mainframes usually require a specialized environment including separate air conditioning, cooling and electrical power. Raised flooring is often built to accommodate many cables connecting the system components underneath. The price range for mainframes is from several hundred thousand rupees to several lakh rupees.

4. Supercomputers

Supercomputers are the most powerful category of computer and accordingly, the most expensive. The capability of these systems to process hundreds of millions of instructions per second is used for such applications as weather forecasting, engineering design and testing, space exploration and other jobs requiring long and complex calculations. These machines cost several lakhs rupees.

COMPUTER ETIQUETTES

A computer should not be kept on the edge of a table. It may fall and get damaged.

Let's Implement

A Tick (✓) the correct option.

1. It refers to data that has been processed into a form that has meaning and is useful.

   a. Data ◯ b. Storage ◯

   c. Information ◯ d. Processing ◯

2. It contains the electronic circuits that actually causes the processing of data to occur.

a. Processor Unit ◯ b. Auxiliary Storage Unit ◯

c. Storage Unit ◯ d. Output Unit ◯

3. This unit store instructions and data when they are not being used by the processor unit.

a. Processor Unit ◯ b. Auxiliary Storage Unit ◯

c. Storage Unit ◯ d. Output Unit ◯

4. They, also called Personal Computers or Micros, are the small desktop sized systems.

a. Minicomputers ◯ b. Macrocomputers ◯

c. Microcomputers ◯ d. Mainframe computers ◯

5. These computer require a specialized environment including separate air conditioning, cooling and electrical power.

a. Minicomputers ◯ b. Macrocomputers ◯

c. Microcomputers ◯ d. Mainframe computers ◯

B Fill in the blanks.

1. ______________ is data that has been processed into a meaningful and useful form.
2. ______________ devices are used to enter data into a computer.
3. The______________ unit contains the electronic circuit that cause processing to take place.
4. ______________ devices are used to print or display data and information.
5. Types of ________________ computers include hand-held notebook, laptop, portable and super microcomputer.

C Write **True** or **False**.

1. Machines save our labour and time and give accurate results. ▭
2. Information refers to data that has been processed into a form that has meaning and is useful. ▭
3. Input devices are used to give output. ▭
4. Main memory is also called primary memory storage. ▭
5. Microcomputers are also called Personal Computers. ▭

D Answer the following questions.

1. Define Computer.
2. Distinguish between Data and Information.

3. What is the Information Processing Cycle?
4. What is the difference between Input and Auxiliary storage?
5. Identify some of the differences among microcomputers, minicomputers, mainframe computers and supercomputers.

Refreshment Corner

A **Name any five input and output devices.**

	Input Devices	Output Devices
a.	____________________	____________________
b.	____________________	____________________
c.	____________________	____________________
d.	____________________	____________________
e.	____________________	____________________

B **Prepare a report for your class describing the use of computer at your school. You may focus on a single department or prepare a general report about computer uses throughout the school.**

C **Prepare a brief report on an individual who made a contribution to the history of computer with the help of his/her guardian or teacher.**

RACKING YOUR BRAIN

Life Skills and Values

Your friend is not able to understand IPO cycle, then what will you do? Justify your answer.

Just Have a Discussion

Communication

Discuss the different types of computers in the class in detail.

Teacher's Note

Explain the concept of IPO cycle to the students in detail.

2

MORE ON WINDOWS 10

Outlines

- Creating a New Desktop
- Working with the Control Panel

Windows 10 is a most secure version of Windows. Introduced by Microsoft with advanced and much improved feature, it is easy-to-use and provides a user-friendly environment. You get more security features and regular updates in Windows 10 to help safeguard against current and future threats. Due to its amazing features, it is gaining popularity, day-by-day. In the previous class, you have learnt about some of its unique features like, Cortana-the Search tool, Snap assist, two Windows store, a bundle of Universal apps, etc. In this chapter, you will be introduced to some more innovative tools that help to get things done quickly.

New Features of Windows 10

Windows 10 is loaded with multi-tasking tools: A custom Start menu, live titles, snap assist, task view and virtual desktop, to keep you organised, focussed and ready for anything. Let us read about its distinctive features.

More Personal

You can enjoy Windows 10 like a personal digital assistant. It is the best Windows interface that provides features to enable your device and phone to remain in synchronisation. You can customise the desktop, change background colour, theme, etc. according to your choice.

Continuum

The new feature Continuum in Windows 10 helps the operating system to work better with devices that support both the mouse and keyboard and touch input.

For convertible devices, there are two modes: Tablet and Desktop. If you wish to use the Tablet mode, follow the given steps:

- Click on the Notification icon present in the Notification Area of the taskbar.
- This will open the 'Action Center Pane' on the right side of the desktop.
- Click on the Tablet Mode button placed at the bottom of the Action Center Pane.

In the Tablet Mode, the Start menu opens in full screen along with the currently active window, if any. Apart from that, the Tablet Mode maximises all the apps, removes your taskbar icons, and leaves only the essential notification tray items. The taskbar in the Tablet Mode displays only the Start button, back arrow, search icon, and the task view button. Windows 10 automatically changes to this mode, if it detects that there is no keyboard attached.

Task View

Task View is another interesting feature of Windows 10 that enables you to organise the applications running on your computer. You can create multiple desktops and view all of them together by clicking on the Task View button located on the taskbar. When all the desktops are in view, you can also drag-and-drop an application from one desktop to another.

Searching made Easy

Windows 10 has the most enhanced and powerful searching tool 'Cortana', which answers all the questions that you ask, either in text or verbally. For verbal search, you need a microphone to talk. Cortana can play music, games, set alarms and reminders, take notes and emails, create and manage lists, find files, open any app on your system, and browse the web to respond to your query.

To turn Cortana on, follow the given steps :

- Click on the Search Box on the taskbar, followed by another click on the Gear icon on the left side of the displayed menu. A Settings menu will appear in the right pane.
- Turn Cortana on by moving the slider to the right in the Settings menu. Cortana will then start gathering information about you.
- Once activated, it becomes visible on the taskbar.
- You can also activate Cortana by giving a voice command like 'Hey Cortana', if

you have a microphone attached to your computer.

Edge

The Windows 10 browser, 'Edge' enables you to conduct a safe and quick search on the internet. It also provides a reading mode that opens a web page, displaying

Action Center

The new Action Center of Windows 10 displays alerts for your device and all the apps in a slide-out pane on the right side of the desktop. The upper part of the pane displays tips and app notifications, while the lower part displays a series of Quick Action buttons. These buttons enable you to perform actions, such as adjusting the screen brightness, turning the Bluetooth on or off, and switching to the Table mode. Along with the notifications, security tips, alarms, reminders, etc. are also displayed in the Action Center.

Note: Quick Actions are a set of tiles that give access to frequently-used settings and tasks.

Windows Hello

Windows Hello is a Microsoft's new built-in biometric security system for Window 10. It allows you to sign in to your system using data, like fingerprint, face or iris recognition. Biometric logins are secure, fast, and easy to create. Windows Hello requires appropriate hardware devices to be connected to the system to ensure that it works properly.

Creating a New Desktop

In Windows 10, you can create multiple desktops for organising the different sets of applications. To create a new desktop, follow the given steps:

- Click on the Task View button located on the taskbar. The Task View pane opens and displays the preview of all the open windows.
- Click on the New Desktop button on the Task View pane. A new desktop thumbnail named 'Desktop 2' appears.
- Click on the thumbnail to view the newly created desktop. It is a replica of Desktop 1 but does not display the applications opened in Desktop 1. You can

open the applications that you want to use on this desktop. You can also group the desktop windows by opening similar kind of applications in one desktop.

- Click on the Task View button again to switch between the desktops.

Moving Apps between Desktops

Follow the given steps to move an app from one desktop to another:

- Open the Task View pane by clicking on the Task View button. Place the mouse pointer over the desktop thumbnail.
- It will display the thumbnails of all the opened apps. Right-click on the app that is to be moved and select the Move to option.
- This will display the desktop list that you have created. Choose the desktop to

Note: You can also drag and drop an application window from one desktop to another.

Closing a Desktop

To close a desktop, open the Task View pane and place the mouse pointer on the desktop you want to close. Click on the Close button present in the upper-right corner of the desktop thumbnail to close the desktop.

Working with the Control Panel

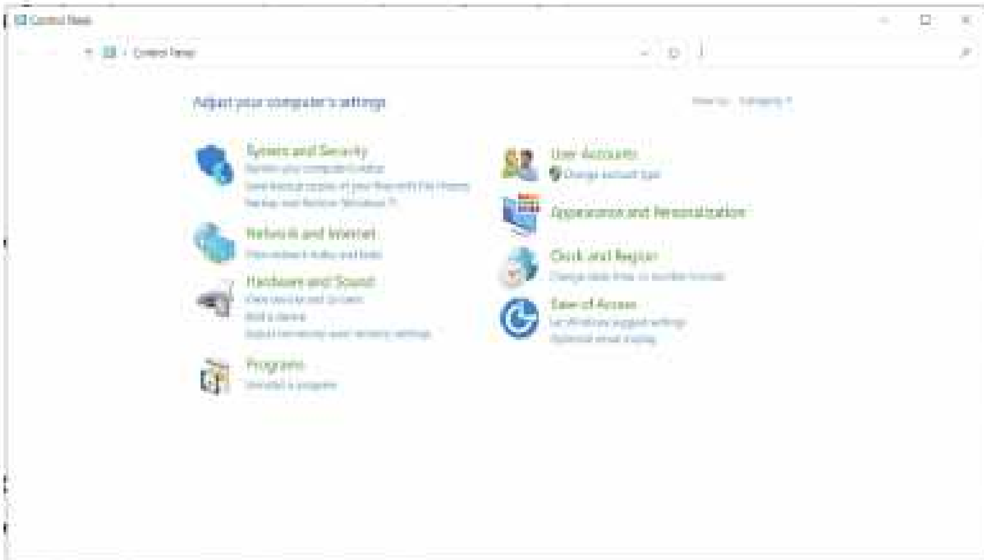
Control Panel is a system folder, using which you can make changes in the appearance and current setting of the Windows. These changes may include the following:

- Appearance and Personalisation of the desktop
- Hardware and Software setup and configuration
- System and Security
- Network and Internet
- User Accounts and Family Safety

- Setting the Clock, Language, and Region

These settings control nearly everything regarding the Windows appearance, internal settings, and personalisation. They allow you to customise the Windows according to your preference.

You can open the Control Panel in the following way:



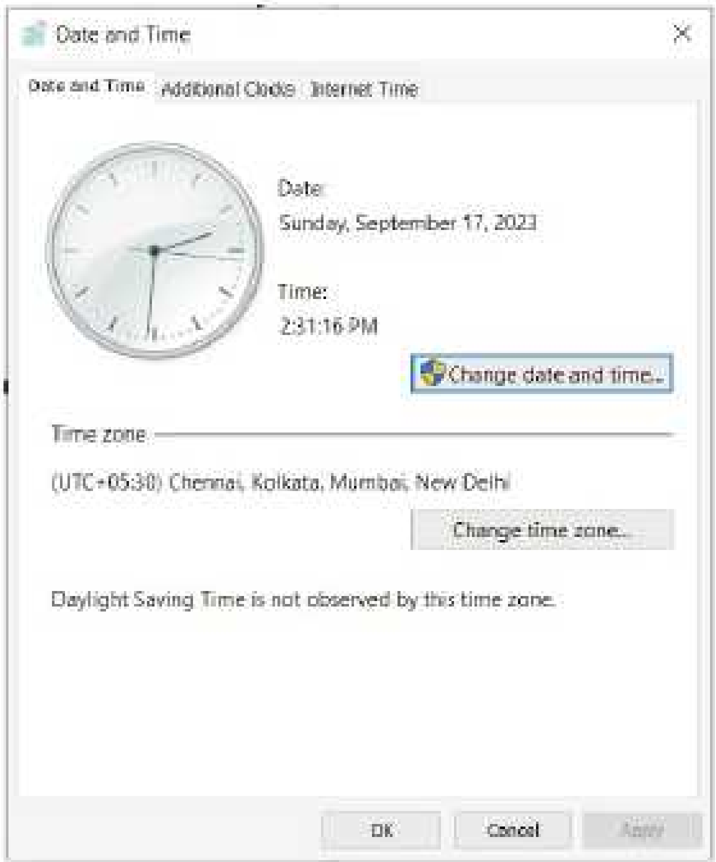
- Click on the Start > scroll down to Windows System. Select the Control Panel from the displayed sub-list.

  Or

- Type 'Control Panel' in the Search box and press Enter.

Changing Date and Time

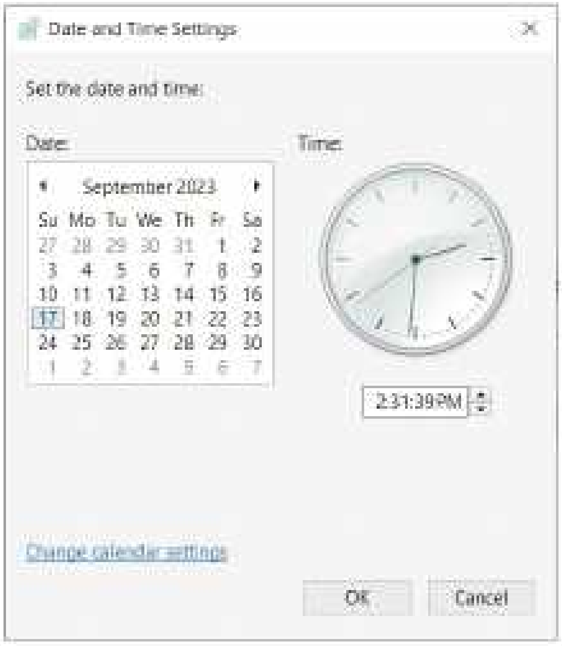
To change a computer's date and time :



- Click on the Date and Time option in the Control Panel. The Date and Time dialog box will appear.
- By default, the Date and Time tab is selected.

To Change Date

- Click on the Change date and time button. The Date and Time Settings dialog box will appear.
- Select the current day in the calendar that is displayed. You can change the month using the



forward/backward arrow buttons present on the calendar.

To Change Time

- To change the hour, select the hour in the text box located below the clock.
- Click on the spin arrows to increase or decrease the hours, as desired. Similarly, you can change the minutes and seconds. Click OK. The Date and time settings of the system will change accordingly.

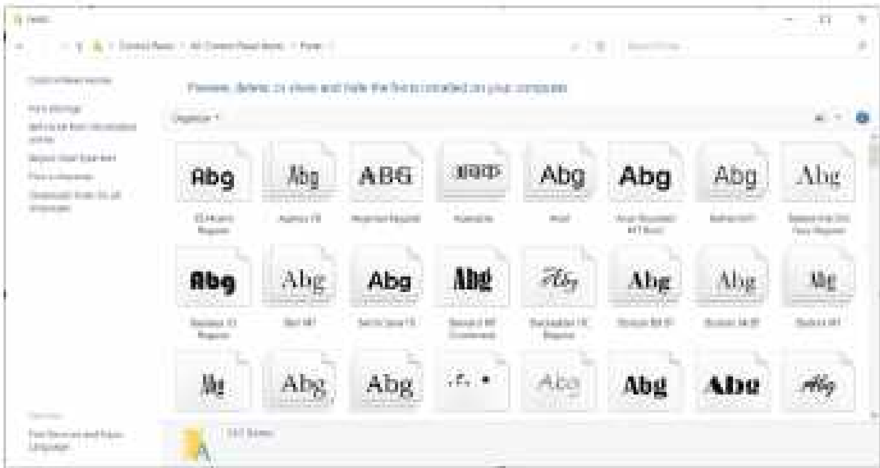
Fonts

Different styles of writing characters of text in a specific size, are known as Fonts. Some of the popular fonts are Arial, Calibri, Verdana, Times New Roman, etc. Font settings are used to change the display of the text on the screen as well as on paper, while printing. You can use the Control Panel to view fonts, add new fonts, or delete the existing ones.

To view the fonts in the Control Panel, follow the given steps:

- Click on the Fonts option in the Control Panel. The Fonts window appears.
- To view the sample of a font, double-click on any given font.

To add new font(s) to the Fonts list, follow the steps given below:



- Open This PC folder by double-clicking its icon on the desktop.
- Select the appropriate drive and then open the folder that has the new fonts.
- Select the font that you

want to add and drag it to the Fonts window. The new font will be added in the list of the existing fonts. You can also copy and paste the desired font in the Fonts window.

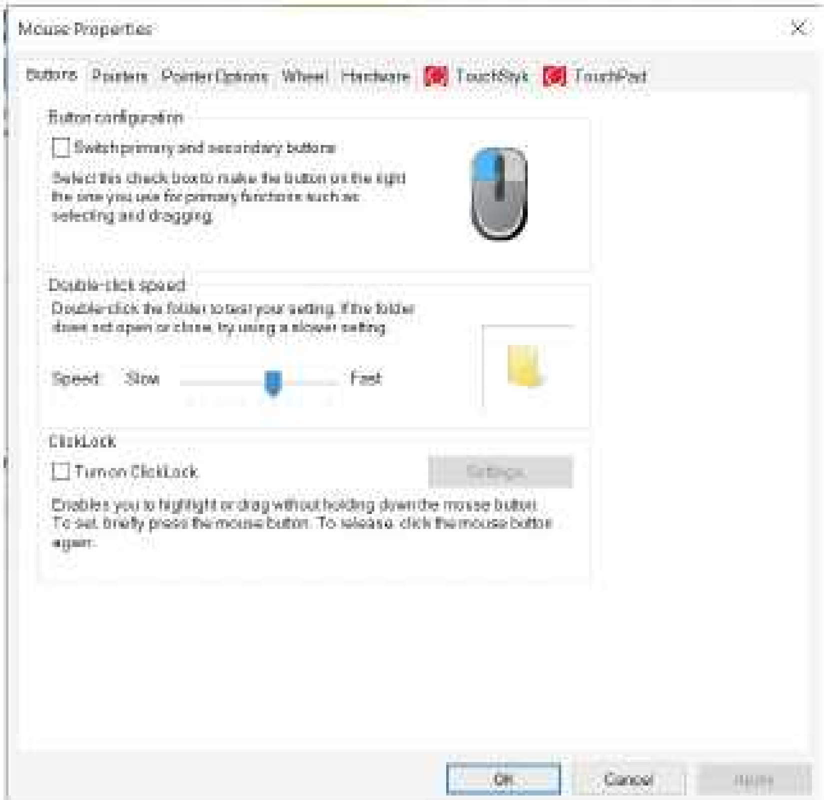
You can also delete an existing font. To do so:

- Select the font that you want to delete.
- Click on the Delete button present above the font thumbnails to remove it from the list.

Changing the Mouse Settings

You can customise the mouse setting in various ways, such as changing the appearance of a mouse pointer, functions of the mouse buttons or altering the scroll speed of the mouse wheel. To do so, follow the given steps:

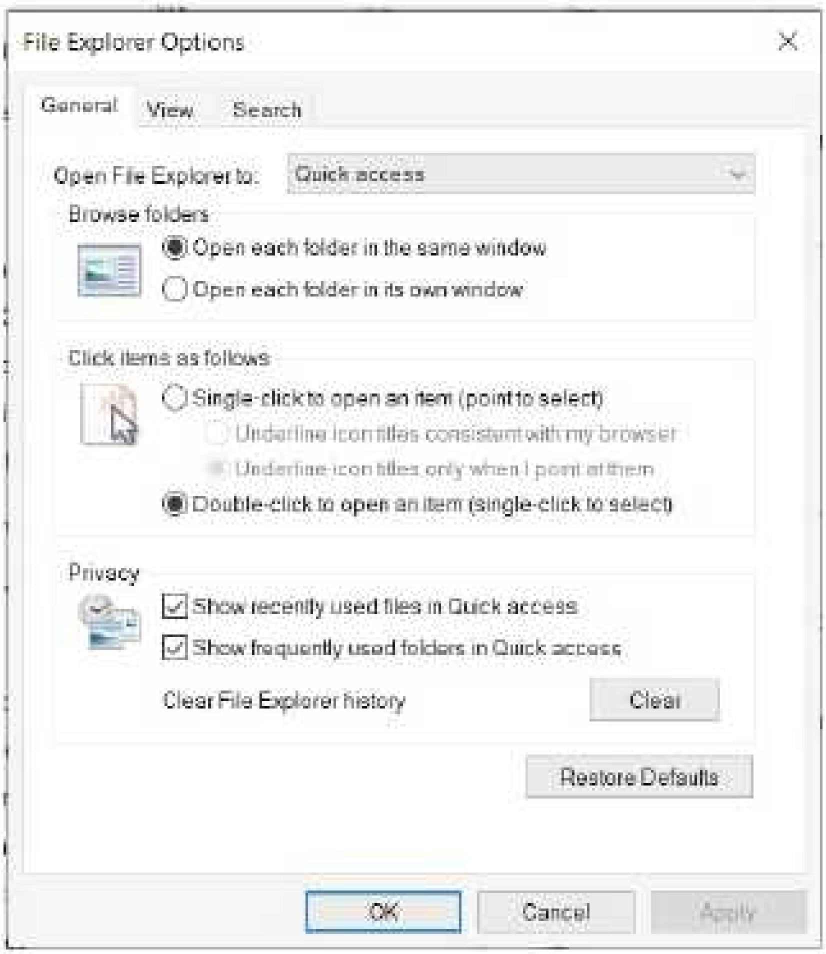
- Select the Mouse option in the Control Panel. The Mouse Properties dialog box will appear.



- The Buttons tab is selected by default. It provides options to change the settings of the mouse buttons, like swapping the functionality of the left and right button, increasing or decreasing the double click speed, etc.
- Under Button configuration section, select the checkbox to make use of the right button for primary functions, like selecting and dragging.
- To change the double-click speed of your mouse, drag the slider under the Double-click speed section, towards the left to make it slow and towards the right to make it fast. To test the speed, double-click the folder given on the right side. If the folder does not open or close, try using a slower speed.
- To change the appearance of your mouse pointer, click on the Pointers tab.

Here, you can change the size, colour, and shape of your mouse pointers.

- To change the pointer image, click on the Scheme drop-down list and choose any mouse pointer scheme. You can also change the pointer images for particular functions, like Normal Select, Busy, etc. To do so, click the pointer function in the Customize section and then click on the Browse button. Double-click on any file name and observe the change.
- Click on the Apply button to incorporate the change or use the Default button to return to the default settings.
- You can also adjust the speed of your mouse pointer by clicking on the Pointer Options tab and dragging the speed slider similar to as shown in the figure.
- Drag the Slider from slow to fast under the Motion section.
- Click on the Apply button and then click OK.



Replacing Double-Click with Single-Click

Windows has the feature to replace the double-click setting of a mouse with single-click to open any file or program. Follow these simple steps and forget the practice of double-clicking the mouse forever:

- Click on the File Explorer Options in the Control panel. The File Explorer Options dialog box appears. Turn on the Single-click to open an item option present under the Click items as follows section.
- From the sub-options, select the Underline icon titles only when I point at them option.
- Click on Apply and then click OK.

COMPUTER ETIQUETTES

Do not play games for long hours on a computer. It is very harmful for your eyes. Try to play outdoor games with your friends to remain fit and healthy.

Let's Implement

A Tick (✓) the correct option.

1. For convertible devices, there are ____________ modes.
   a. three ◯ b. two ◯
   c. four ◯ d. None of these ◯
2. You can customise the mouse settings by selecting the ____________ in the Control Panel.
   a. File Explorer ◯ b. Mouse option ◯
   c. Action Center ◯ d. Cortana ◯
3. The ____________ displays alerts for your device and all the apps in a slide-out pane on the right side of the desktop.
   a. Tablet Mode ◯ b. Cortana ◯
   c. File Explorer ◯ d. Action Center ◯
4. The ____________ is a new feature in Windows 10, which enables you to organise the applications running on your computer.
   a. Task View ◯ b. File Explorer ◯
   c. Cortana ◯ d. Action Center ◯
5. ____________ in Windows 10 helps you to quickly find information on the internet.
   a. Control Panel ◯ b. Action Center ◯
   c. Task View ◯ d. Cortana ◯

B Fill in the blanks.

1. ____________ is easy-to-use and provides a user-friendly environment.
2. ____________ allows you to sign in to your system using data, like fingerprint, face or Iris recognition.
3. Different styles of writing characters in a specific style are known as ____________.
4. You can copy and paste the desired font in the ____________ window.
5. ____________ allows you to change Windows 10 into a touch-friendly interface called the Tablet Mode.

C Write **True** or **False**.

1. You can drag and drop an app from one desktop to another in the Task View pane.
2. Windows 10 automatically changes to the Tablet mode if it detects that there is no keyboard attached.
3. The reading mode of Microsoft Edge opens a web page with only the pictures displayed.
4. Windows Hello enables you to perform quick actions, such as adjusting the screen brightness.
5. Some of the popular fonts are Arial, Calibri, Verdana, Times New Roman, etc.

D Answer the following questions.

1. What is the use of the Windows Media Player?
2. How can we move an app from one desktop to another?
3. What is the use of Windows Hello feature?
4. Write the steps to create a new desktop.
5. How is the 'Edge' feature useful?
6. How can we close a desktop?

Refreshment Corner

- **Draw and colour the picture of Windows 10 logo in the space given below.**

Practical Session

Experiential Learning

A Adding a New Font

- Open Control Panel and click on the Fonts option. The Fonts window appears.
- Now, open This PC folder by double-clicking its icon on the desktop.

B Deleting an Existing Font

- Select the font that you want to delete.
- Click on the Delete button present above the font thumbnails to remove it from the list.

C Creating a New Desktop

- Click on the Task View button located on the taskbar.
- In the Task View pane, click on the New desktop button located on the bottom-right corner. A new desktop icon named Desktop 2 will appear at the bottom.
- Click on the Desktop 2 icon to open it. Open the application Word and PowerPoint in Desktop 2.
- Move the PowerPoint window from Desktop 2 to Desktop 1.

Competency Based Questions

1. Sia wants to create multiple desktops and view all of them together. Which feature of Windows 10 should she use?
2. Kriti wants to make her system more secure through biometric login. Which feature of Windows 10 can help her to do so?
3. Manish wants as powerful searching tool to get answers of all his questions either in text or verbally. Which searching tool will help him.

RACKING YOUR BRAIN

Life Skills and Values

Your friend purchased a new computer having Windows 10 operating system. He is not familiar with Windows 10. How will you help him/her? Justify your answer.

Just Have a Discussion

Communication

Discuss the following topic:

- Impact of Windows 10 on the Windows Ecosystem

TEACHER'S NOTE

The teacher should demonstrate to the students, how to change the display settings using the control panel.

3

ADVANCED FEATURES OF MS WORD 2016

Outlines

- Tables
- Creating a Table
- Entering Data in a Table
- Modifying a Table
- Formatting a Table
- Calculations in a Table

Tables

Tables are useful for various tasks such as presenting text information and numerical data. In Word, you can create a blank table, convert text to a table, and apply a variety of styles and formats to existing tables.

A table is a set of data (text and/or numbers) arranged in rows and columns. The intersection of rows and columns forms rectangular boxes called cells.

The vertical series of cells in a table is called a column. The horizontal series of cells in a table is called a row.

Table is useful to:

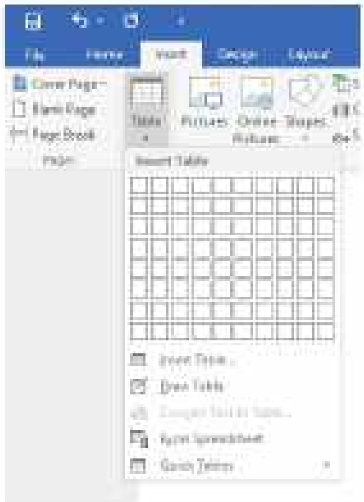
- Organize a collection of related data in rows and columns.
- Find the information easily.
- Compare the information on different categories.

Creating a Table

In MS Word 2016 document, a table can be created in different ways that are discussed here one by one:

Using a Grid

To insert a table using a grid, follow these steps:



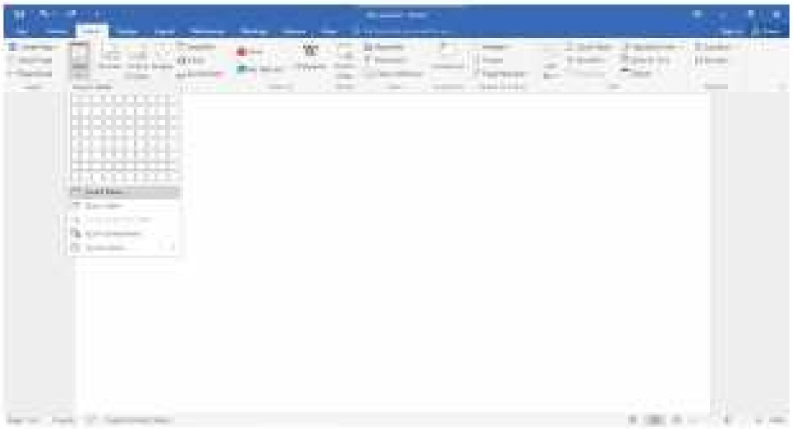
1. Place the insertion point in the document where you want the table to appear.
2. Click on the Insert tab.
3. Click on the Table option in the Tables group. A table grid appears.
4. Drag the mouse button to highlight the desired number of rows and columns. Click the mouse button.

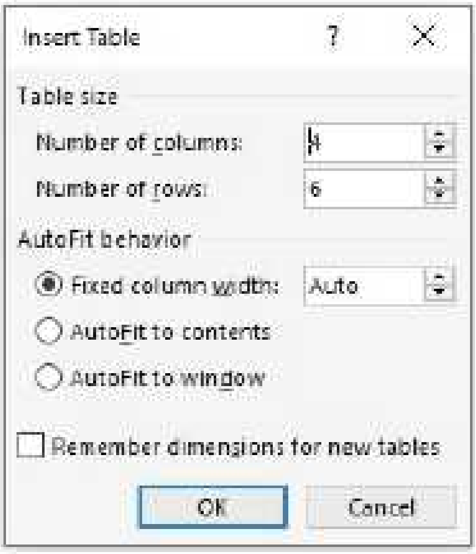
An empty table is inserted in the document.

Using the Insert Table Option

The steps to insert a table in a document using the Insert Table option are as follows:

1. Click on the Insert tab.
2. Click on the Table option in the Tables group.
3. Select the Insert Table... option. The Insert Table dialog box appears.
4. Specify the number of rows and columns.
5. Click on the OK button.



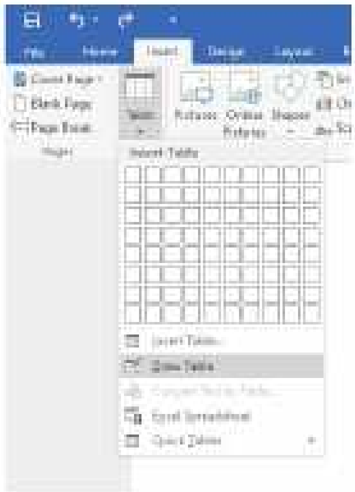


Drawing a Custom Table

To draw a custom table, follow these steps:

1. Click on the Table option in the Tables group on the Insert tab.
2. Click on the Draw Table option.

3. Click and drag to draw a rectangle in the document. Draw horizontal lines to create rows, and draw vertical lines to create columns.



Converting Text to a Table

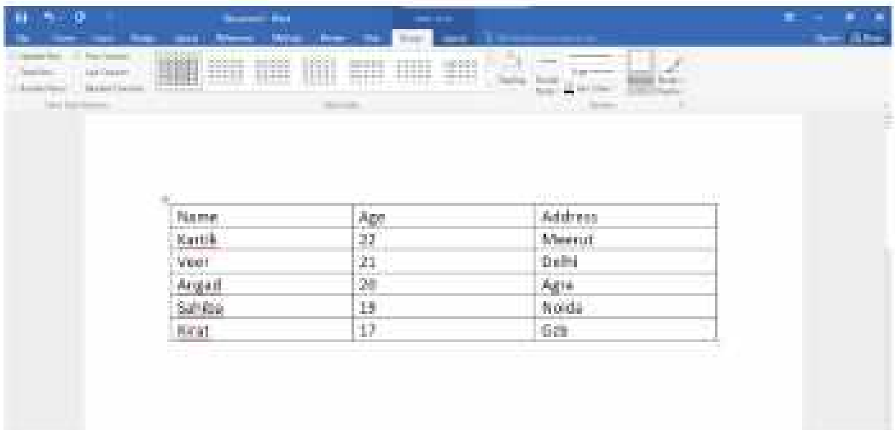
In Word 2016, you can convert the text to a table. In this example given below, each row of information contains an item name and price, separated by spaces.

1. Select the text.
2. Click on the Insert tab.
3. Click on the Table option in the Tables group.
4. Click on the Convert Text to Table... option. The Convert Text to Table dialog box appears.
5. Click on the OK button.

Note: The Convert Text to Table option gets highlighted in the Table drop-down menu only when the text is selected.

Entering Data in a Table

You can enter text in a table in the same way as you enter text in a document. To enter data in a cell, click in it and then type the text.



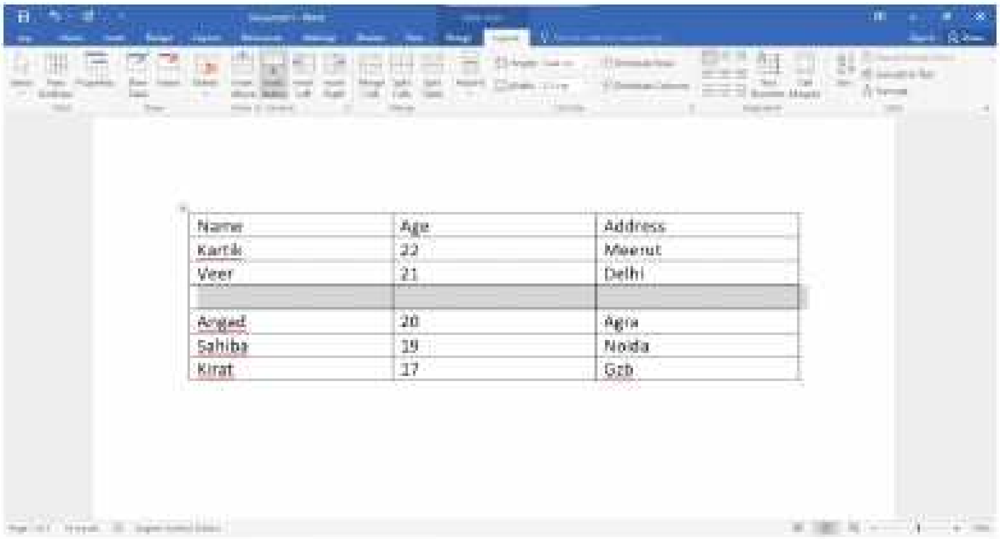
- To move to the next cell, press Tab key.
- To move to the previous cell, press Shift + Tab keys.
- To move through the cells, press arrow keys.

Modifying a Table

Inserting a New Row

To insert a new row in a table, follow these steps:

1. Place the cursor in the column to the left or right of which a new column is to be inserted.
2. Click on the Layout tab under Table Tools.
3. Click on the Insert Above or Insert Below option in the Rows & Columns group.



Info Zone

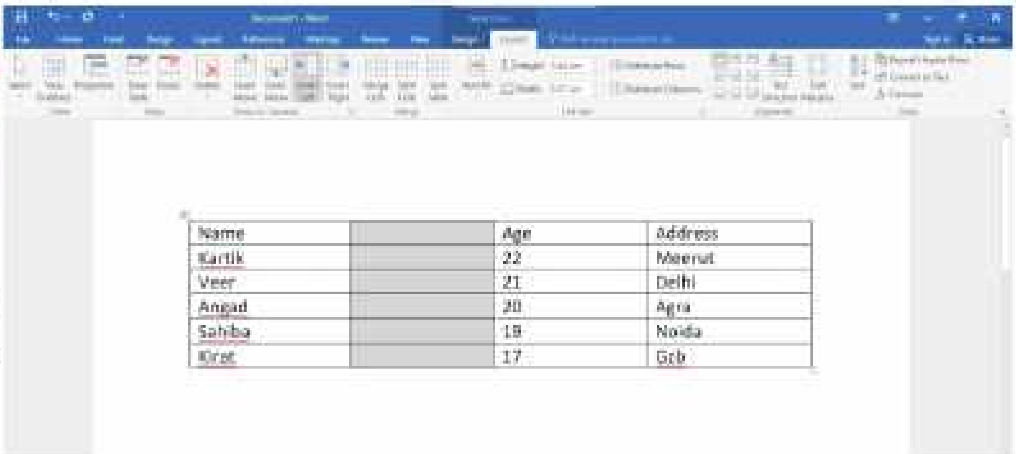
You can add a row quickly at the end of a table. For this, click on the last cell of the last row and then press the Tab key.

Inserting a New Column

To insert a new column in a table, follow these steps:

1. Place the cursor in the column to the left or right of which a new column is to be inserted.
2. Click on the Layout tab under Table Tools.
3. Click on the Insert Left or Insert Right option in the Rows and Columns group.

   A new row gets inserted in the table.



Smart Tip

Alternatively, you can right-click the table, then hover the mouse over Insert to see various rows and columns options. Then, select the desired option.

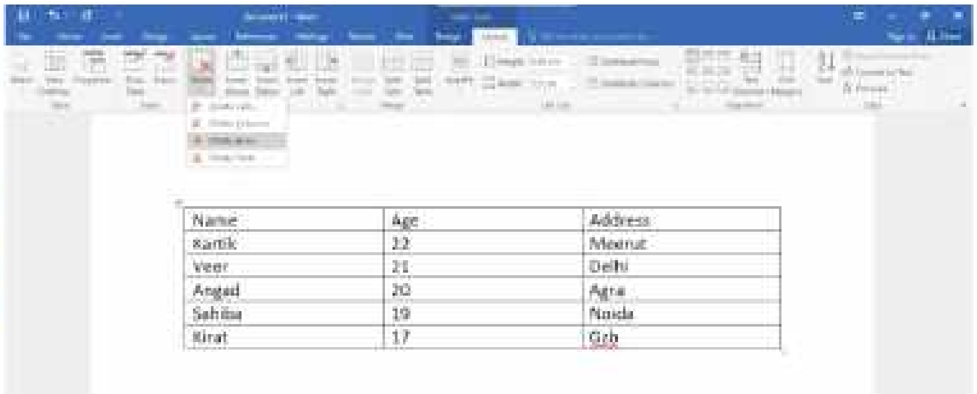
Info Zone

We can add a row or a column in a table by clicking on the Plus + sign, which appears when we place a mouse pointer over the intersecting line between a row and a column.

Deleting Rows or Columns

To delete rows or columns, follow these steps:

1. Place the cursor in the row or column to be deleted.
2. Click on the Layout tab under Table Tools.
3. Click on the Delete option in the Rows and Columns group. A list of options appears.

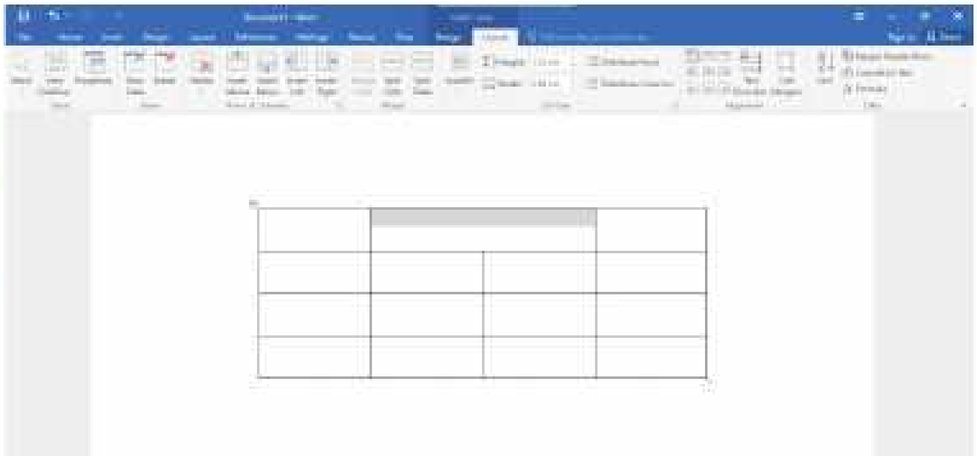


4. Click on the Delete Columns or Delete Rows option to delete a row or column.

Merging Cells

Merging cells refers to combining two or more cells in a row or a column into a single cell. To merge cells, follow these steps:

1. Select the cells to be merged.
2. Click on the Layout tab under Table Tools.
3. Click on the Merge Cells option in the Merge group.

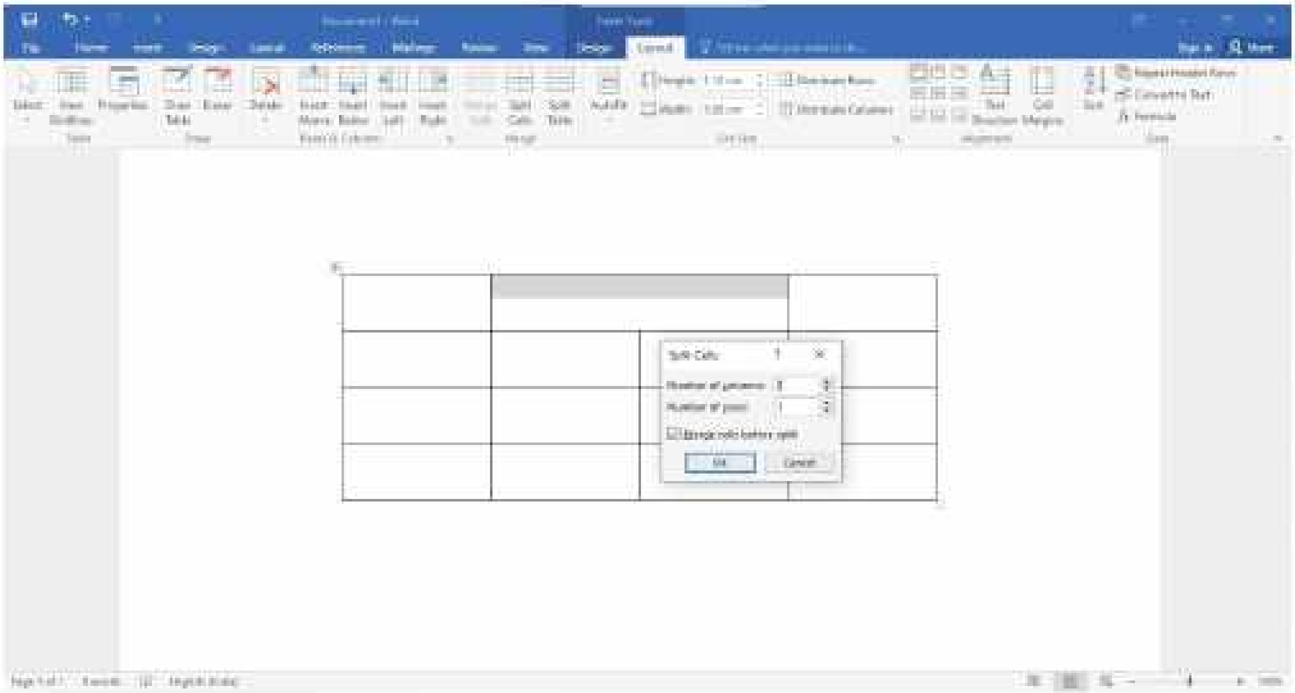


The selected range of cells get merged into a single cell.

Splitting Cells

Splitting cells refers to dividing a single cell into multiple cells. To split a cell, follow these steps:

1. Click on the cell that you want to split.
2. Click on the Layout tab under Table Tools.
3. Click on the Split Cells option in the Merge group.
4. Specify the number of columns and rows.
5. Click on the OK button.



Formatting a Table

After creating a table, you can format it, *i.e.*, apply different styles or apply borders and shading on it.

Table Styles

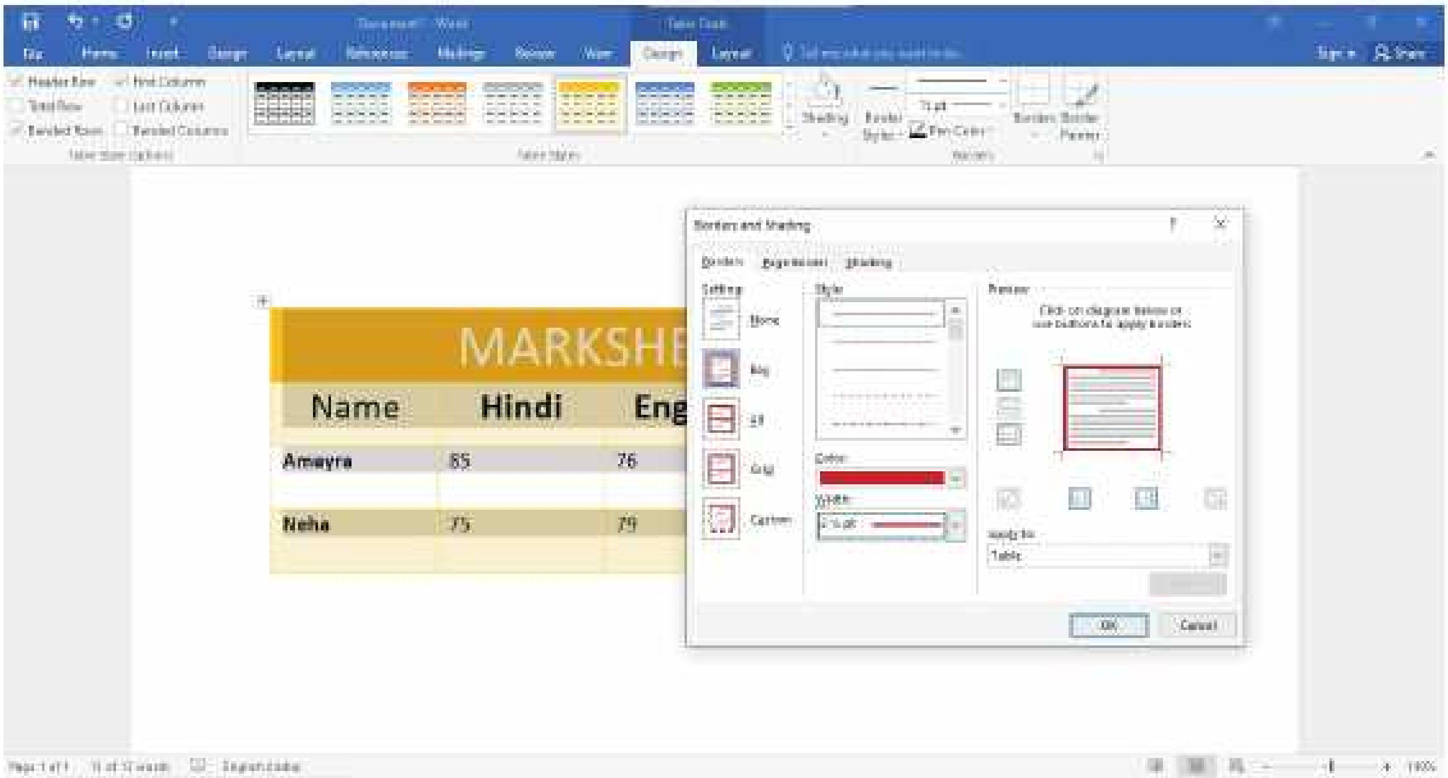
MS Word provides many inbuilt table styles that let you quickly change the borders, shading, alignment and fonts in tables.

1. Select the table or click anywhere inside the table.
2. Click on the Design tab under Table Tools.
3. Click on the More button in the Table Styles group to see all the table styles.
4. Click and select the desired style.

Applying Borders to the Table

To apply borders, follow these steps:

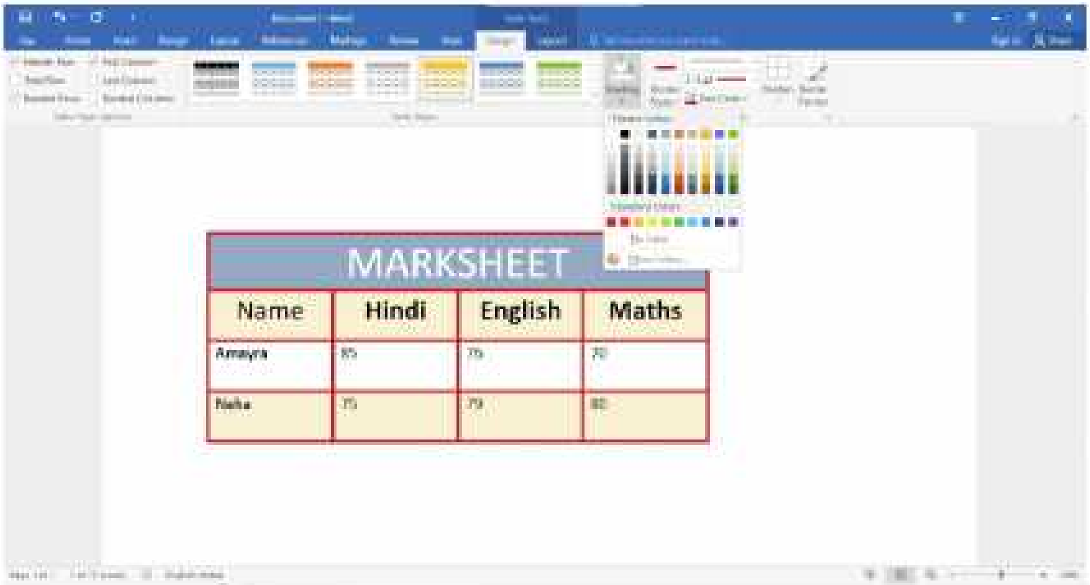
1. Select the table or position the cursor in any cell of the table to apply table border.
2. Click on the Design tab under Table Tools.
3. Click on the arrow next to the Borders option in the Table Styles group. A drop-down list appears.
4. Click on the Borders and Shading... option.
5. Select the desired Setting, Style, Color and Width for the border under Borders tab.
6. Click on the OK button.



Applying Shading to the Table

To apply background colour on the table or on selected cells, follow these steps:

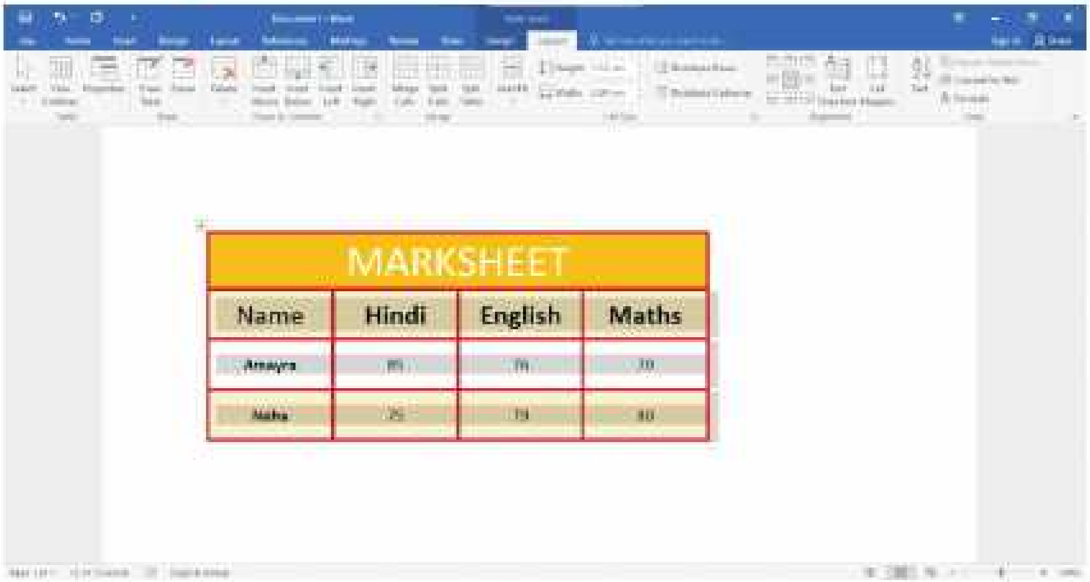
1. Select the table or select the cells.
2. Click on the Design tab under Table Tools.
3. Click on the arrow next to the Shading option in the Table Styles group. A list of colour options appears.
4. Click and select the colour of your choice.



Alignment of Text in Table

To change the alignment of text, follow these steps:

1. Select the cell(s) that contains the text you want to align.
2. Click on the Layout tab under Table Tools.

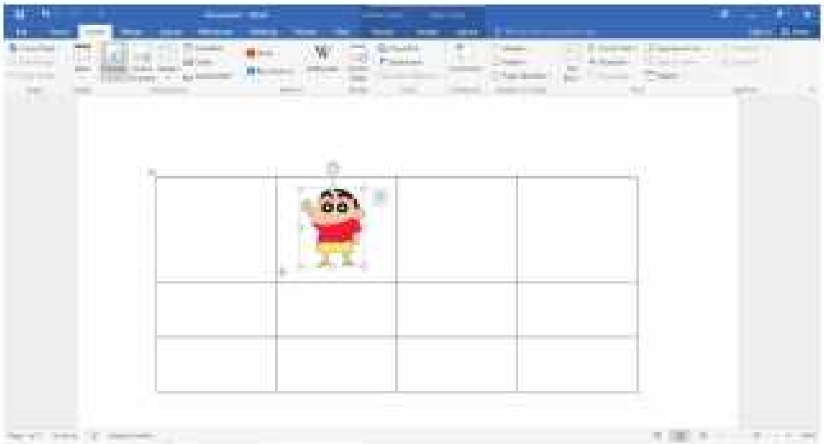


Formatting Text Inside the Table

You can format the text inside a table in the same way as you format text in a document, using various options in the Font group on the Home tab.

Inserting Picture in a Table

You can also insert a picture in a table cell. To insert a picture in a table cell, follow these steps:

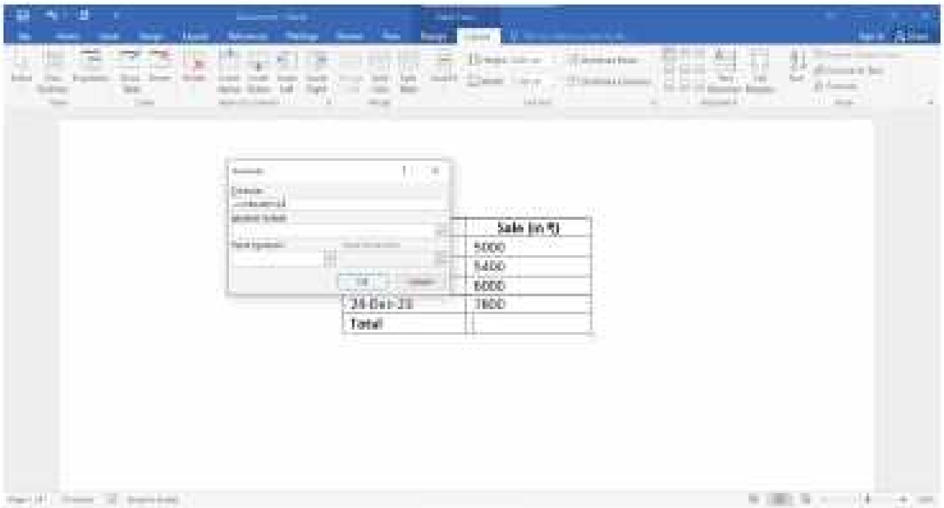


1. Click on the cell where you want to insert a picture.
2. Click on the Pictures option in the Illustrations group on the Insert tab. The Insert Picture dialog box appears.
3. Browse and select the picture you want to insert.
4. Click on the Insert button.

Calculations in a Table

You can perform calculations on the numeric data entered in a table in Word. To perform calculation, follow these steps:

1. Click in the cell where the result is to be displayed.
2. Click on the Layout tab under Table Tools.
3. Click on the Formula option. The Formula dialog box appears.
4. The SUM formula is already displayed. Click on the OK button.



COMPUTER ETIQUETTES

Do not press the keys of the keyboard hard.

Let's Implement

A Tick (✓) the correct option.

1. To insert a row in a table, click on the ____________ tab.
   a. Home ◯ b. Insert ◯ c. Design ◯ d. Layout ◯
2. To move to the next cell, press ____________ key.
   a. Tab ◯ b. Shift ◯ c. arrow ◯ d. delete ◯
3. ________________ to move through the cells press.
   a. Tab ◯ b. arrow ◯
   c. Both a and b ◯ d. None of these ◯
4. Insert above option is in the ________________ group.
   a. Merge ◯ b. Table Styles ◯
   c. Rows and Columns ◯ d. None of these ◯
5. The horizontal series of cells in a table is called a ____________.
   a. cell ◯ b. row ◯ c. table ◯ d. column ◯

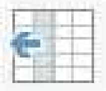
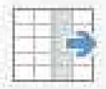
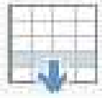
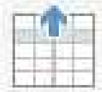
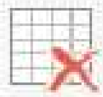
B Fill in the blanks.

1. The intersection of rows and columns forms rectangular boxes called ____________.
2. To move to the next cell in a table, press the ____________ key.
3. The Merge Cells option is present in the ____________ group on the Layout tab.
4. ____________ cells refer to dividing a single cell into multiple cells.
5. We can position the cursor in any cell of the table to apply ____________.

C Write **True** or **False**.

1. You cannot merge the cells of a row in a table. ▭
2. The vertical series of cells in a table is called a row. ▭
3. The Borders option is available on the Design tab. ▭
4. Tables are useful for presenting numerical data. ▭
5. We can enter text in a table in the same way as we enter text in a document. ▭

D Match the following.

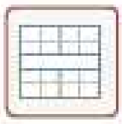
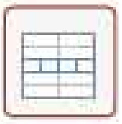
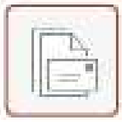
1.		a.	Insert Above
2.		b.	Insert Left
3.		c.	Delete
4.		d.	Insert Below
5.		e.	Insert Right

E Answer the following questions.

1. What is the use of a table?
2. How can we insert a table using a grid?
3. How can we convert text to a table?
4. How can we delete rows or columns?
5. Write the steps to merge the cells of a column.
6. What should we do to change the alignment of text in a table?

Refreshment Corner

Identify the following options and write their names.

1.  ____________________
2.  ____________________
3.  ____________________
4.  ____________________
5.  ____________________
6.  ____________________

Practical Session

Experiential Learning

A **Create the given tables in MS Word 2016 as shown below.**

Marksheet 2024				
Name	**Hindi**	**English**	**Maths**	**Total**
Kanak	89	84	75	248
Ritika	78	79	75	232
Shikha	75	72	79	226
Monty	84	75	76	235

Cartoon Characters	Images
Motu Patlu	
Pepa Pig	
Shin-Chan	

B **Insert a new column in a table using the given steps.**

- Place the cursor in the column to the left or right of which a new column is to be inserted.
- Click on the Layout tab under Table Tools.
- Click on the Insert Left or Insert Right option in the Rows and Columns group.

Competency Based Questions

Critical Thinking

1. Rajeev is a computer operator in ABC Company. This month his company has made ten new clients. How can he maintain the contact details of these new clients using MS Word 2016?
2. Observe the dialog box given alongside and answer the following questions:
   (a) Name the tab, group and option you will click to get this dialog box.
   (b) How many rows and columns are specified for the table?

RACKING YOUR BRAIN

Life Skills and Values

Your friend wants to make a table in MS Word 2016 using all options of table. He/She is trying again and again but not able to do so. What will you do? Justify your answer.

Just Have a Discussion

Communication

Discuss the different ways of creating a table in MS Word 2016 in the class.

Teacher's Note

- Demonstrate to the students how the appearance of a Word document can be improved.
- Demonstrate the use of mail various options of table to the students in detail.

4

EDITING AND FORMATTING IN EXCEL 2016

Outlines

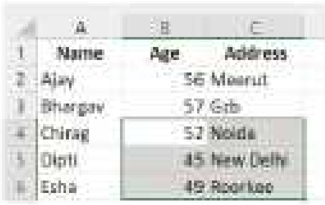
- Selecting Cells
- Deleting Cell's Content
- Inserting Cells, Rows and Columns
- Changing Row Height
- Font Group
- Editing Cell's Content
- Copying and Moving Data
- Deleting Cells, Rows and Columns
- Changing Column Width
- Alignment Group

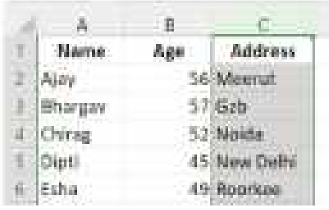
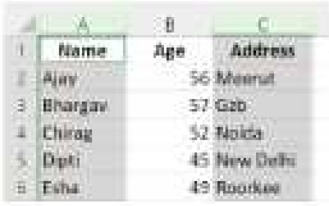
A spreadsheet software is used to organise and display data in a form as required by the user. The software uses worksheets for data entry and manipulation.

Let us learn about some of the editing and formatting operations that we can perform on a worksheet.

Selecting Cells

Some operations, for example, copying, moving, etc. may need to be performed on a range of cells.

To select a single cell	Click on the cell or use the arrow keys to move to the desired cell.	
A range of cells	Select the first cell in the range and then drag the selection to the last cell in the range, which we need to select.	

An entire row/column	▪ Click the row heading to select a complete row.	
	▪ Click the column heading to select a complete column.	
To select multiple ranges simultaneously	Select the first range of cells and hold the Ctrl key. Then, select another range of cells.	
To select an entire worksheet	Click on the Select All button.	

Smart Tip

- To select an entire row, press the Shift + Spacebar keys together.
- To select an entire column, press the Ctrl + Spacebar keys together.
- To select an entire worksheet, press the Ctrl + A keys together.

Editing Cell's Content

After we have entered data in a cell, we might need to change it in some way or the other. There are two ways in which we can edit the cell's content.

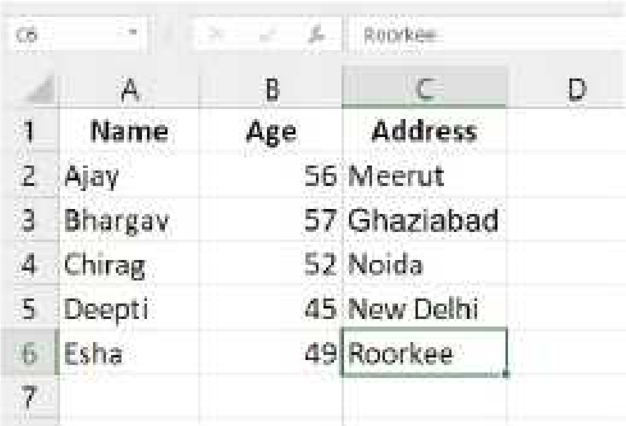
- Changing the content completely
- Modifying the content partially

Changing the Content Completely

To change the content of a cell completely, follow these steps :

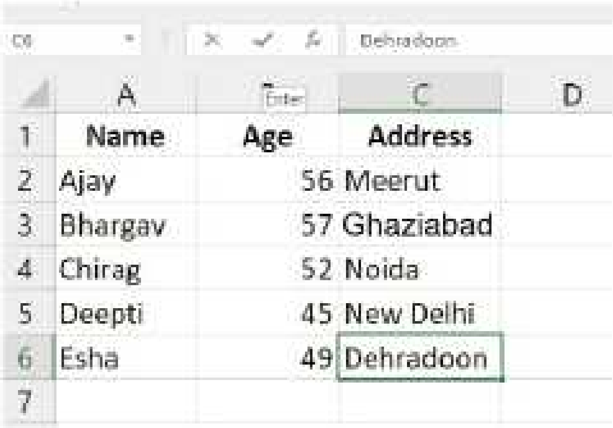
1. Select the cell whose content has to be changed.
2. Type in the new data into the cell.

3. Then, press the Enter key or the Enter button on the Formula Bar to accept the changes.



C6 | Roorkee

	A	B	C	D
1	Name	Age	Address	
2	Ajay	56	Meerut	
3	Bhargav	57	Ghaziabad	
4	Chirag	52	Noida	
5	Deepti	45	New Delhi	
6	Esha	49	Roorkee	
7				



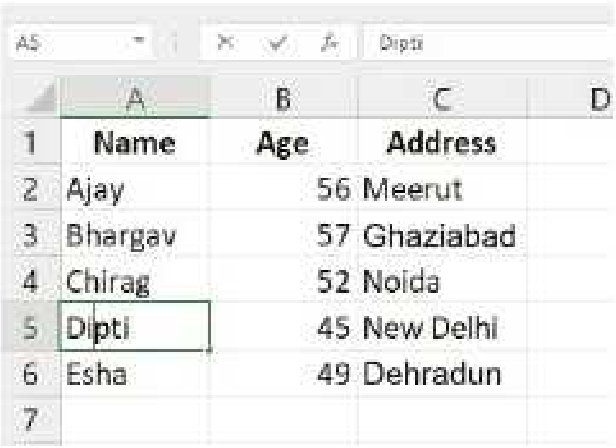
C6 | Dehradoon

	A	B	C	D
1	Name	Age	Address	
2	Ajay	56	Meerut	
3	Bhargav	57	Ghaziabad	
4	Chirag	52	Noida	
5	Deepti	45	New Delhi	
6	Esha	49	Dehradoon	
7				

Modifying the Content partially

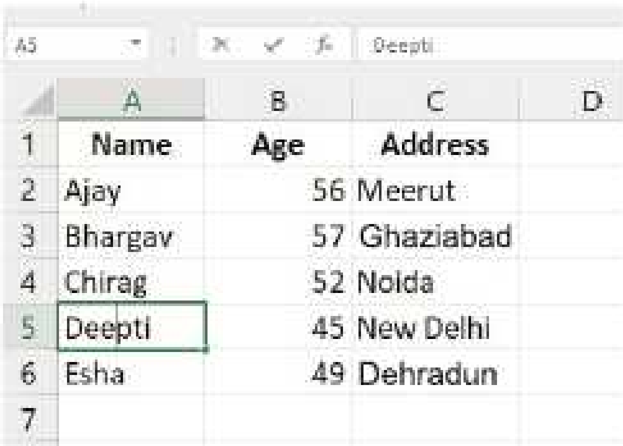
To modify the content of a cell partially, follow these steps:

1. Double-click the cell to be edited. The cursor will appear in the cell.
2. Edit the content of the cell in the cell itself, or in the Formula Bar.
3. Then, press the Enter key or the Enter button ✓ on the Formula Bar to accept the changes.



A5 | Dipti

	A	B	C	D
1	Name	Age	Address	
2	Ajay	56	Meerut	
3	Bhargav	57	Ghaziabad	
4	Chirag	52	Noida	
5	Dipti	45	New Delhi	
6	Esha	49	Dehradun	
7				



A5 | Deepti

	A	B	C	D
1	Name	Age	Address	
2	Ajay	56	Meerut	
3	Bhargav	57	Ghaziabad	
4	Chirag	52	Noida	
5	Deepti	45	New Delhi	
6	Esha	49	Dehradun	
7				

Smart Tip

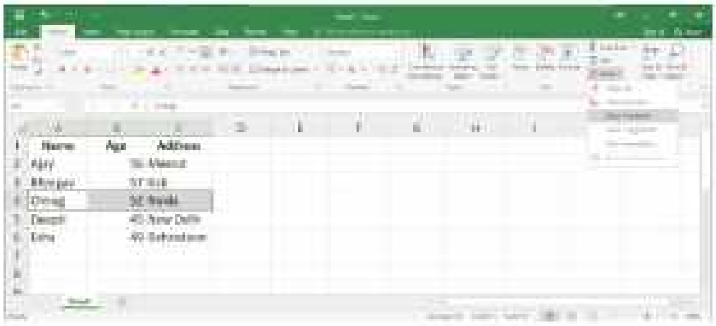
Press the F2 key after selecting the cell to change the cell's content.

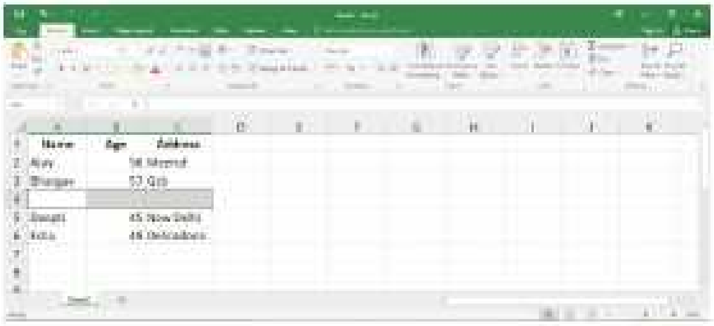
Deleting Cell's Content

To delete the cell's content, follow these steps:

1. Select the cell(s).

2. Click on the drop-down menu arrow of the Clear button in the Editing group on the Home tab. A menu appears.
3. Select the Clear Contents option.





The selected content will be deleted.

Smart Tip 

- To delete the cell's content, select the cell(s) and press the Delete key.

Or

- Select the cell(s). Right-click and select the Clear Contents from the shortcut menu.

Copying and Moving Data

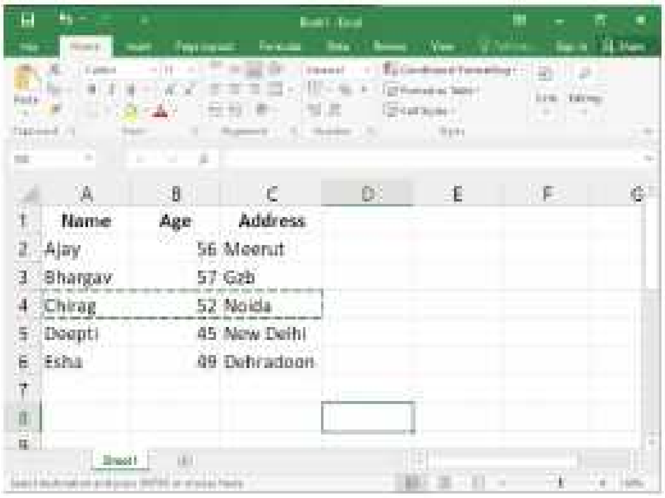
The content of the worksheet can be copied or moved to a different position on the same worksheet or on a different worksheet.

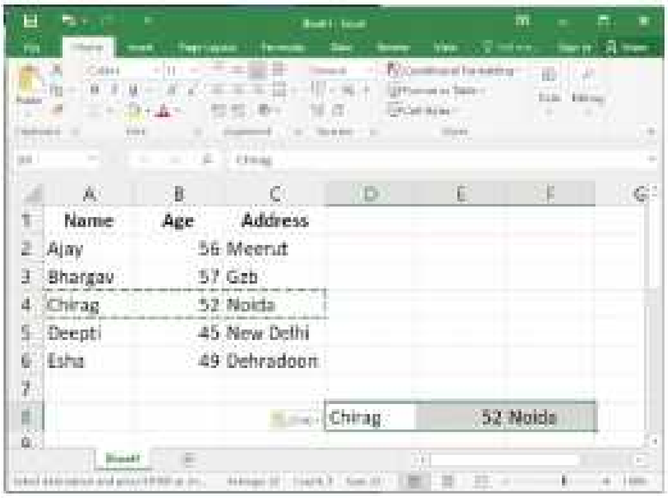
Copying Cell's Content

You can copy data from one location to another using the Copy and Paste options. To copy the cell's content, follow these steps :

1. First, select the range of cells to be copied.
2. Click on the Copy option in the Clipboard group on the Home tab. We will see a dotted line around the selected cells.

3. Select the destination cell where we want to paste the data.
4. Click on the Paste option in the Clipboard group on the Home tab.



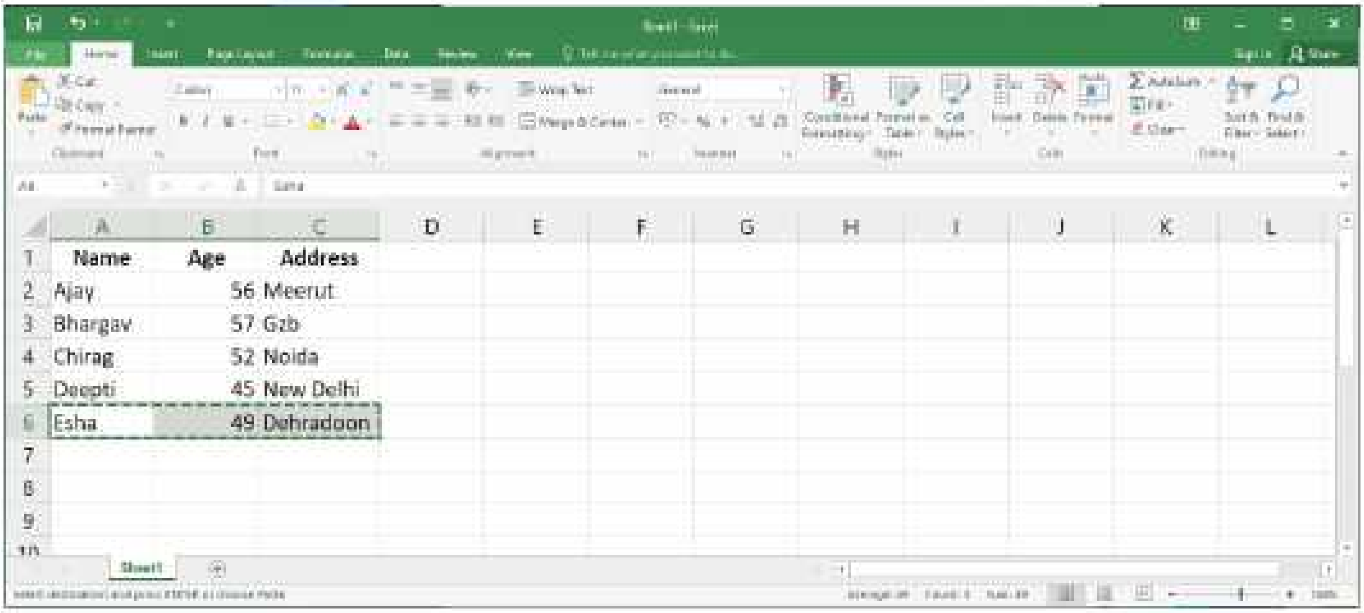


The selected cells are copied to the new location.

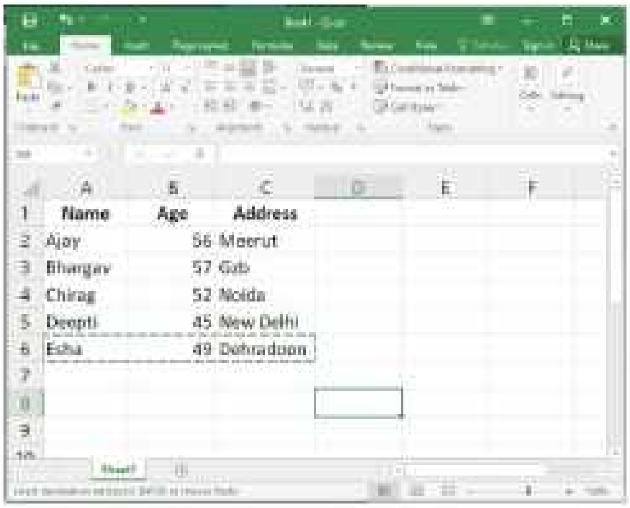
Moving Cell's Content

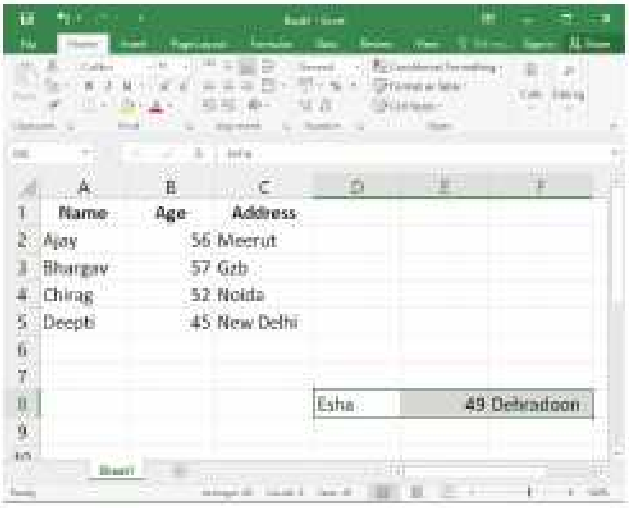
We can move cell's content from one location to another by using Cut and Paste. To move the cell's content, follow these steps:

1. Select the range of cells to be moved.
2. Click on the Cut option in the Clipboard group on the Home tab. We will see a dotted line around the selected cells.



3. Select the destination cell where we want to paste the data.
4. Click on the Paste option in the Clipboard group on the Home tab.





The selected cells are moved to the new location.

Smart Tip

- To copy the cell's content, press Ctrl + C keys together.
- To cut the cell's content, press Ctrl + X keys together.
- Press Ctrl + V keys together for the Paste option.

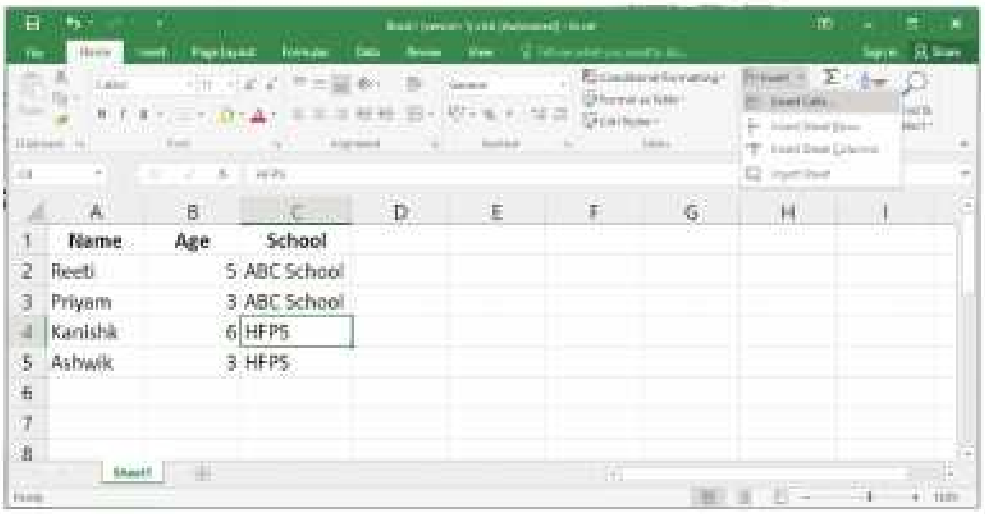
Inserting Cells, Rows and Columns

You can insert more cells, rows and columns in the table. Let us see how.

Inserting Cells

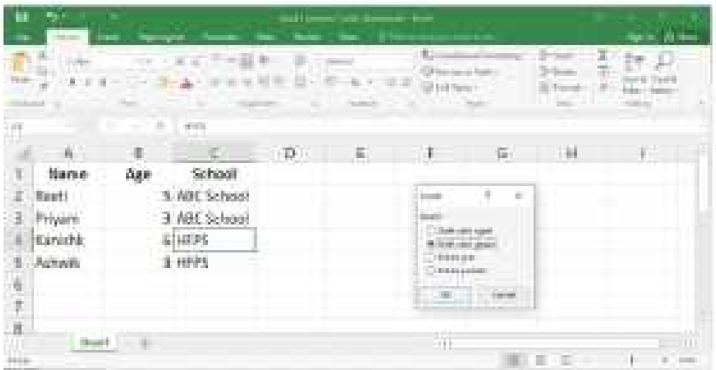
To insert a cell or a block of cells, follow these steps:

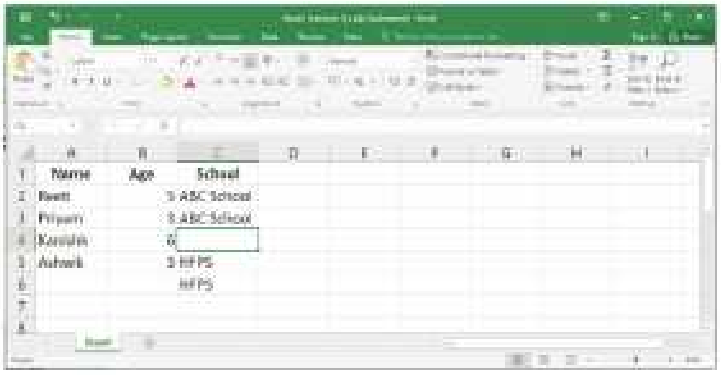
1. Select the range of cells where you want to insert the cell.
2. Click on the drop-down menu arrow of the Insert button in the Cells group on the Home tab. A menu appears.



3. Select Insert Cells.

4. The Insert dialog box is displayed. Select the appropriate option. Here we select Shift cells down.
5. Click on the OK button.





Inserting Rows

To insert a new row, follow these steps:

1. Click on the row heading to select the row above where you want to insert the new row.
2. Click on the arrow next to the Insert option in the Cells group on the Home tab. A drop-down list is displayed.
3. Click on the Insert Sheet Rows option.

The selected row is shifted down and a new row is inserted at its place.

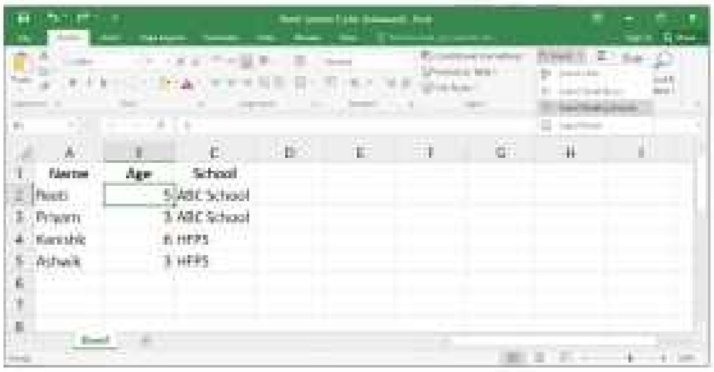
Info Zone

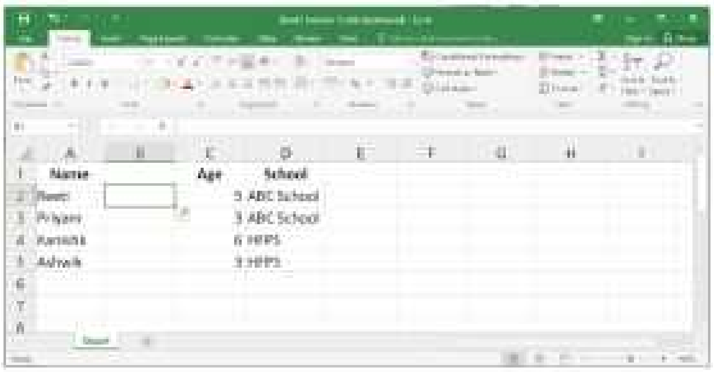
To insert multiple rows at the same time, select more than one row and then choose Insert Sheet Rows. The number of rows inserted will be the same as selected by you.

Inserting Columns

To insert a new column, follow these steps:

1. Click on the column heading to select the column to the left of which a new column is to be inserted.
2. Click on the arrow next to the Insert option in the Cells group on the Home tab. A drop-down list is displayed.
3. Click on the Insert Sheet Columns option.





The selected column is shifted to the right and a new column is inserted at its place.

Info Zone

To insert multiple columns at the same time, select more than one column and then choose Insert Sheets Column. The number of columns inserted will be the same as selected by you.

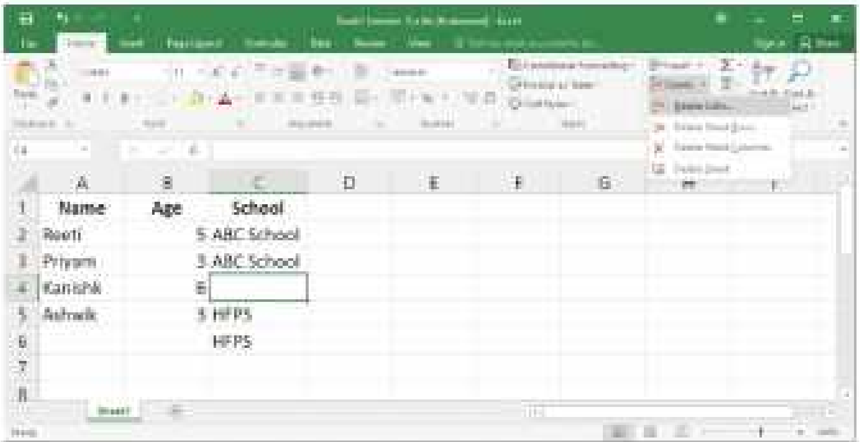
Deleting Cells, Rows and Columns

Cells, rows and columns can also be deleted. Let us learn the steps.

Deleting Cells

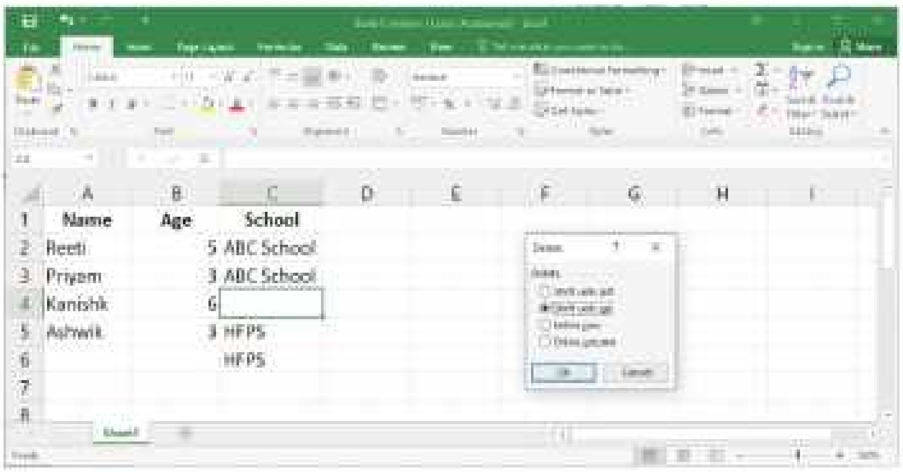
To delete cells, follow these steps:

1. Select the range of cells you want to delete.
2. Click on the drop-down menu arrow of the Delete button in the Cells group on the Home tab. A menu appears.
3. Select Delete Cells. The Delete dialog box is displayed.

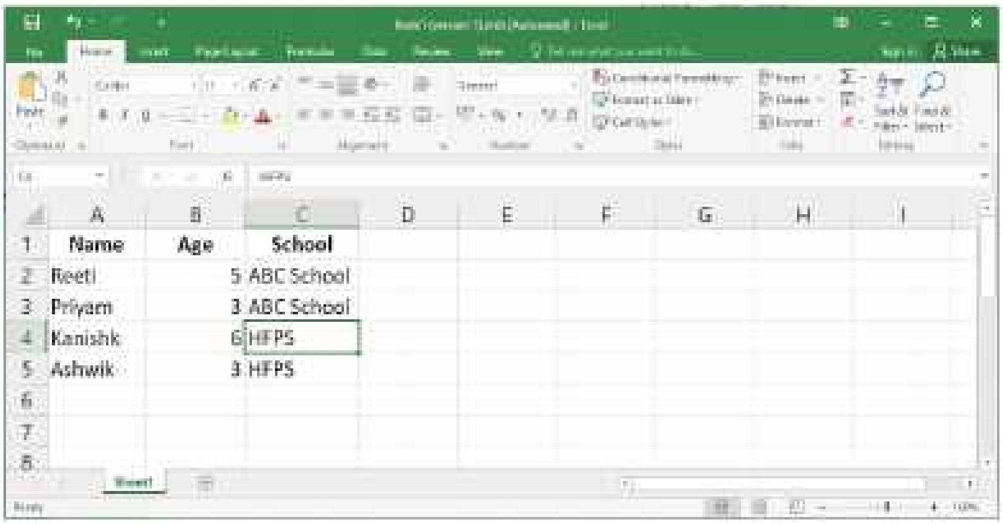


4. Select the appropriate option.

5. Click on the OK button.



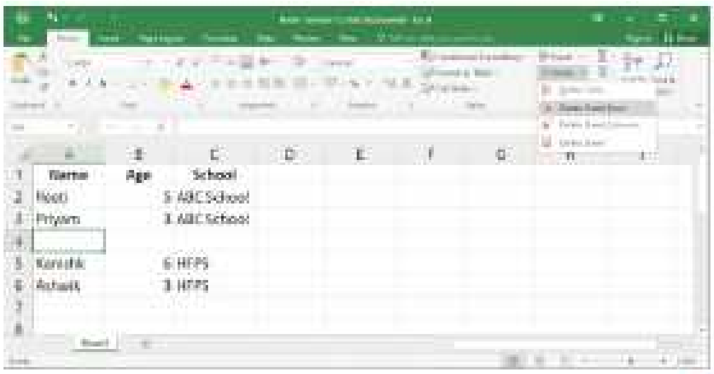
It shows the output after selecting the Shift cells up option.

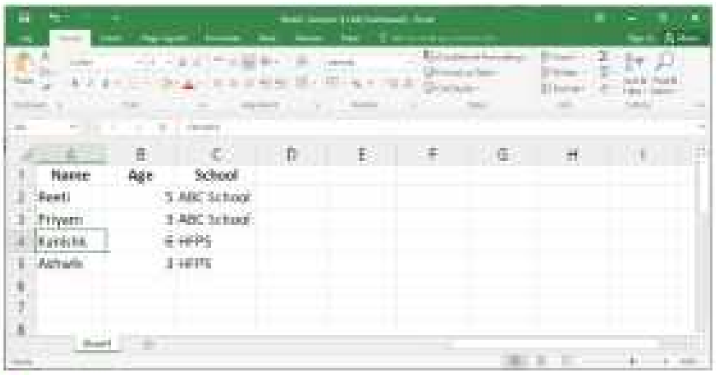


Deleting Rows/Columns

To delete rows or columns, follow these steps:

1. Select the rows/columns to be deleted.
2. Click the arrow next to the Delete option in the Cells group on the Home tab. A menu appears.
3. Select Delete Sheet Rows option to delete rows or Delete Sheet Columns option to delete columns.





The selected rows/columns will be deleted. The remaining rows are shifted up to fill the space left by deleted rows and remaining columns shift to the left to fill the space left by deleted columns.

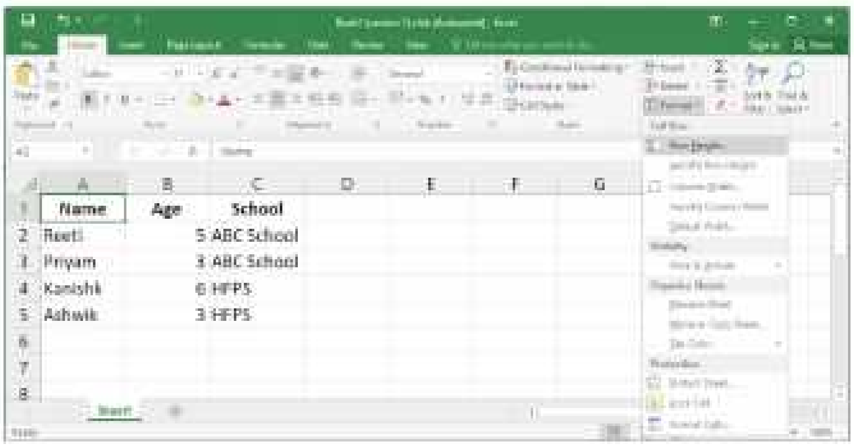
Info Zone

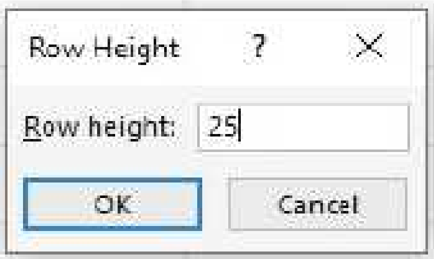
An easier method by which you can delete cells is by right-clicking anywhere in the selected cells, rows or columns and choosing Delete from the shortcut menu. Select the desired option from the dialog box that appears.

Changing Row Height

To change the row height, follow these steps:

1. Select the row(s).
2. Click on the down arrow next to the Format option in the Cells group on the Home tab. A drop-down list is displayed.
3. Click on the Row Height option for specifying the row height. The Row Height dialog box appears.
4. Enter a value in the Row Height text box.
5. Click on the OK button.





The row height is adjusted according to the value specified.

Smart Tip

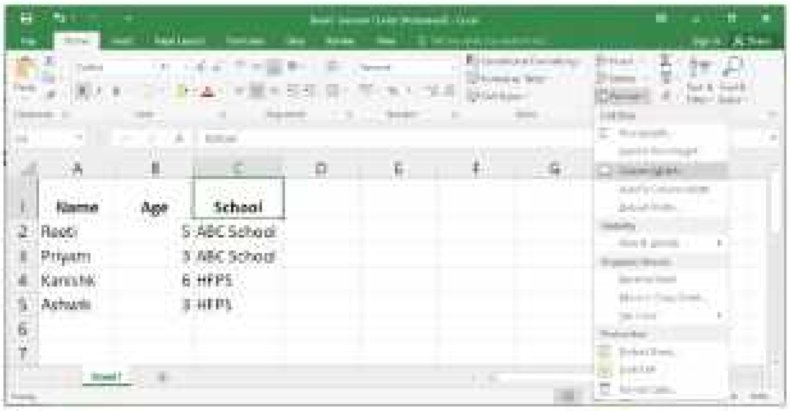
You can also change the row height by dragging the border. Place the mouse pointer at the bottom of the border of the row heading. The pointer shape changes to a double-headed arrow . Then, drag the bottom border up or down to decrease or increase the height until the row is of desired height.

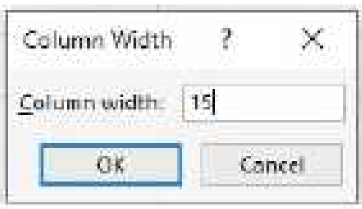
Info Zone: The row height is measured in points (72 points is equal to 1 inch).

Changing Column Width

To change the column width, follow these steps:

1. Select the column(s).
2. Click on the down arrow next to the Format option in the Cells group on the Home tab. A drop-down list is displayed.
3. Click on the Column Width option for specifying the column width. The Column Width dialog box appears.
4. Enter a value in the Column Width text box.
5. Click on the OK button.





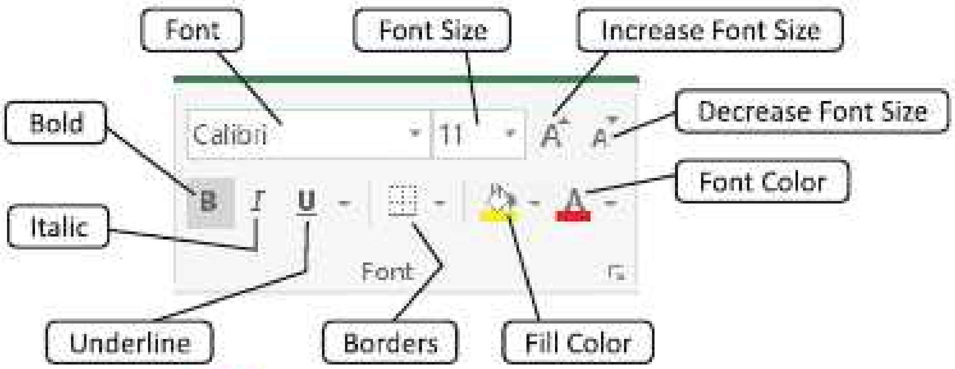
The column width is adjusted according to the value specified.

Smart Tip

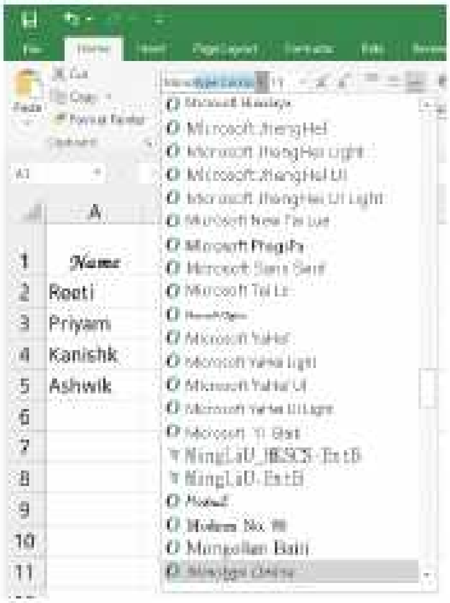
- By double-clicking the right border of the column, you can quickly resize the column to fit the content.
- You can also change the column width by dragging the border. Place the mouse pointer at the right border of the column heading. The pointer shape changes to a double-headed arrow ✛. Then, drag the right border left or right to decrease or increase the column width.

Font Group

Like MS Word and MS PowerPoint, MS Excel also has a Font group. The Font group is found under the Home tab of Excel.



Changing Font



To change the font, follow these steps:

1. Type the text in any cell and select it.
2. Click the arrow next to the Font. A list of fonts appears.
3. You will get preview of the font in the cell as you scroll down the list. Click on the desired font.

Changing Font Size

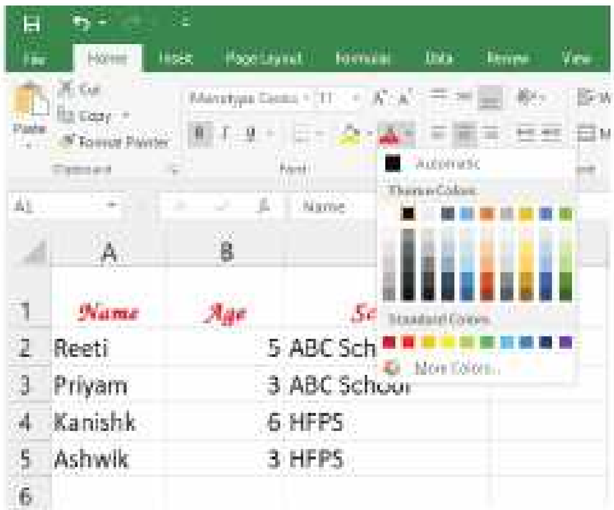
To change the font size, follow these steps:

1. Select the cell.
2. Click the arrow next to the Font Size. A list of font sizes appears.
3. You will get a preview of the font in the cell as you scroll down the list. Click on the desired font size.

Smart Tip

You can also change the font size by using the Decrease Font Size and Increase Font Size buttons in the Font group on the Home tab.

Changing Font Colour



To change the font, follow these steps:

1. Select the cell.
2. Click the arrow next to the Font Color. A list of colours appears.
3. Select the colour of your choice and the text colour changes.

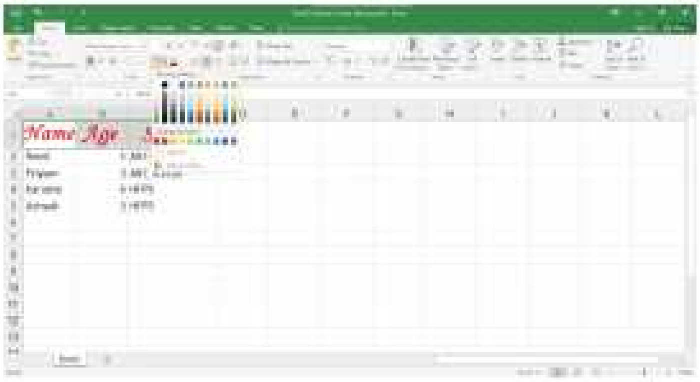
Bold, Italics and Underline

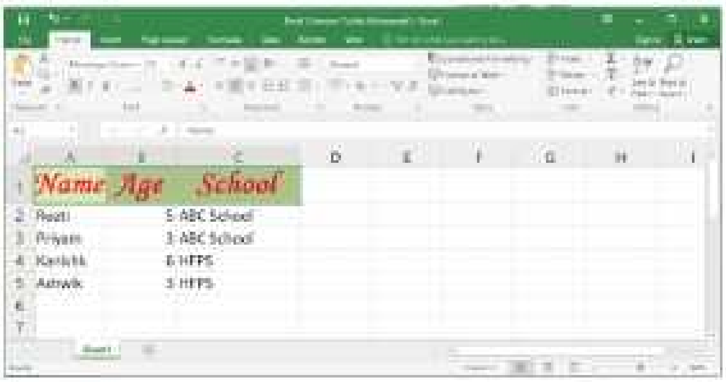
You can emphasize the content of a cell by making the text bold, by using italics, or by underlining the text. Just select the cell and click the Bold **B**, Italic *I* or U Underline button.

Applying Background Colours

To add a background colour, follow these steps:

1. Select the heading cells (Row1). Here we will add a background colour to the heading of a table.
2. Click on the arrow next to the Fill Color option.
3. Select the colour of your choice. That colour will be set as the background of the selected cells.

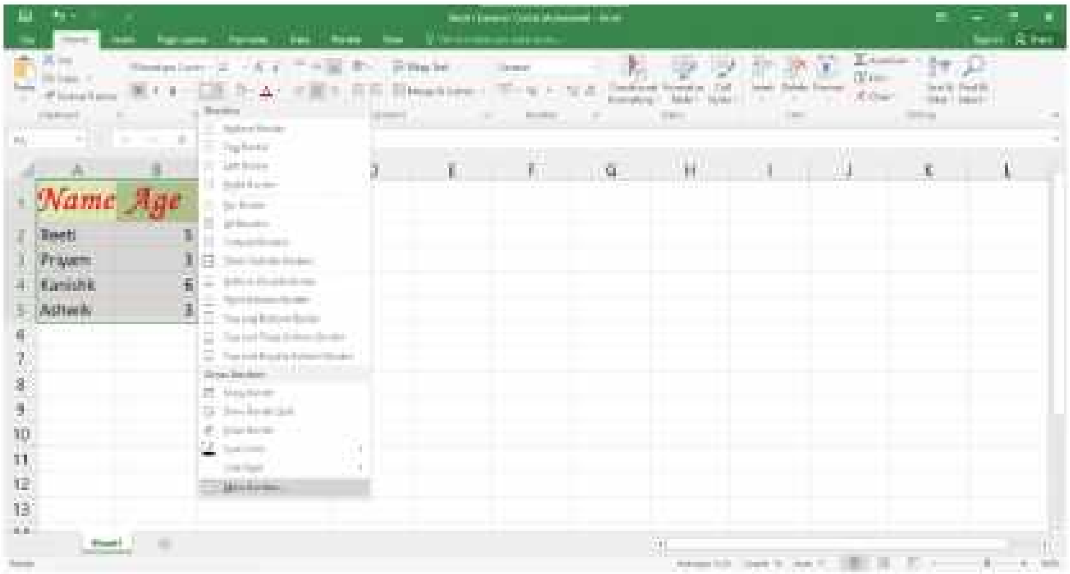




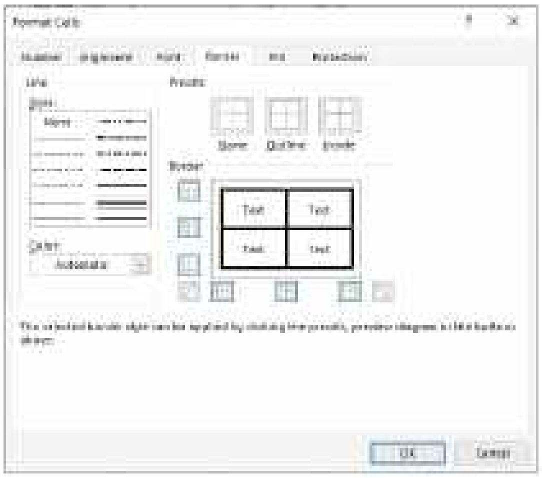
Applying Borders

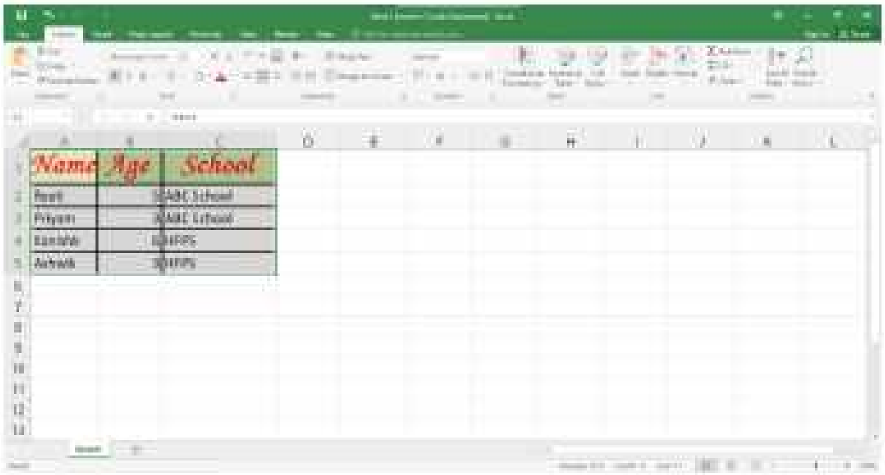
To apply a border, follow these steps:

1. Select the whole table.
2. Click the arrow next to Borders. A drop-down menu appears.
3. Click on More Borders... in the Borders menu.



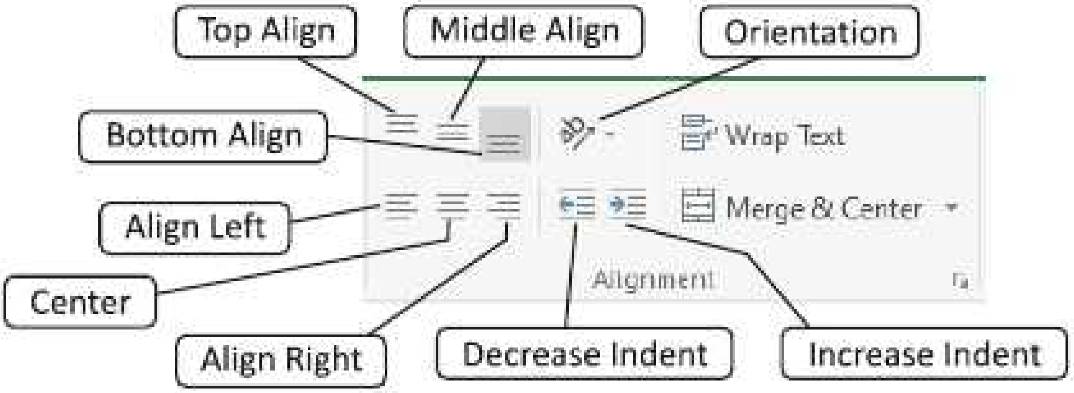
4. The Format Cells dialog box will appear. Select the border style.
5. Select the border colour.
6. Click on the OK button.





Alignment Group

The Alignment group provides options for changing the alignment, orientation and indentation of data inside a cell.

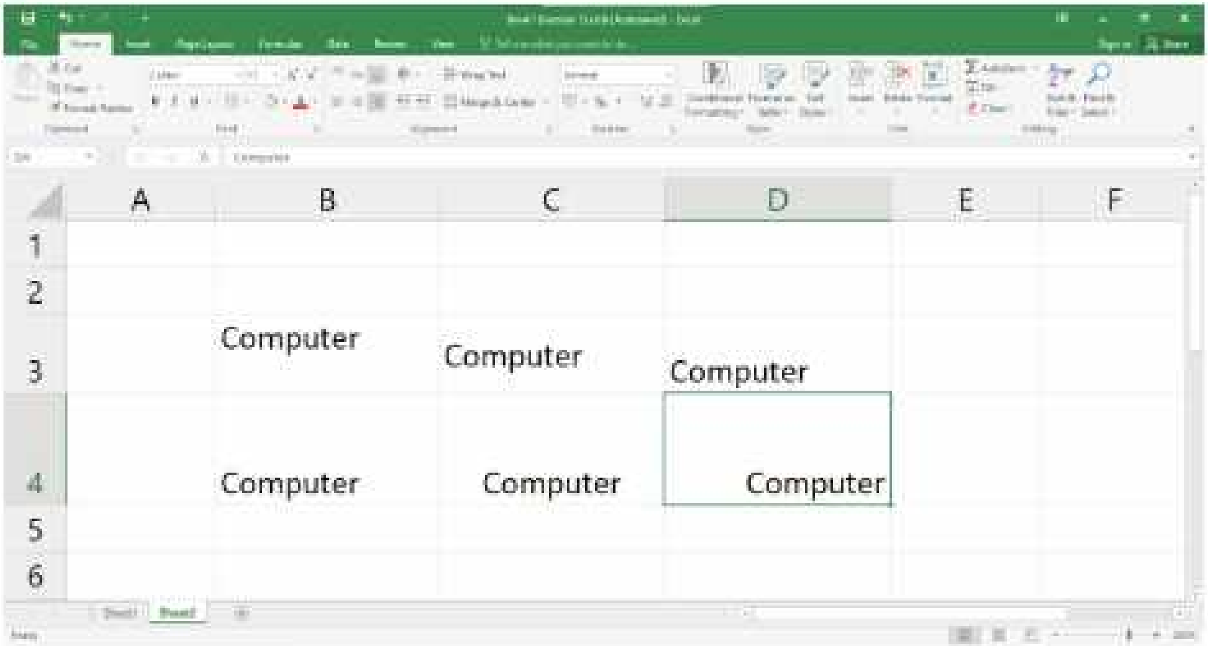


Changing Alignment of Text

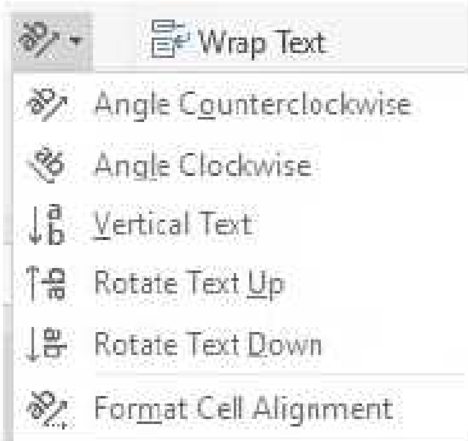
Alignment refers to the position where data is placed within the boundary of a cell. By default, Excel applies the appropriate horizontal alignment to each data type, for example to numbers and text. By default, the numbers are right-aligned and text is left-aligned in a cell. You can also control the vertical alignment within a cell.

- To change the vertical alignment of cell's content, click the Top Align, Middle Align or the Bottom Align option.
- To change the horizontal alignment of cell's content, click the Align Left, the Center or the Align Right option.

The worksheet given below highlights the use of various alignment options for formatting the cell's content.

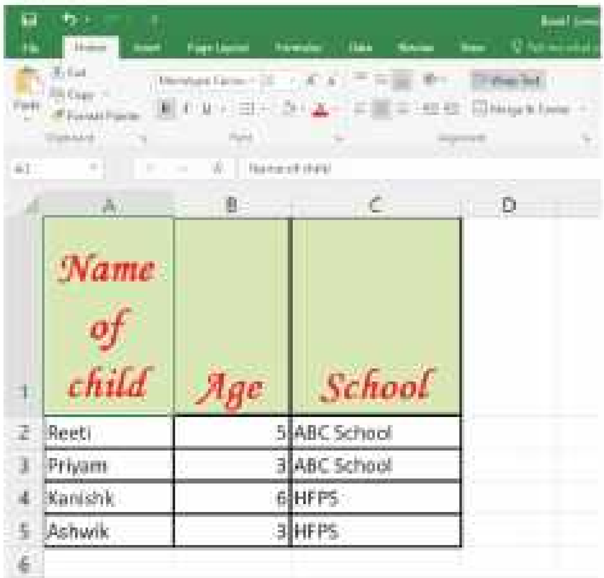


Changing Orientation of Text



Orientation refers to the rotation of text in different angles inside the cell. The various orientation options in the Alignment group are— Angle Counterclockwise, Angle Clockwise, Vertical Text, Rotate Text Up, and Rotate Text Down.

The worksheet given alongside illustrates the use of various orientation options for formatting the cell's content.

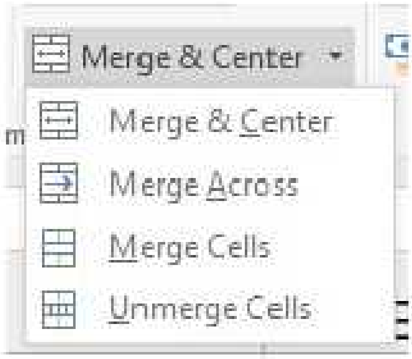


Wrap Text Option

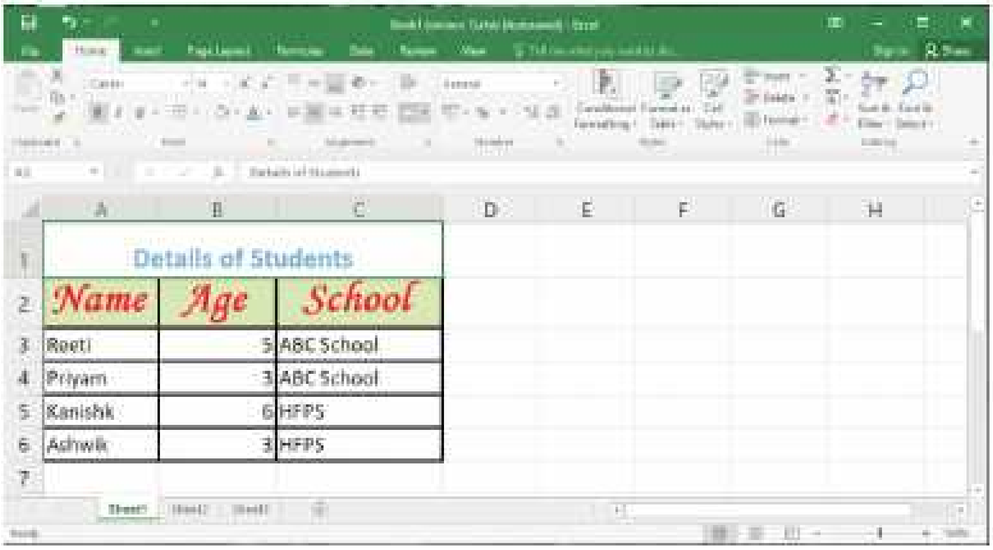
This option wraps up text into multiple lines inside a cell. Select the cell and click Wrap Text option in the Alignment group. The text will be confined to the cell and displayed in multiple lines.

Merge and Center Option

Merging allows you to combine a cell with adjacent empty cells to create one large cell. For merging, select the cells in a row or a column and choose from the following options:



- **Merge and Center :** It combines and centre the content of the selected cells in a new larger cell.
- **Merge Across :** It merges the selected cells in the same row into one large cell.
- **Merge Cells :** It merges the selected cells into one cell.
- **Unmerge Cells :** It splits the current cell into multiple cells.



COMPUTER ETIQUETTES

Always keep your mouse on the mouse pad.

Let's Implement

A Tick (✓) the correct option.

1. The options for inserting or deleting cells, rows or columns are present in the ____________ group on the Home tab.

   a. Editing ◯ b. Font ◯ c. Cells ◯ d. Clipboard ◯

2. We can copy data from one location to another using ____________ and Paste options.

   a. Copy ◯ b. Cut ◯ c. Both a and b ◯ d. None of these ◯

3. ______________ button is used to change the background colour of cells.
   a. Fill Color ◯ b. Background Color ◯
   c. Font Color ◯ d. None of these ◯
4. To select an entire row, press the ______________ keys together.
   a. Shift + Spacebar ◯ b. Ctrl + Spacebar ◯
   c. Spacebar ◯ d. None of these ◯
5. The row height is measured in __________.
   a. points ◯ b. cm ◯ c. m ◯ d. dm ◯
6. By default, the numbers are __________.
   a. left-aligned ◯ b. right-aligned ◯
   c. center-aligned ◯ d. None of these ◯

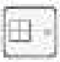
B Fill in the blanks.

1. By default, text is __________.
2. To delete the cell's content, select the cell and press the __________ key.
3. We can move cell's content from one location to another by using __________ and __________.
4. To insert a new column, click on the Insert __________ option.
5. __________ refers to the position of data within the boundary of a cell.
6. __________ refers to the rotation of the text in different angles inside the cell.

C Write **True** or **False**.

1. To select a complete row, click on the row heading. ☐
2. Press Ctrl + A keys to select an entire worksheet. ☐
3. Insert option is present in the Editing group on the Home tab. ☐
4. Merging refers to the rotation of the text in different angles inside the cell. ☐
5. By default, Excel applies the appropriate horizontal alignment to each data type. ☐
6. Orientation allows you to combine a cell with adjacent empty cells to create one large cell. ☐

D Match the following.

1. 	a. Wrap Text
2. 	b. Font Color
3. 	c. Fill Color
4. 	d. Borders
5. 	e. Merge and Center

E Answer the following questions.

1. What are the three options for horizontal and vertical alignments in a cell?
2. What is the function of Comma style and Accounting Number Format option in the Number group?
3. List out the steps to copy data from one location to another.
4. List out the steps to insert a new row in a worksheet.
5. List out the steps to change the column width.
6. What is the function of Wrap Text button in the Alignment group?

Refreshment Corner

Identify the following options and write their names. Problem Solving

1. A ________________
2. A ________________
3. , ________________
4. % ________________
5. ←.0 .00 ________________
6. .00 →.0 ________________

Practical Session

Experiential Learning

A Create a worksheet given below and perform the following tasks.

1. Copy the data in cells C1:C9 and paste it to the cells F1:F9
2. Move the data from cells A1:E9 to G1:G9.
3. Delete the data of cells F1:F4.
4. Change the row height of the first row to 24 points and other rows to 21 points.
5. Change the column width of all the rows to 21 points.

	A	B	C	D	E
1	**Name**	**Hindi**	**English**	**Maths**	**Computer**
2	Advit	89	89	88	89
3	Chandni	85	78	84	90
4	Daksh	86	79	86	94
5	Tanuj	84	76	85	86
6	Palak	87	79	88	83
7	Seerat	88	90	82	88
8	Kajal	79	69	79	66
9	Simran	75	68	75	78

B Create a worksheet given below and perform the following tasks.

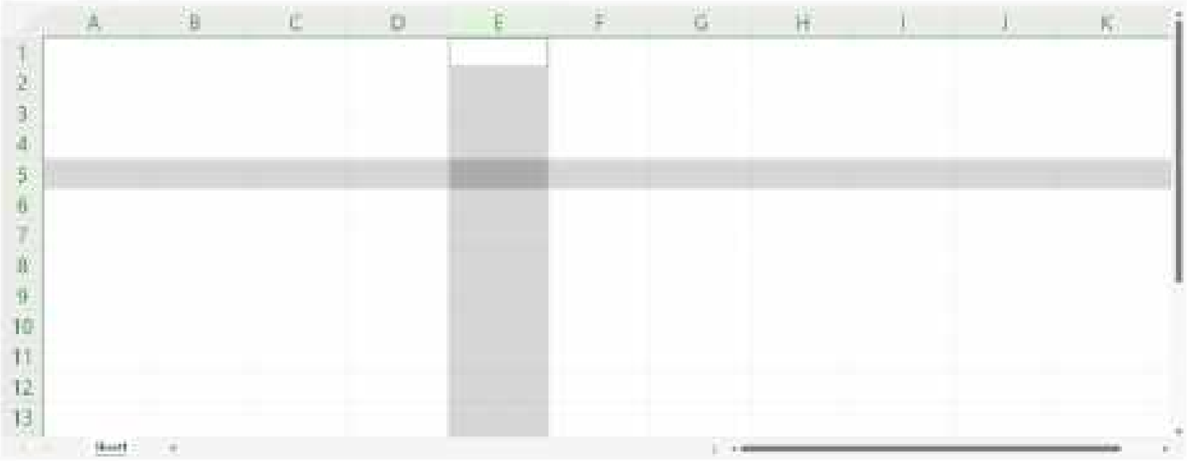
1. Move the Place column (column B) after the Sale (in column C)
2. Increase the row height of all the rows to 22 points.
3. Increase the column height of all columns to 27 points.
4. Include the row between LED and Mouse :
   Headphones ⟶ 96000 ⟶ New Delhi
5. Move the row 'Laptop' before 'Monitor'.

	A	B	C
1	IT Hub Sales Report		
2	Product	Sale	Place
3	LED	300000	Meerut
4	Mouse	100000	Noida
5	Printer	250000	Noida
6	Scanner	300000	Gzb
7	BCR	250000	New Delhi
8	MICR	360000	Agra
9	Monitor	790000	New Delhi
10	Keyboard	90000	Meerut
11	Laptop	600000	Meerut

Competency Based Questions

Critical Thinking

1. If you want to change the content of a cell completely, then what will you do?
2. Consider the worksheet given alongside and answer the questions that follow.
   (a) Name the cell formed by the intersection of row 5 and column E.
   (b) How will you select the entire column E and the row 5 simultaneously?



RACKING YOUR BRAIN

Life Skills and Values

Should we use someone else's data while formatting in Excel? Justify your answer.

Just Have a Discussion

Communication

Discuss the different options of Font group of Home tab in the class.

Teacher's Note

- The teacher should demonstrate to the students various formatting features, using examples.
- It should be explained to the students that too many formatting elements used together can distract the reader.

QUICK REVIEW 1

Tick (✓) the correct option.

1. It refers to data that has been processed into a form that has meaning and is useful.
   a. Data ◯ b. Storage ◯ c. Information ◯ d. Processing ◯
2. The ____________ displays alerts for your device and all the apps in a slide-out pane on the right side of the desktop.
   a. Tablet Mode ◯ b. Cortana ◯
   c. File Explorer ◯ d. Action Center ◯
3. To move through the cells, press ________________ keys.
   a. Tab ◯ b. arrow ◯
   c. Both a and b ◯ d. None of these ◯
4. To select an entire row, press the ________________ keys together.
   a. Shift + Spacebar ◯ b. Ctrl + Spacebar ◯
   c. Spacebar ◯ d. None of these ◯
5. These computers require a specialized environment including separate air conditioning, cooling and electrical power.
   a. Minicomputers ◯ b. Macrocomputers ◯
   c. Microcomputers ◯ d. Mainframe computers ◯
6. ____________ in Windows 10 helps you to quickly find information on the internet.
   a. Control Panel ◯ b. Action Center ◯ c. Task View ◯ d. Cortana ◯
7. To move to the next cells, press ____________ keys
   a. Tab ◯ b. Shift ◯ c. arrow ◯ d. delete ◯
8. ________________ button is used to change the background colour of cells.
   a. Fill Color ◯ b. Background Color ◯
   c. Font Color ◯ d. None of these ◯
9. It contains the electronic circuits that actually causes the processing of data to occur.
   a. Processor Unit ◯ b. Auxiliary Storage Unit ◯
   c. Storage Unit ◯ d. Output Unit ◯
10. The ____________ is a new feature in Windows 10, which enables you to organise the applications running on your computer.
    a. Task View ◯ b. File Explorer ◯
    c. Cortana ◯ d. Action Center ◯

5

FUNCTIONS AND FORMULAS IN EXCEL 2016

Outlines

- Functions in Excel
- Function Library
- Elements of Formulas in Excel
- Types of Cell References
- AutoSum Feature
- Formulas in Excel
- Performing Calculations in Excel

Functions in Excel

A function is a predefined formula that acts on a cell or range of cells and performs calculations. Functions save time and eliminate the chance of writing wrong formulas. All spreadsheet programs include common functions that can be used for quickly finding the sum, average, count, maximum value, and minimum value for a range of cells. The function consists of two parts— the function name and the arguments. The arguments are included within the round brackets.

The cell references passed to a function are called arguments.

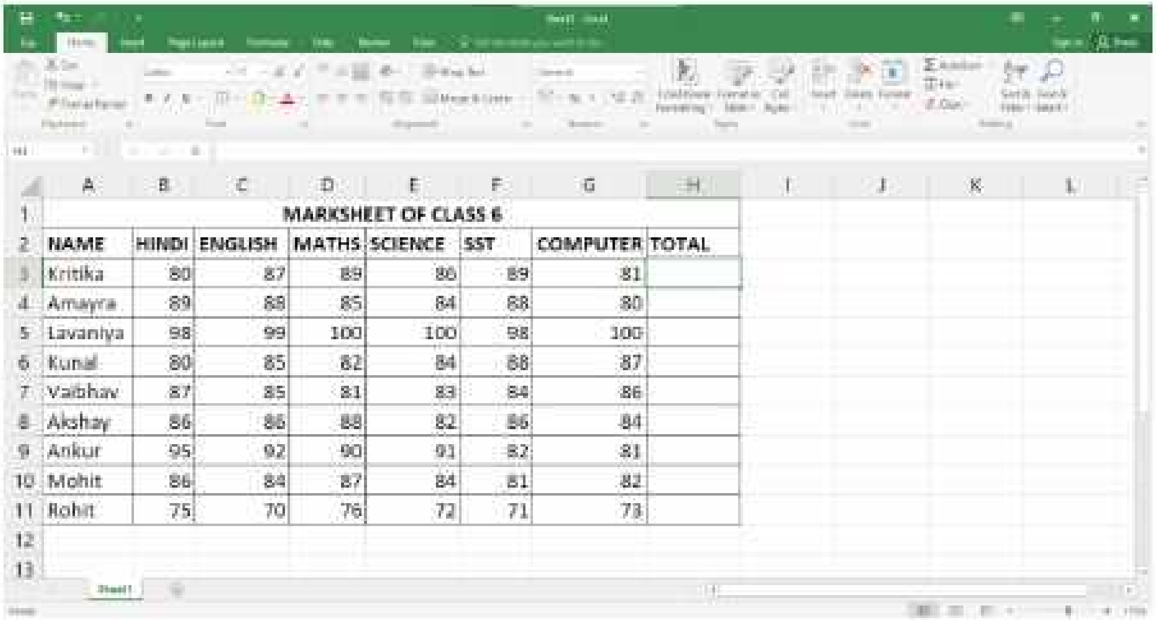
In order to work correctly, a function must be written in a specific way, which is called the syntax. The basic syntax for a function is an equals sign (=), the function name (SUM, for example), and one or more arguments. The function name must be followed by an opening and closing parenthesis. Arguments contain the information you want to calculate. Arguments are enclosed in the parenthesis. The function in the example below would add the values of the cell range A1:A20.

AutoSum Feature

The AutoSum feature is used to add numbers in the selected range of cells quickly. This feature is particularly useful when you want to add all numbers in a row or a column. The AutoSum option is present in the Editing group on the Home tab. It is also present in the Function Library group on the Formulas tab.

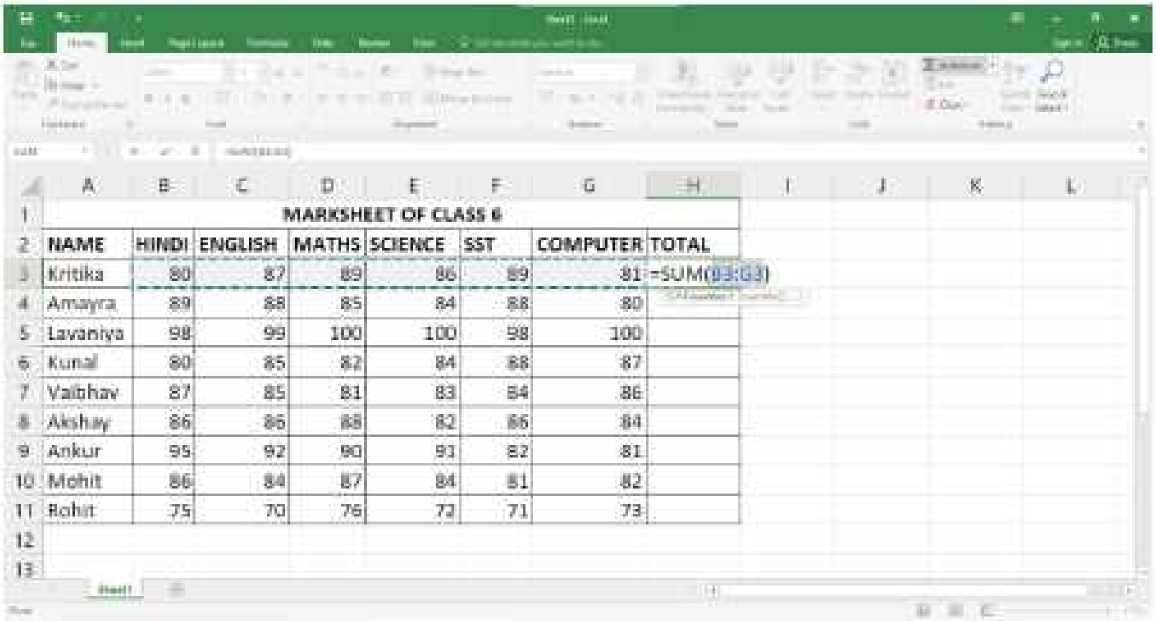
To use the AutoSum feature, follow these steps:

1. Type the data as shown.



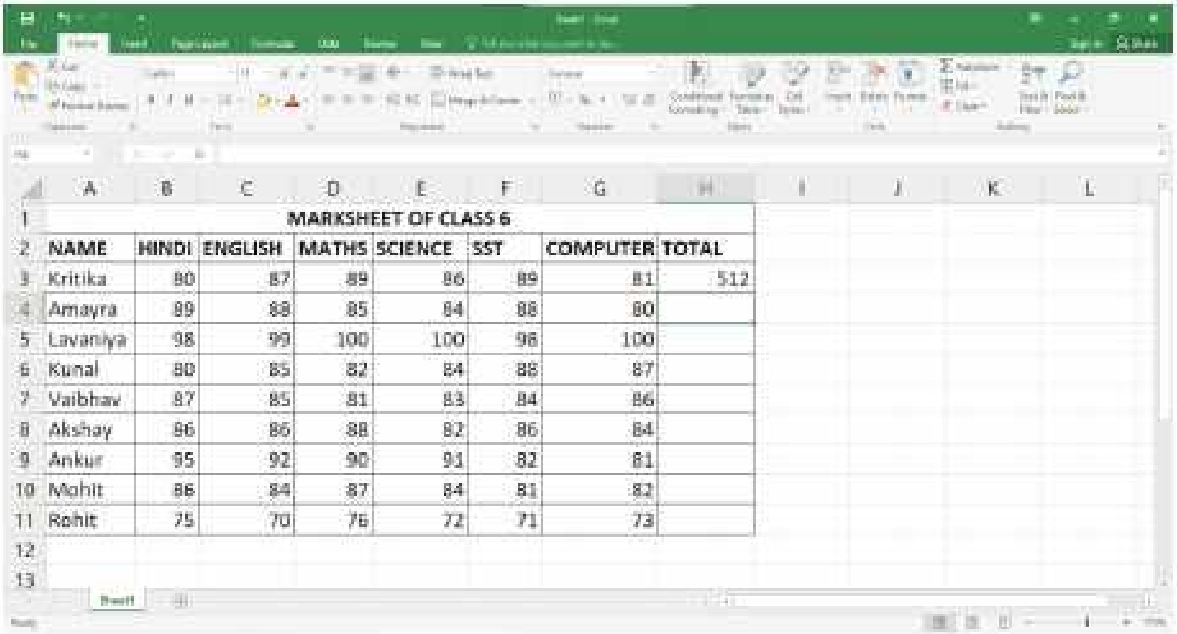
MARKSHEET OF CLASS 6							
NAME	HINDI	ENGLISH	MATHS	SCIENCE	SST	COMPUTER	TOTAL
Kritika	80	87	89	86	89	81	
Amayra	89	88	85	84	88	80	
Lavaniya	98	99	100	100	98	100	
Kunal	80	85	82	84	88	87	
Vaibhav	87	85	81	83	84	86	
Akshay	86	86	88	82	86	84	
Ankur	95	92	90	91	82	81	
Mohit	86	84	87	84	81	82	
Rohit	75	70	76	72	71	73	

2. Select the cell H3 and click on the AutoSum option in the Editing group on the Home tab. Clicking on the AutoSum option automatically creates a SUM formula along with cell references in the selected cell.



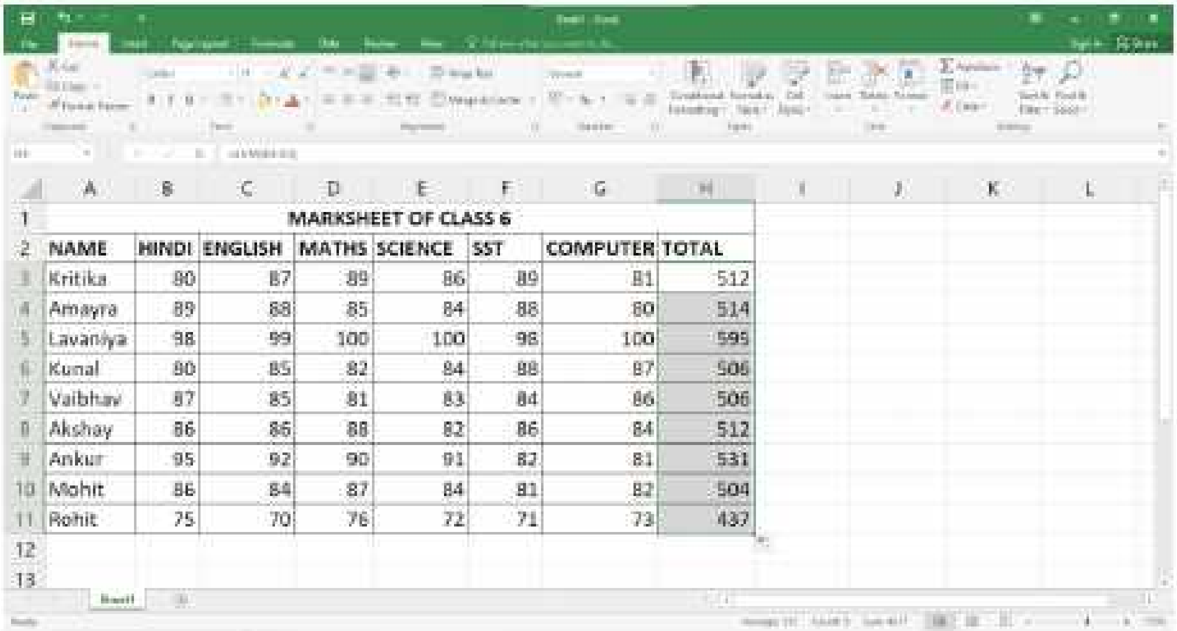
MARKSHEET OF CLASS 6							
NAME	HINDI	ENGLISH	MATHS	SCIENCE	SST	COMPUTER	TOTAL
Kritika	80	87	89	86	89	81	=SUM(B3:G3)
Amayra	89	88	85	84	88	80	
Lavaniya	98	99	100	100	98	100	
Kunal	80	85	82	84	88	87	
Vaibhav	87	85	81	83	84	86	
Akshay	86	86	88	82	86	84	
Ankur	95	92	90	91	82	81	
Mohit	86	84	87	84	81	82	
Rohit	75	70	76	72	71	73	

3. Press the Enter key.



	A	B	C	D	E	F	G	H
1	MARKSHEET OF CLASS 6							
2	NAME	HINDI	ENGLISH	MATHS	SCIENCE	SST	COMPUTER	TOTAL
3	Kritika	80	87	89	86	89	81	512
4	Amayra	89	88	85	84	88	80	
5	Lavaniya	98	99	100	100	98	100	
6	Kunal	80	85	82	84	88	87	
7	Vaibhav	87	85	81	83	84	86	
8	Akshay	86	86	88	82	86	84	
9	Ankur	95	92	90	91	82	81	
10	Mohit	86	84	87	84	81	82	
11	Rohit	75	70	76	72	71	73	

4. Click, hold and drag the fill handle over the cells you wish to fill.



	A	B	C	D	E	F	G	H
1	MARKSHEET OF CLASS 6							
2	NAME	HINDI	ENGLISH	MATHS	SCIENCE	SST	COMPUTER	TOTAL
3	Kritika	80	87	89	86	89	81	512
4	Amayra	89	88	85	84	88	80	514
5	Lavaniya	98	99	100	100	98	100	595
6	Kunal	80	85	82	84	88	87	506
7	Vaibhav	87	85	81	83	84	86	506
8	Akshay	86	86	88	82	86	84	512
9	Ankur	95	92	90	91	82	81	531
10	Mohit	86	84	87	84	81	82	504
11	Rohit	75	70	76	72	71	73	437

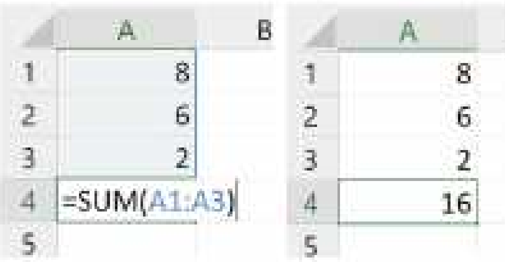
Commonly used functions in Excel

The commonly used functions in Excel are as follows:

SUM ( ) : It is used to add all the numbers in a range of cells.

= SUM (A1: A3)

The sum of the range A1:A3 will be displayed in the cell A4.



	A	B
1	8	
2	6	
3	2	
4	=SUM(A1:A3)	
5		

	A
1	8
2	6
3	2
4	16
5	

PRODUCT ( ) : It is used to multiply all the numbers in a range of cells.

= PRODUCT (A1: A3)

The product of the range A1:A3 will be displayed in the cell A4. It is used to multiply all the numbers in a range of cells.

	A	B
1	2	
2	5	
3	6	
4	=PRODUCT(A1:A3)	
5		

	A
1	2
2	5
3	6
4	60
5	

= PRODUCT (A1: A3)

The product of the range A1:A3 will be displayed in the cell A4.

AVERAGE ( ) : It returns the average (arithmetic mean) of all the numbers in a range of cells.

= AVERAGE (A1: A3)

The average of the range A1:A3 will be displayed in the cell A4.

	A	B
1	9	
2	8	
3	10	
4	=AVERAGE(A1:A3)	
5		

	A
1	9
2	8
3	10
4	9
5	

MAX ( ) : It returns the largest value from the numbers in a range of cells.

= MAX (A1: A3)

The maximum value in the range A1:A3 will be displayed in the cell A4.

	A	B
1	78	
2	98	
3	75	
4	=MAX(A1:A3)	

	A
1	78
2	98
3	75
4	98

MIN ( ) : It returns the smallest value from the numbers in a range of cells.

= MIN (A1: A3)

The minimum value in the range A1:A3 will be displayed in the cell A4.

	A	B
1	78	
2	98	
3	75	
4	=MIN(A1:A3)	
5		

	A
1	78
2	98
3	75
4	75
5	

COUNT ( ) : It counts the number of cells that contain numbers and counts numbers within the list of arguments.

= COUNT (A1: A3)

It counts the number of cells that contain numbers and returns the result in cell A4.

	A	B
1	78	
2	98	
3	75	
4	=COUNT(A1:A3)	
5		

	A
1	78
2	98
3	75
4	3
5	

Function Library

While there are hundreds of functions in Excel, the ones you use most frequently will depend on the type of data your workbooks contain. There is no need to learn every single function, but exploring some of the different types of functions will be helpful as you create new projects. You can search for functions by category, such as Financial, Logical, Text, Date and Time, and more from the Function Library group on the Formulas tab.

To access the Function Library, select the Formulas tab on the Ribbon. The Function Library will appear.

Some of the Math & Trig category formulas are as follows:

Functions	Purpose
LCM	It returns the least common multiple of integers.
GCD	It calculates the greatest common factor or highest common factor of two or more integers.
ODD (number)	It returns the number rounded up to the nearest odd integer.
INT (number)	It rounds a number to the nearest integer.
ROUND (number, num_digit)	It rounds a number to the specified digits.
EXP (number)	It returns (natural logarithm) raised to the power of a number.
SQRT (number)	It returns a square root.
POWER (number)	It returns the result of a number raised to some number.
MOD (number, divisor)	It returns the remainder after a number and is divided by the divisor.

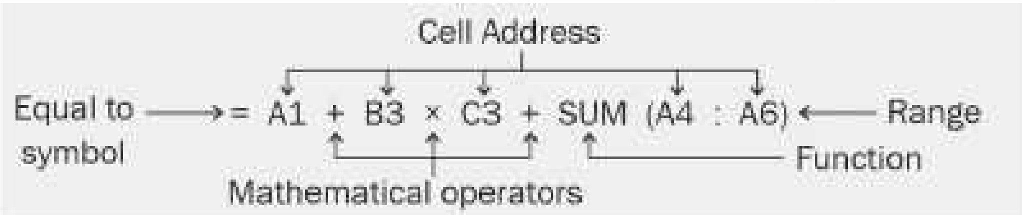
Formulas in Excel

In Excel, formulas are written differently from how you write them in Maths.

An Excel formula always starts with an equals (=) sign and it can contain values, cell references, functions and operators.

- **Values :** They are the numbers or text values that can be entered directly in a formula. Values are also known as constants. For example: 2510, 'Priyam' or 12.9.
- **Cell References :** While you can create simple formulas in Excel manually (for example, =2+2 or =5*5), most of the time you will use cell addresses to create a formula. This is known as making a cell reference. Using cell references will ensure that your formulas are always accurate because you can change the value of referenced cells without having to rewrite the formula. For example, C8, L7: L18, or A5: H5.
- **Functions :** They are built in formulas in MS Excel such as sum, average, etc.
- **Operators :** Excel uses standard operators for formulas, such as a plus sign for addition (+), a minus sign for subtraction (-), an asterisk for multiplication (*), a forward slash for division (/) and a caret (^) for exponents. By combining a mathematical operator with cell references, you can create a variety of

Elements of Formulas in Excel

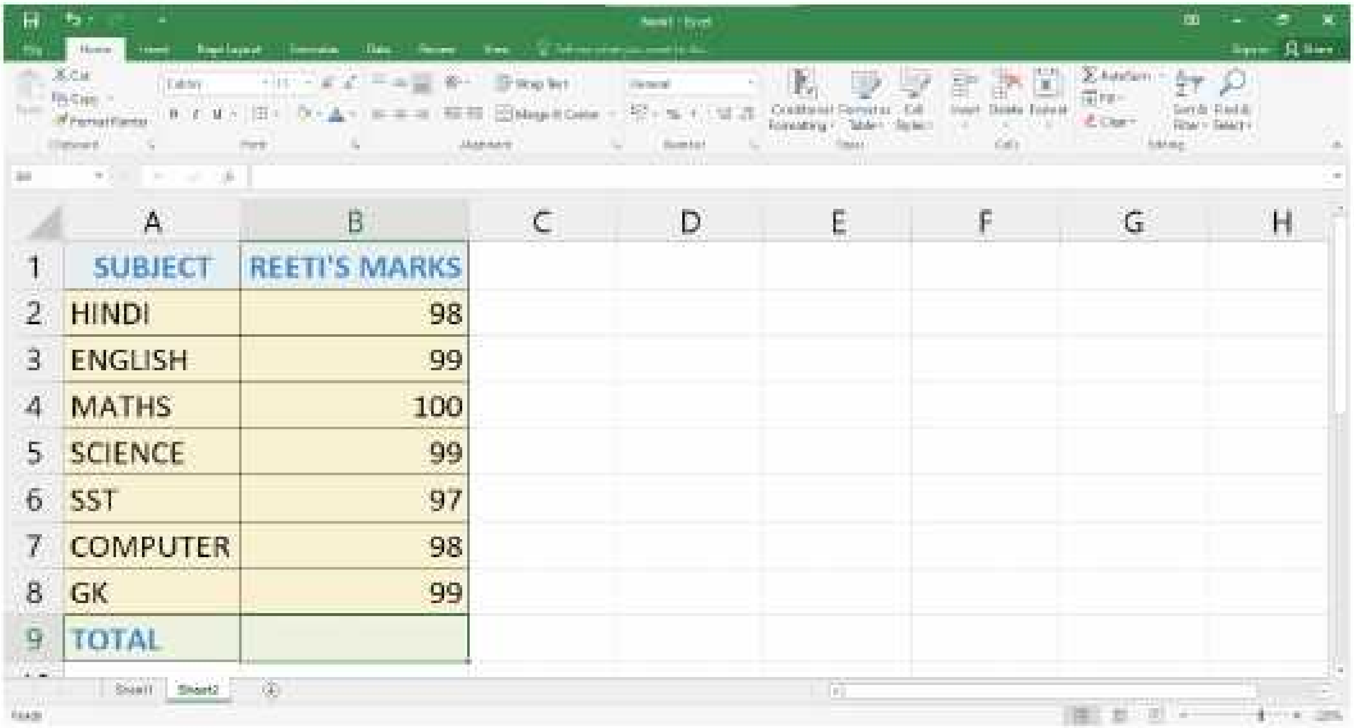


Thus, a formula in Excel is a sequence of values, cell references, functions and/ or operators in a cell that produces a new value from existing values, and is used to perform calculations involving addition, subtraction, division and multiplication.

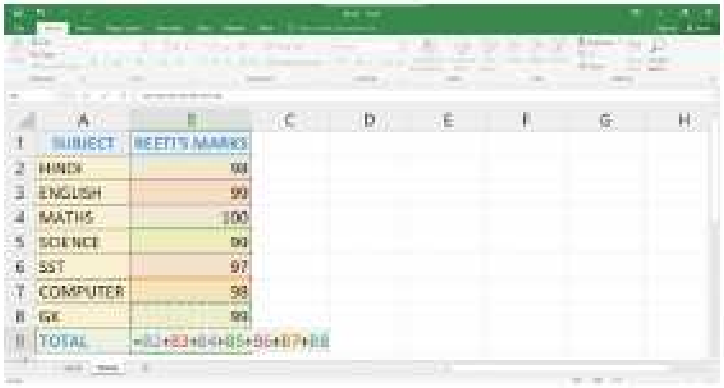
Performing Calculations in Excel

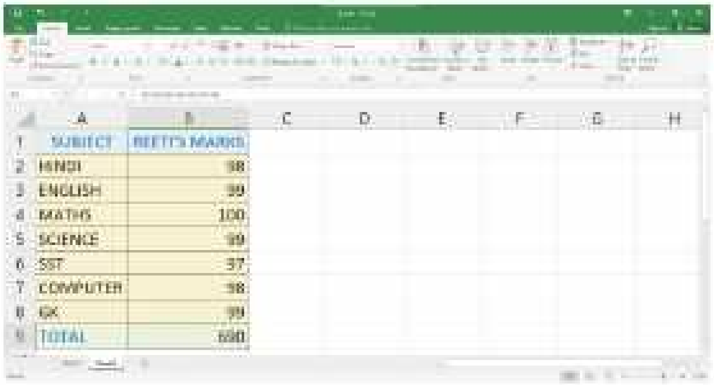
You can perform calculations with ease using formulas in Excel. Let us calculate the sum of the numbers in B2, B3, B4,..., B8 and display the result in B8.

1. Enter the data as shown.



2. Click on the cell B2. The address will appear in cell B9. Type + sign after that. Further repeat the steps till B7 or Type: = B2+B3+B4+B5+B6+B7+B8.
3. Press the Enter key. Cell B9 will display the total of all the values.





Some other examples of formulas are:

= (A1 + A2 + A3) /3

= (A1 * B2) + (B3 * D3) E3/5

= (A2 * B3) (C8/3)

Info Zone

Formulas in Microsoft Excel are not case sensitive, i.e., A1 is same as a1.

Using Text Operator

The text operator (&) is used to join two or more text strings. Joining two or more text values is called concatenation.

Follow the steps given below:

1. Enter two strings, type values in cell A1 and B1.

	A	B
1	PREETI	RAJPUT
2		
3		
4		
5		

2. Enter the formula = A1 & B1 in cell A4.
3. Press the Enter key.

	A	B
1	PREETI	RAJPUT
2		
3		
4	=A1&B1	
5		

	A	B
1	PREETI	RAJPUT
2		
3		
4	PREETIRAJPUT	
5		

To display a space between the text values of A1 and B1 in the B4 cell, the formula can be modified to include a space.

1. Type = A1 & " " & B1.

	A	B
1	PREETI	RAJPUT
2		
3		
4	=A1&" "&B1	
5		

2. Press the Enter key.

	A	B
1	PREETI	RAJPUT
2		
3		
4	PREETI RAJPUT	
5		

Types of Cell References

There are three types of cell references:

- Relative Reference
- Absolute Reference

> **Info Zone**
> MS Excel follows the BEDMAS rule to evaluate mathematical expressions. Excel uses this method automatically when a formula contains more than one operator.

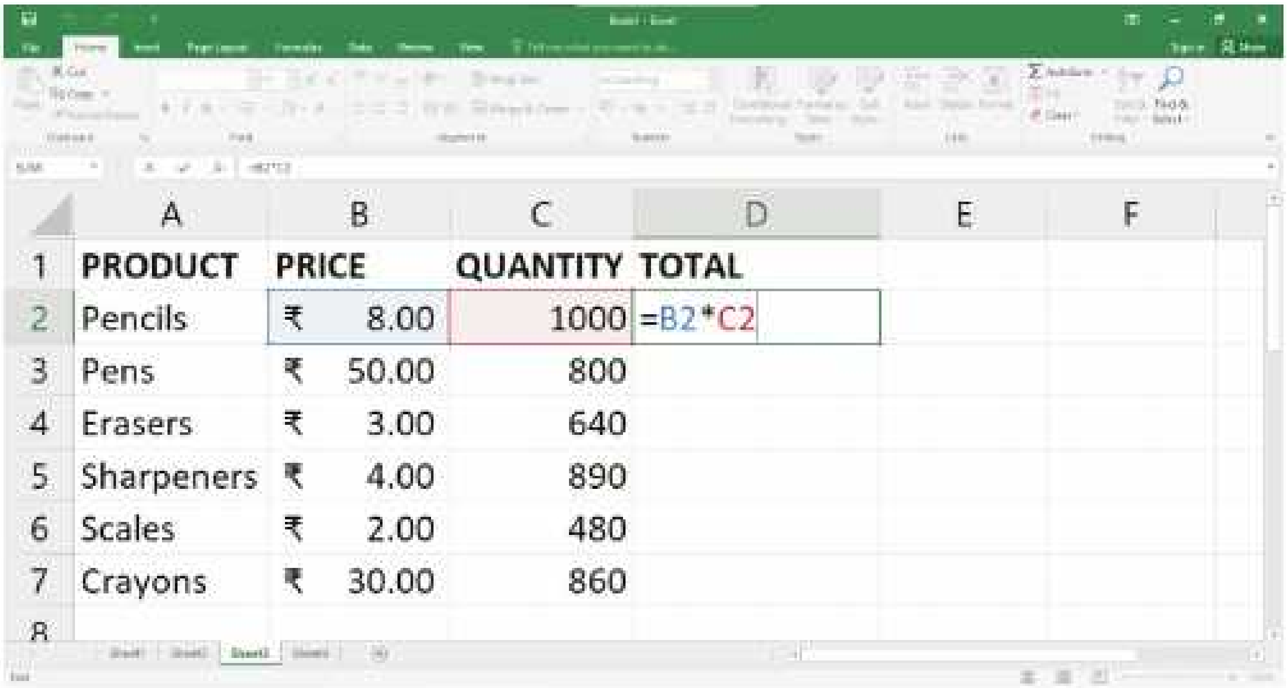
Relative Reference

By default, all cell references are relative references. When copied across multiple cells, they change based on the relative position of rows and columns. For example, if you copy the formula =A1+B1 from row 1 to row 2, Excel automatically changes the reference to match the location of cells, i.e., =A2+B2. Relative references are especially convenient whenever you need to repeat the same calculation across multiple rows or columns.

When you copy or move the formula to other cells, the reference cell automatically gets changed. This is known as relative reference.

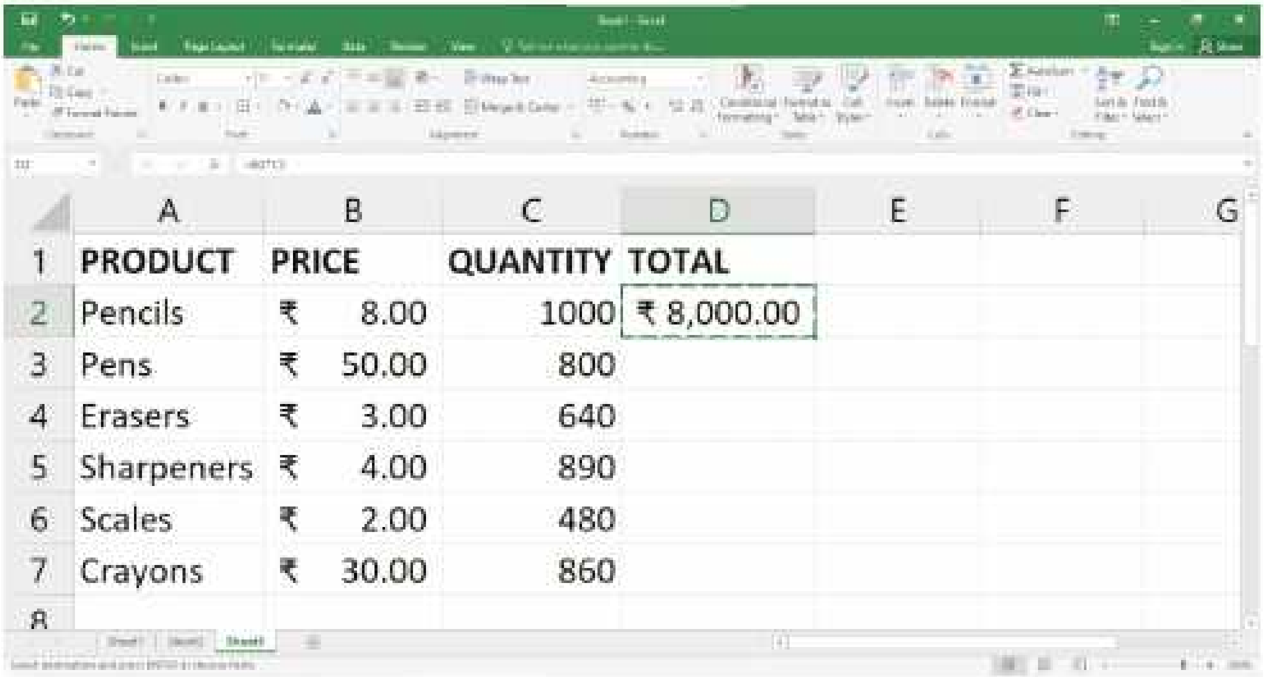
Follow these steps:

1. Enter the data as shown.
2. Type the formula = B2 * C2 in cell D2.

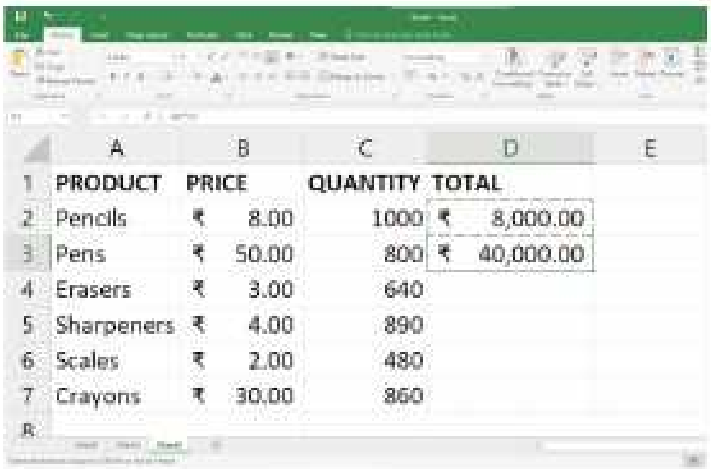


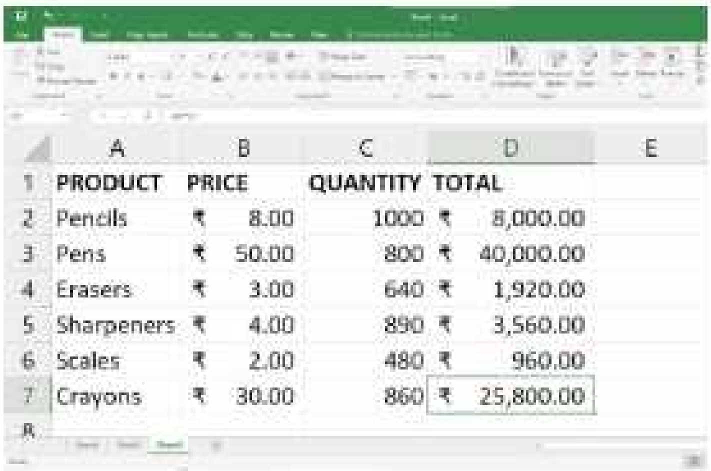
3. Press the Enter key.

4. Select the cell D2 and click on the Copy option in the Clipboard group on the Home tab.



5. Select the cell D3 and click on the Paste option in the Clipboard group on the Home tab.
6. Similarly, you can calculate the total in D4, D5, D6, D7, and D8.





Note that the formula = B2 * C2 changes to =B3 * C3 in cell D3, =B4 * C4 in cell D4, =B5 * C5 in cell D5, =B6 * C6 in cell D6, =B7 * C7 in cell D7 and =B8 * C8 in cell D8.

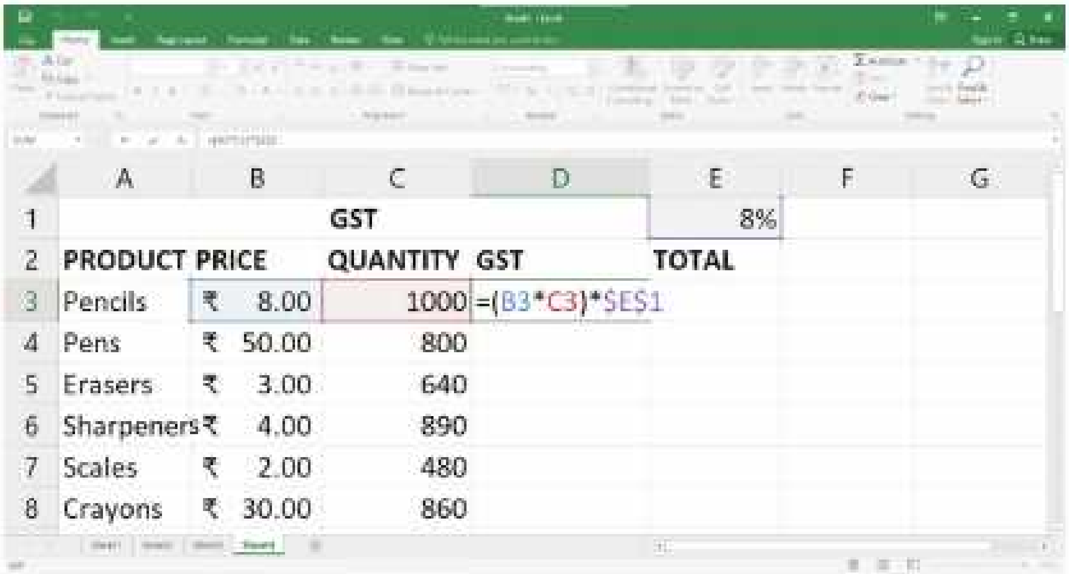
Absolute Reference

There may be times when you do not want a cell reference to change when filling cells. Unlike relative references, absolute references do not change when copied or filled. You can use an absolute reference to keep a row and/or column constant. So, we can say that, when we do not want to change the address of the cell on copying the formula to another cell, the absolute reference is used.

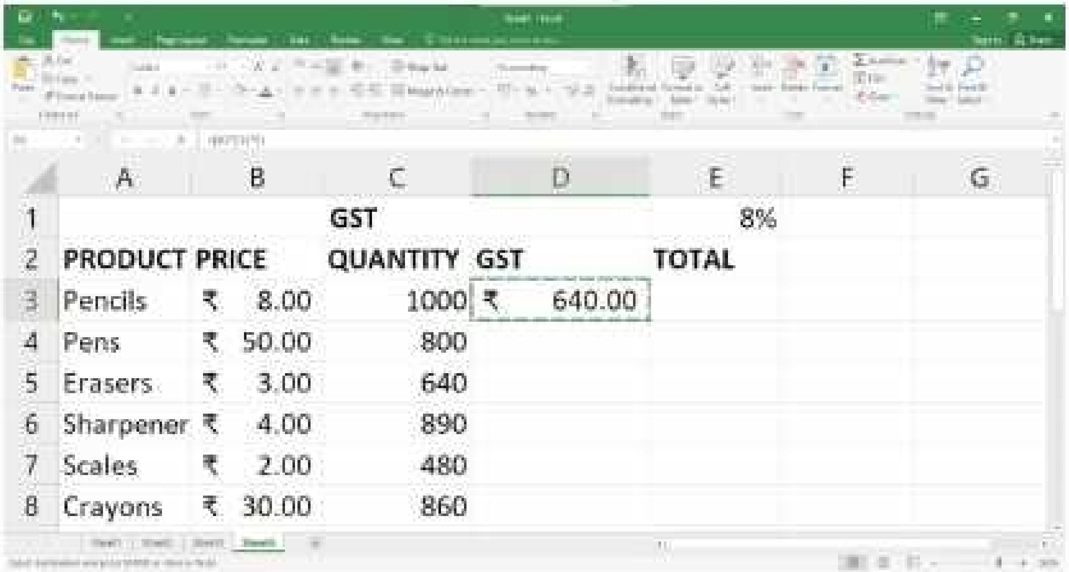
An absolute reference is designated in a formula by the addition of a dollar sign ($). It can precede the column reference, the row reference or both. For example, = $A$1+$A$2+$A$3.

Let us consider an example:

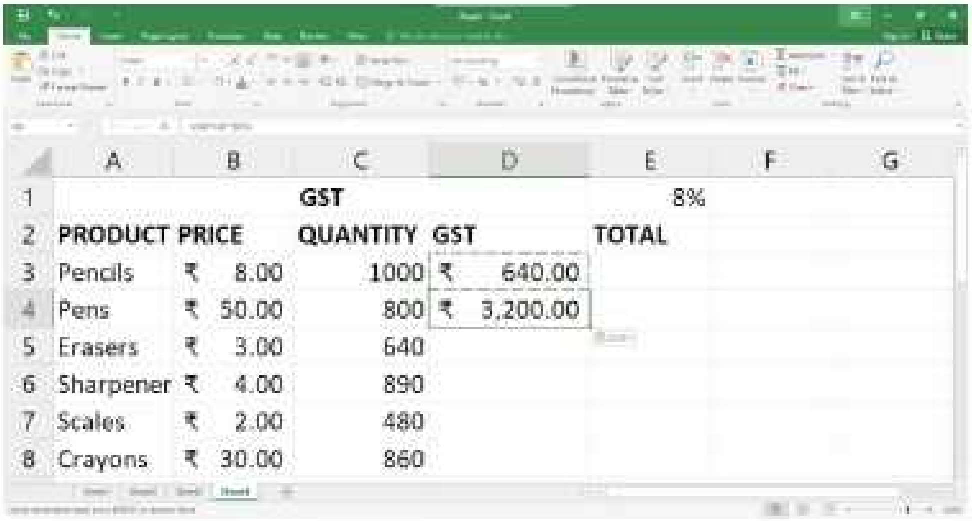
1. Enter the data as shown.
2. Click on the cell D3 and type = (B3 * C3) * $E$1.



3. Press the Enter key.
4. Select the cell D3 and click on the Copy option in the Clipboard group on the Home tab.



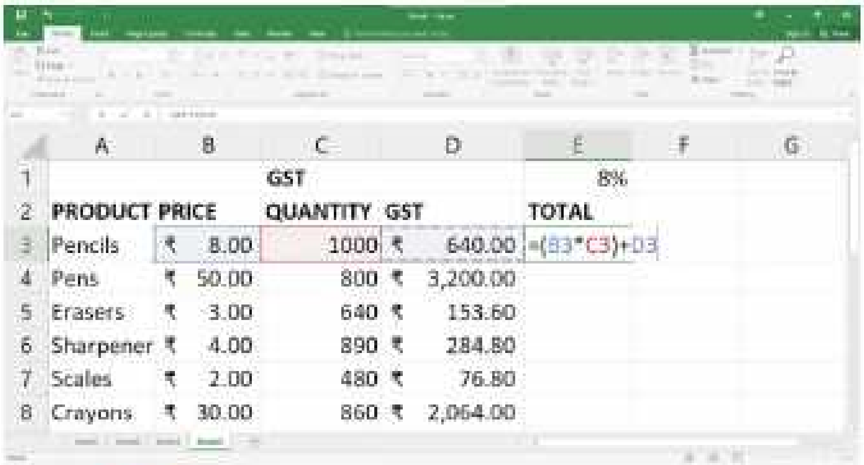
5. Select the cell D4 and click on the Paste option in the Clipboard group on the Home tab.



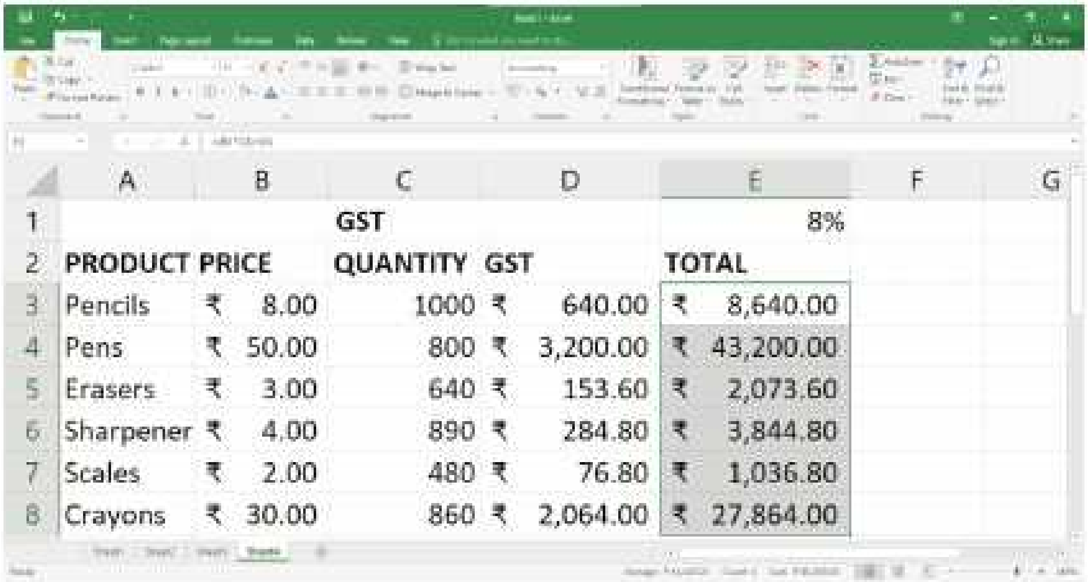
6. Click, hold, and drag the fill handle over the cells you wish to fill, cells D4:D8 in our example.

7. Release the mouse. The formula will be copied to the selected cells with an absolute reference, and the values will be calculated in each cell.

8. Click on the cell E3 and type = (B3 * C3) + D3.



9. Press the Enter key. Now, click, hold, and drag the fill handle over the cells you wish to fill, cells E3:E8 in our example.



	A	B	C	D	E
1			GST		8%
2	PRODUCT	PRICE	QUANTITY	GST	TOTAL
3	Pencils	₹ 8.00	1000	₹ 640.00	₹ 8,640.00
4	Pens	₹ 50.00	800	₹ 3,200.00	₹ 43,200.00
5	Erasers	₹ 3.00	640	₹ 153.60	₹ 2,073.60
6	Sharpener	₹ 4.00	890	₹ 284.80	₹ 3,844.80
7	Scales	₹ 2.00	480	₹ 76.80	₹ 1,036.80
8	Crayons	₹ 30.00	860	₹ 2,064.00	₹ 27,864.00

Note that, in the formula, the first cell address has no $ sign, so it is relative reference and the second cell address is absolute reference. Therefore, when the cell is copied, the first one will change the reference in the new location, but the second value will remain the same, i.e., $E$1.

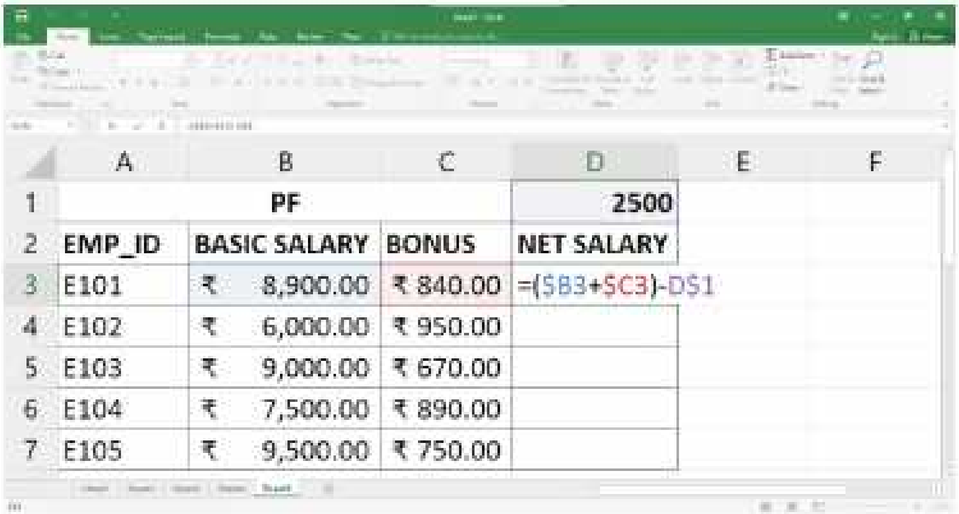
Info Zone

Be sure to include the dollar sign ($) whenever you're making an absolute reference across multiple cells. Otherwise, this will cause the spreadsheet to interpret it as a relative reference, producing an incorrect result when copied to other cells.

Mixed Reference

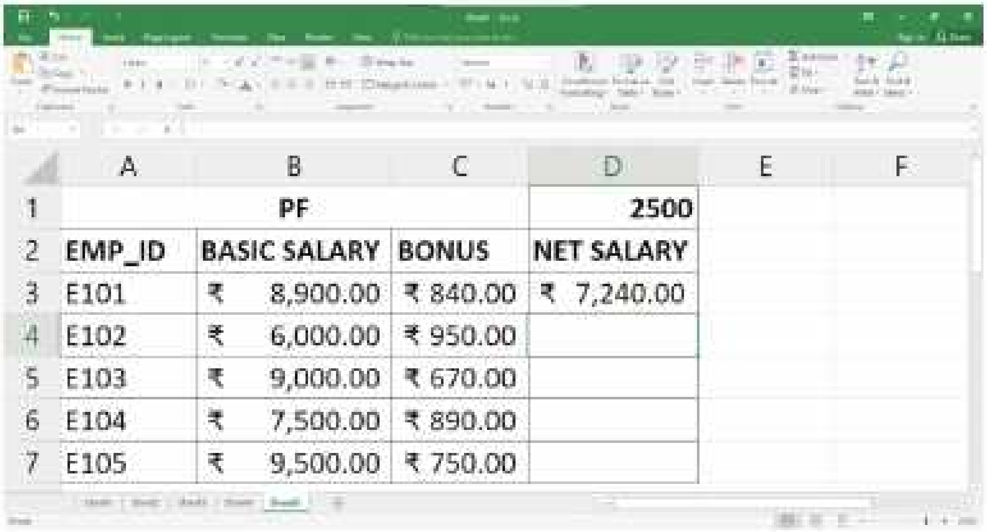
Mixed Reference is a combination of relative and absolute reference. In this type of reference either a row or a column has to remain fixed. For example, $A1 + A$2.

1. Enter the data as shown.
2. Click on the cell D3 and type = ($B3 * $C3) - D$1.



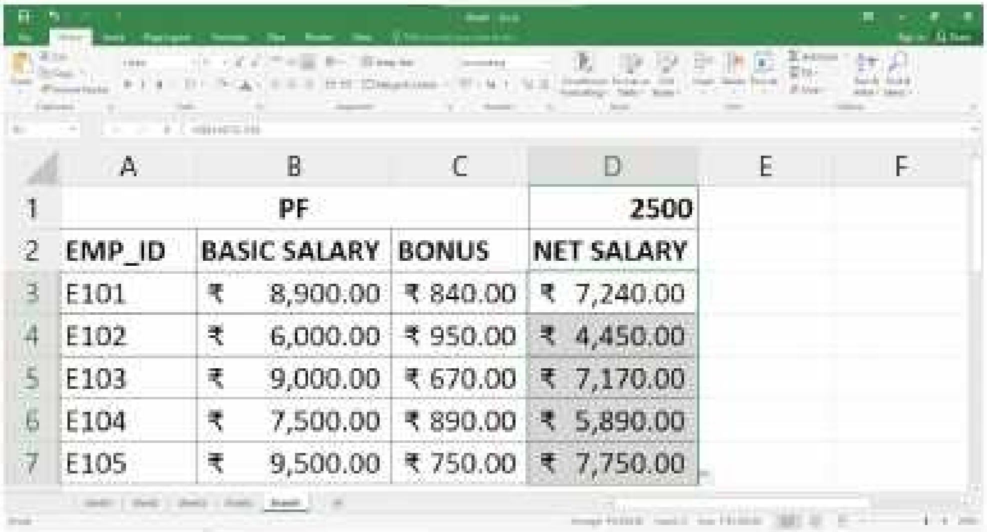
	A	B	C	D
1		PF		2500
2	EMP_ID	BASIC SALARY	BONUS	NET SALARY
3	E101	₹ 8,900.00	₹ 840.00	=($B3+$C3)-D$1
4	E102	₹ 6,000.00	₹ 950.00	
5	E103	₹ 9,000.00	₹ 670.00	
6	E104	₹ 7,500.00	₹ 890.00	
7	E105	₹ 9,500.00	₹ 750.00	

3. Press the Enter key.



	A	B	C	D	E	F
1		PF		2500		
2	EMP_ID	BASIC SALARY	BONUS	NET SALARY		
3	E101	₹ 8,900.00	₹ 840.00	₹ 7,240.00		
4	E102	₹ 6,000.00	₹ 950.00			
5	E103	₹ 9,000.00	₹ 670.00			
6	E104	₹ 7,500.00	₹ 890.00			
7	E105	₹ 9,500.00	₹ 750.00			

4. Click, hold, and drag the fill handle over the cells you wish to fill.



	A	B	C	D	E	F
1		PF		2500		
2	EMP_ID	BASIC SALARY	BONUS	NET SALARY		
3	E101	₹ 8,900.00	₹ 840.00	₹ 7,240.00		
4	E102	₹ 6,000.00	₹ 950.00	₹ 4,450.00		
5	E103	₹ 9,000.00	₹ 670.00	₹ 7,170.00		
6	E104	₹ 7,500.00	₹ 890.00	₹ 5,890.00		
7	E105	₹ 9,500.00	₹ 750.00	₹ 7,750.00		

COMPUTER ETIQUETTES

Never play with the wires of the computer.

Let's Implement

A Tick (✓) the correct option.

1. The cell reference __________ can be included in relative reference.

   a. $B8 ◯ b. B$8 ◯ c. B8$ ◯ d. B8 ◯

2. The function consists of ____________ parts.

   a. four ◯ b. three ◯ c. two ◯ d. None of these ◯

3. ____________ function finds the largest number in a range.

a. Largest ◯ b. Maximum ◯ c. Max ◯ d. None of these ◯

4. ____________ calculates the greatest common factor or highest common factor of two or more integers.

a. LCM ◯ b. GCD ◯ c. EXP ◯ d. INT ◯

5. ____________ is not in the AutoSum menu.

a. Add ◯ b. Sum ◯ c. Total ◯ d. Average ◯

6. Formulas always starts with ________.

a. + ◯ b. – ◯ c. × ◯ d. = ◯

B Fill in the blanks.

1. ____________ reference is a combination of relative and absolute reference.
2. The ____________ feature is used to quickly add numbers in the selected range of cells.
3. There are ____________ types of cell references.
4. ____________ function is used to add all the numbers in a range of cells.
5. The cell address in a formula that does not change on copying is considered as ____________.
6. A function must be followed by opening and closing ____________.

C Write **True** or **False**.

1. The Count Numbers function counts the cells in a range having numeric data. ☐
2. The AutoSum option is present in the Editing group on the Home tab. ☐
3. Microsoft Excel formulas can start with the '+' sign. ☐
4. There are five types of cell references. ☐
5. Functions save time and eliminate the chance to write wrong formulas. ☐
6. Formulas in Microsoft Excel are case sensitive. ☐

D Match the following.

1. LCM	a. returns a square root
2. ROUND (number, num-digit)	b. returns the result of a number raised to some number
3. INT (number)	c. rounds a number to the specified digits
4. SQRT (number)	d. return the least common multiple of integers
5. POWER (number)	e. rounds a number to the nearest integer

E Answer the following questions.

1. What do you mean by function? Name some of the commonly used functions.
2. What is the use of AutoSum feature?
3. What is cell reference? Mention its types.
4. Differentiate between relative, absolute and mixed references.
5. What is formula?
6. Explain the Math and Trig category formulas.

Refreshment Corner

- **Consider the worksheet as shown in the figure given below.**

	A	B	C	D
1	Number of Students of Class 6th			
2	Section	BS House	MPS House	RS House
3	A	20	30	25
4	B	33	43	44
5	C	25	22	28
6	D	45	47	48
7	E	65	68	57
8	F	45	40	55

What will be the output of the following formulas?

(a) =SUM(C3:C8) ____________________

(b) =AVERAGE(B3:B5) ____________________

(c) = MAX (D3:D8) ____________________

(d) = PRODUCT (B3:B4) ____________________

(e) = MIN (B3:B8) ____________________

(f) = COUNT (C3:C8) ____________________

Practical Session

Experiential Learning

A Create the worksheet given below. Then, calculate the Sales tax and Total.

	A	B	C	D	E
1	*Sales Tax*				8.0%
2	**Books of Class 6**	**Price**	**Quantity**	**Sales Tax**	**Total**
3	Hindi Reader	₹ 245.00	100		
4	Hindi Grammar	₹ 298.00	200		
5	English Reader	₹ 250.00	800		
6	English Grammar	₹ 300.00	500		
7	Mathematics	₹ 345.00	600		
8	Science	₹ 350.00	900		
9	Social Science	₹ 320.00	1000		
10	Computer Science	₹ 280.00	1500		
11	Drawing	₹ 200.00	800		

B **Consider the given worksheet data and calculate the following based on the given worksheet data.**

(a) Amount for all the items. (Amount = Price Per Unit * Quantity)

(b) Total amount payable after adding amount for all the items.

	A	B	C	D
1	DIGITAL ELECTRONICS			
2	PRODUCTS	PRICE PER UNIT	QUANTITY	TOTAL AMOUNT
3	Refrigerator	₹ 50,000.00	17	
4	Washing Machine	₹ 45,000.00	18	
5	TV	₹ 48,000.00	15	
6	Laptop	₹ 60,000.00	10	
7	LED	₹ 12,000.00	15	
8	Juicer	₹ 6,500.00	16	
9	AC	₹ 48,000.00	17	
10	Mixer	₹ 7,500.00	12	
11	Digital Camera	₹ 5,000.00	13	
12	Total Sale			

Competency Based Questions

Critical Thinking

1. Do the formulas in MS Excel follow BEDMAS rule while doing a long calculation? Do you need to enclose operations in brackets if more than one operation has to be done?
2. Consider the worksheet and answer the questions.
    (a) What will be the result in cell B4 if you press Enter?
    (b) If the formula in cell B4 is copied and pasted to cell D2, what will be the formula and the resultant value?
    (c) Which type of cell reference is this?

	A	B	C	D
1	5	2	9	4
2	6	4	5	????
3	7	6	8	3
4	3	=A4+B3	1	6

RACKING YOUR BRAIN

Life Skills and Values

Your friend wants to learn formula and functions in MS Excel. He copies the notes from you and simply starts mugging them up. Is he doing the right thing? Justify your answer.

Just Have a Discussion

Communication

Divide the class into small group and discuss on different functions in the class.

Teacher's Note

- Demonstrate the uses of some commonly used functions with appropriate examples.
- Discuss various real-life applications of Excel formulas.

Scan Me

6

GOOGLE DRIVE

Outlines

- What is Google Drive?
- Getting Started with Google Drive
- Sharing a File or Folder
- Advantages of Google Drive
- Organising Files and Folders

What is Google Drive?

Google Drive is a free service from Google that enables you to store and share files online and access them anytime and anywhere, using the cloud. It was launched on April 24, 2012 It offers 15 GB of free storage space that facilitates you to store all kinds of files and documents on it. You can store unlimited amount of photos, drawings, recordings, videos, text documents and much more. The files stored on Google Drive can be opened and edited from any device.

Along with storing files, you can create, share and manage various types of files on Google Drive, such as Documents, Spreadsheets, Presentations, Forms, Drawings, etc.

Info Zone

The Google drive can be accessed offline on the Google Chrome browser via a Chrome app, which can be installed from the Chrome Web Store.

Google also offers various storage plans with a monthly subscription. The storage capacity of Google Drive can be increased from 100 GB to 30 TB.

Advantages of Google Drive

The advantages of Google Drive are as follows :

1. **Easy Uploading and Storing :** With Google Drive, one can easily upload any type of file, such as documents, drawings, photos, videos, etc., online. The advantage of keeping your files online is that it eliminates the need to send a file as an e-mail attachment or save it on any secondary storage device.
2. **Easy Editing of the Files :** Google Drive gives you an instant access to Google Docs, a suite of editing tools that allows you to edit and make changes to your files.
3. **Remote Access from Any Device :** The files stored on Google Drive can be easily accessed from any computer, tablet or a Smartphone, with an Internet connection.
4. **Easy Searching :** Google Drive has its own efficient search engine that allows you to find information on any topic easily. Google Drive provides the feature of searching by keyword and filtering by file type, that helps in accessing the files quickly.
5. **Sharing Files with Your Friends :** Users can easily share the created and uploaded files with others. This, in turn, enables them to collaborate on their work and manage a single document with the participation of each one in making the modifications.
6. **Open Discussion :** You can create and reply to comments to get the feedback and make files more collaborative. It can be done by commenting directly on the files.
7. **Free of Cost :** Google provides 15 GB of storage space with the facility to upload, access and share files, free of cost. However, users can buy a larger storage plan, if required.
8. **Free Web-based Applications :** Google Drive also gives access to free web-based applications for creating the documents, spreadsheets, presentations, forms, drawings and much more.

Info Zone

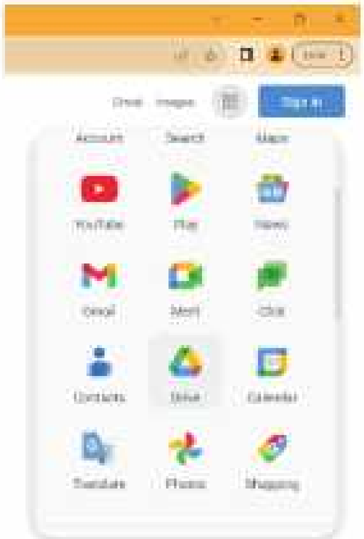
To edit an uploaded file online, it is first required to convert it to Google Drive format. To do so, click the Settings icon on the top right corner of the web page. In the Settings dialog box, select the checkbox in front of Convert uploaded files to Google Docs editor format under the Convert uploads section and click on Done.

Getting Started with Google Drive

Accessing Google Drive

To use the Google Drive, the user is required to have a Google account. To access the Google Drive, follow the given steps:

1. Visit www.google.com and sign in with your Gmail account by giving the valid Username and Password. Your Gmail account window opens.
2. Now, click on the Google Apps Launcher button located at the top right corner of the web page. A list of options appears.



3. Select the Google Drive icon 

from the list. The Google Drive window opens.

Uploading a File or Folder

1. To upload a file to your drive, choose the File upload option from the My Drive drop-down list. Or

   Click on the New button and select File upload.
2. An Open dialog box appears. Browse and select the file(s) you want to upload and then click on the Open button. A pop-up window appears notifying the uploading status of your document.
3. Once completed, the uploaded files will appear in your Google Drive folder. Similarly, you can upload a folder either by clicking on the Folder upload option from the MyDrive drop-down list or by selecting the Folder upload option from the New button.

4. The Select Folder to Upload dialog box opens. Select the folder that you want to upload and click on OK.

Once you upload the files to Google Drive, you can manage, organise, share and access them from anywhere. You can also edit your files online, but for that it is required to convert them to Google Drive format.

Info Zone

You can easily download a file for offline access. To do so, locate and right-click the file and select Download from the displayed list of options.

Organising Files and Folders

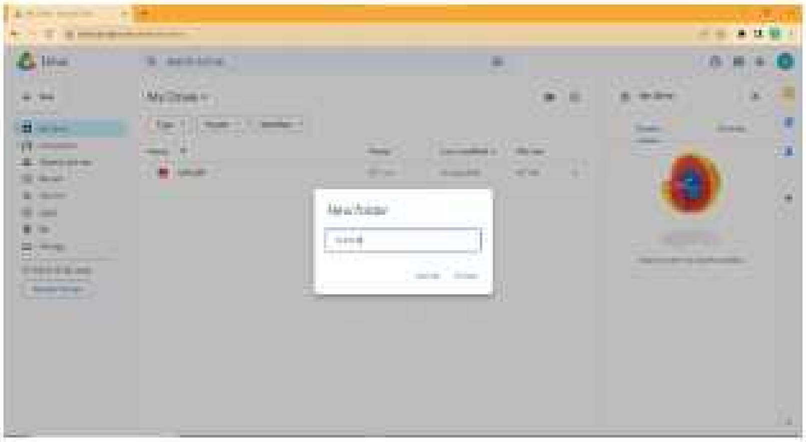
After uploading the files, you can organise them in the relevant folders. Google Drive provides you with the facility to create as well as to upload folders.

Creating a New Folder

The following steps are used to create a new folder:

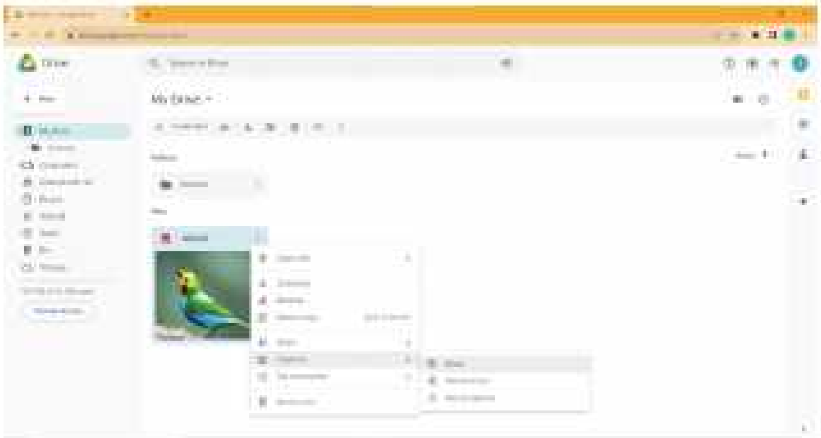
1. Click on the New button. A list of options appears.
2. Select the New folder option.
3. A small window appears. Specify the name of the folder and click on the Create button.

   The folder gets created.



Moving a File to a Folder

You can move a file to a folder by following the given steps:



1. Click on the file that is to be moved.
2. Click on the More actions button on the top right corner of the web page.
3. Select the Organise option. A list appears.
4. Select the Move option from the displayed list.

5. Select the folder from the displayed list of folders and click on the Move button. The file moves to the particular folder.

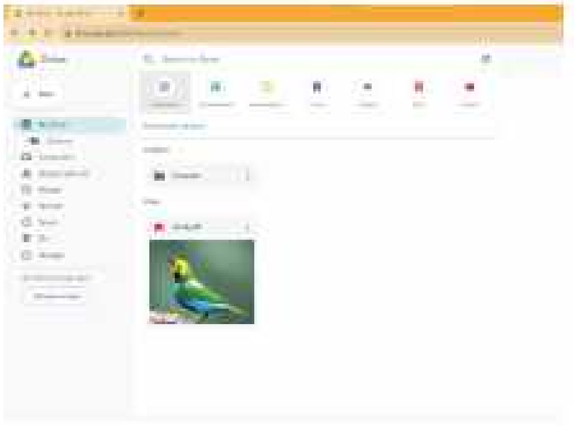
Searching a File or Folder

Users can store a large number of files on the massive storage capacity that Google provides. Sometimes, it becomes difficult to locate a file among so many files. The searching feature of Google Drive enables you to look for a file, using keywords that exist within the file or file name. You can search a file using three methods.

Method 1 : To search a file by name:

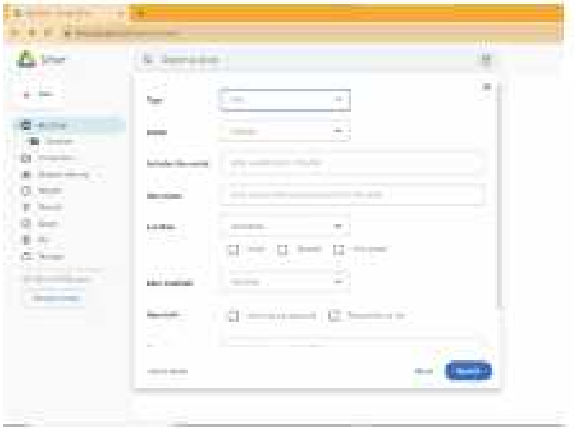
1. Click on the Search box.
2. Enter the name of the file to be searched. A list of file names appears.
3. Select the required file. The selected file opens.

Method 2 : To search a file using file type:



1. Click on the Search box. A list of file types, such as PDFs, Text documents, Spreadsheets, etc. appears.
2. Select the type of file that you are looking for. All the files of the selected type appear.

Method 3 : To search a file using Search options:



1. Click on the Advanced search button located at the right end of the Search box. A form appears.
2. Fill in the details on the form to search the required file and click on the Search button.

The required file appears in the Search results pane below the Search box.

Viewing the Files

Google Drive displays your files and folders in two different ways:

List View ( ) : This view displays a list of files and folders along with their application icons and other details, such as owner of the file, date when it was last modified, file size, etc. By default, all the files and folders appear in a list.

Grid View ( ) : The Grid view displays the files and folders in the form of thumbnails. To use this view, click on the Grid view button on the top right corner of the window.

Sharing a File or Folder

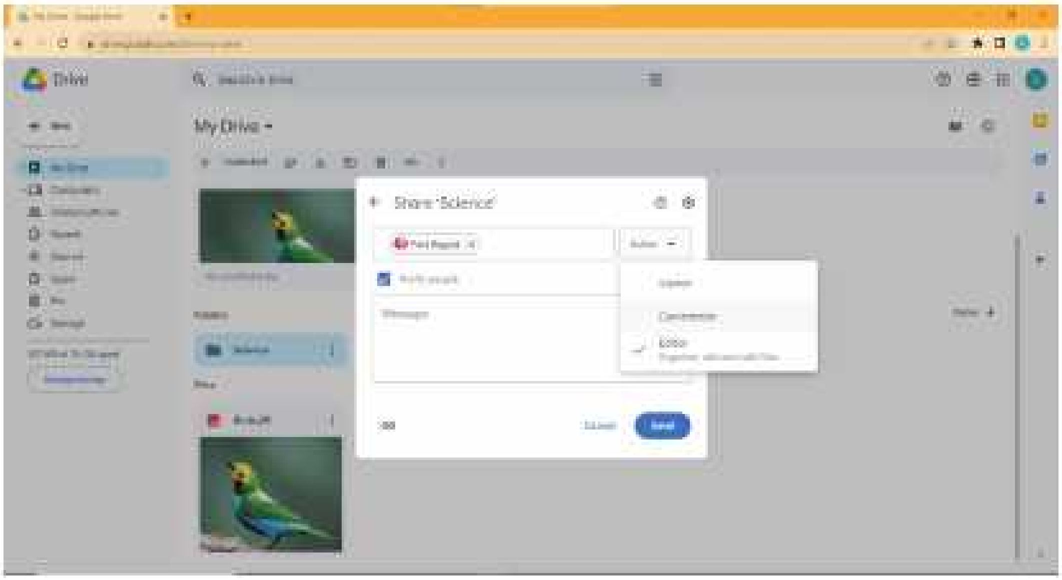
Using Google Drive, you can also share your files or folders with others. This enables multiple people together on a file. To share a file, follow the given steps:

1. Select the file that is to be shared. Click on the More actions button located at the upper right corner of the web page and select Share option. A list appears. Again, select Share. Or

   Right-click on the selected file. Select the Share option from the displayed list of options.
2. A dialog box appears. Enter the e-mail addresses of the people with whom you want to collaborate on the file.
3. For more control over the files, you can select one of the following authorisation options by clicking on the drop-down arrow of the Editor button to decide whether people can edit, comment on, or simply view the file:
   - **Editor :** This option is selected by default, which allows multiple people to work on the same project in real time. When you share a file with people, they receive an e-mail. It contains a link to the shared file, which allows them to edit it along with other users. Each editor who is working in the file will be visible as icons at the top right corner of the screen.
   - **Commenter :** By using this option you can restrict people to only comment on or view the file.
   - **Viewer :** This option allows others, only to view a file.

4. Add a note in the Add a note box (if required) and send it along with your file.

5. Finally, click on the Send button.



COMPUTER ETIQUETTES

Do not touch the computer with wet hands.

Let's Implement

A Tick (✓) the correct option.

1. Google Drive was launched on:
   a. April 12, 2012 ◯ b. April 24, 2012 ◯
   c. April 12, 2009 ◯ d. April 24, 2009 ◯
2. The files stored on Google Drive ____________ opened from any device.
   a. can be ◯ b. cannot be ◯ c. Both a and b ◯ d. None of these ◯
3. You can easily ____________ a file from Google Drive for offline access.
   a. upload ◯ b. download ◯ c. delete ◯ d. share ◯
4. ____________ this authorization option allows others, only to view a file.
   a. Viewer ◯ b. Commentor ◯ c. Editor ◯ d. All of these ◯
5. Which authorisation option restricts people to only commentor or view the file?
   a. Editor ◯ b. Viewer ◯ c. Commentor ◯ d. None of these ◯

B Fill in the blanks.

1. Google Drive offers ____________ of free storage space to its users.
2. Google Dive gives you an instant access to ____________.
3. ____________ option allows you to move a file to a different folder.
4. ____________ option allows multiple people to work on the same project.
5. ____________ view displays a list of files and folders along with application icons and other details.

C Write **True** or **False**.

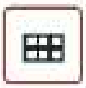
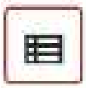
1. The storage capacity of Google Drive cannot be increased beyond 15 GB.
2. The files stored on Google Drive can be easily accessed from a computer having Internet connection.
3. To use Google Drive, the user is required to have a Google account.
4. You cannot create a folder in Google Drive.
5. List view displays the files and folders in the form of thumbnails.

D Answer the following questions.

1. Write a short note on Google Drive.
2. List any three features of Google Drive.
3. How can you upload a file on Google Drive?
4. Explain the process of Sharing a file.
5. Differentiate between the three authorisation options provided by Google Drive, while sharing a file.

Refreshment Corner

- **Identify the following pictures and write their names.**

1.  ______________ 2.  ______________ 3.  ______________

Practical Session

Experiential Learning

- Create a PowerPoint presentation on Five Latest IT Inventions and upload the same on Google Drive.

Competency Based Question

Critical Thinking

- Riya has created a word document on her system. She wants to upload and edit the same on Google Drive. Suggest her the suitable option to upload the file.

RACKING YOUR BRAIN

Life Skills and Values

One of your friend is unable to move a file to a folder. What will you do? Justify your answer.

Just Have a Discussion

Communication

Discuss the use of Google Drive in the class in detail.

Teacher's Note

Demonstrate the Google Drive practically to the students in detail.

7

WHAT ARE AI ETHICS?

Outlines

- What are AI Ethics?
- Why are AI Ethics Important?
- Ethical Concerns in AI
- Implementation of AI Ethics
- AI is Imperfect — What if It Makes a Mistake?

Human beings are intelligent by nature. They have the ability to think, to retain knowledge, to learn from past experience, use reasoning to solve problems and to adapt to new situations. Our complex brain helps us to achieve these capabilities. Animals like chimpanzees, elephants or dolphins show some intelligent traits, but are in no way comparable to humans.

As a child you may have slipped over a wet floor while running and hurt yourself. This experience taught you to walk carefully over wet floors. You remembered your past experience, and this prevented you from facing similar falls in future. This is human intelligence— the ability to learn from past experience, that prevents you from taking wrong decisions that may harm you.

Since the invention of computer machines, their capability to perform various tasks increased exponentially. Humans have developed the strength of computer systems in terms of their diverse working domains, their increased speed and reduced size, with respect to time. With the advancement in technology, now we have chess-playing computers, self-driving cars, personal digital assistant apps, voice recognition apps, flying drones, etc. These applications use various technological tools to learn and make decisions on their own. This ability of machines is known as Artificial Intelligence or AI. It is a science and technology based on disciplines such as Computer Science, Biology, Psychology, Philosophy, Mathematics and Sociology.

Ethics is a set of moral principles which help us discern between right and wrong. Ethics are moral principles that govern a person's behaviour or the conduct of an activity. These are based on well-founded standards of right and wrong that prescribe what humans should do and what they should not.

When we see an injustice happening, our inner voice tells us that something is not right. Our ethics depend on our upbringing and environment, but most of us can usually differentiate between right and wrong.

What are AI Ethics?

AI ethics is a set of guidelines that advise on the design and outcomes of artificial intelligence. AI ethics are the moral principles and techniques that form the guidelines for the development and responsible use of artificial intelligence technology. Ethical AI ensures that the AI initiatives of an organisation maintain human dignity, practice fairness, and do not cause any harm to people.

Artificial intelligence powers Google's search engine, enables Facebook to target advertising, and allows Alexa and Siri to do their jobs. AI is also behind self-driving cars, predictive policing, and autonomous weapons that can kill without human intervention. These and other AI applications raise complex ethical issues. Unfortunately, artificial intelligence lacks the ability to judge right from wrong. AI can only separate right from wrong based on data that has been labelled 'right' and 'wrong' by the developer. When a developer fails to consider the outcome of every algorithm that has been implemented, it may lead to undesirable or unethical results. As program-based machinery, AI systems are not morally accountable agents. So the question is who is to be blamed if AI makes a mistake or if AI gives incorrect results.

Why are AI Ethics Important?

AI is a technology designed by humans to replicate, enhance, or even replace human intelligence. AI relies on large volumes of data to produce output. Poorly designed AI algorithms built on data that is faulty, inadequate or biased, can have unintended or potentially harmful results.

Because of the rapid advancements in AI algorithms, it is not always clear how the AI reached its conclusions. You have to study AI ethical concerns and develop an ethics framework for AI. The AI ethics framework is important because it helps in considering the risks and benefits of AI and establishes guidelines for its responsible use.

Ethical Concerns in AI

Let us see some of the ethical concerns of AI:

Bias, Social Injustice, and Inclusiveness

Instances of bias and discrimination across a number of intelligent systems have raised many ethical questions regarding the use of artificial intelligence. AI bias occurs when an algorithm produces results that are prejudiced because they are trained on biased data.

The data fed into the algorithm, could cause bias because of three reasons— the data does not reflect the main population, it is based on historic data which itself is biased, or the data has been unethically manipulated.

This may lead to social injustice and create a barrier to building an inclusive society.

For example, in 2019, a team from the University of California discovered a problem with an AI that was being used to allocate healthcare to 200 million patients in the U.S. Black people were assigned lower risk scores and therefore lower standards of healthcare than white people, though black patients in fact experienced higher levels of health risks.

This problem occurred since the AI system was allocating risk values based on the cost of healthcare, and black patients were less able to pay for higher standards of care.

Technological Singularity

The idea of singularity is that if AI reaches up to a human level of intelligence, humans will someday be overtaken by AI machines. The fear that 'robots will take over the world' has been discussed in science fiction, but with the rapid development in AI, it is turning into a serious debate. This idea assumes that such super intelligent AI systems would quickly self-improve, develop even more intelligent systems and may either completely control humans or may cause the extinction of the human species, which is called an 'existential risk'.

Impact of AI on Jobs and Wealth Inequality

One of the primary concerns people have with AI is future loss of jobs. Should we strive to fully develop and integrate AI into society if it means many people will lose their jobs— and quite possibly their livelihood?

As more and more jobs are being automated, humans will move from physical work to strategic and administrative work. This of course is an ideal situation if humans get enough employment opportunities, but there is still a concern if there will be sufficient jobs for both humans and AI.

As a job shift occurs from humans to AI, an employer could cut down on costs by using AI. An ethical concern is that revenues will no longer be distributed fairly among people, but will mainly go to AI driven companies and countries with AI technology.

Privacy and Security

Privacy includes the right not to have one's personal matters disclosed publicly, the right to be left alone, information privacy, etc. Controlling who collects which data, and who has access is difficult in the digital world. This has led to security concerns. For example, face recognition in photos and videos allows identification, profiling, and searching for individuals.

Manipulation of Behaviour

You avail of many free services on the net, leaving behind a trail of data, but you are often not made aware of it. The big five companies, Amazon, Google(Alphabet), Microsoft, Apple, and Facebook (Meta) use AI to collect this data in order to gain, maintain, and direct your attention. You can even say that these AI algorithms may know you better than you know yourself.

Your data can be used to change your behaviour through manipulation and deception. Many advertisers, marketers and online sellers use this data to maximise profit, but this data also has the potential of being misused by third parties.

For example, in September 2019, Twitter admitted letting advertisers access its users' personal data to improve their marketing campaigns.

Accountability

One of the biggest problems brought about by AI decision-making, is who should be blamed or held accountable when an AI causes harm?

There is an 'accountability gap' since the legal system is built for humans. When humans are replaced by autonomous agents, like self-driving cars, social media algorithms and others, the same system and rules cannot be applied.

For example, if a self-driving car makes an autonomous decision to leave a highway at high speed to avoid an obstacle and crashes into another vehicle, you cannot take the self-driving car in front of a court to face justice. And even if you did, there are no legal rules that can be applied to the case.

Human Interaction

This is the beginning of an age where there is an increase in interaction with machines as if they are humans. AI chatbots and digital assistants are being designed to show emotions and mimic human behaviour to build friendly relationships with humans. This gives rise to the possibility that some humans may interact more with AI, which will affect their relations with other humans.

For example, a child can develop a friendship with an AI pet since it is designed to please humans but may not want to interact with human friends who may not always act as per the child's wishes.

Implementation of AI Ethics

The implementation of ethics is crucial for AI systems to provide safety guidelines that can prevent risks for humans, to solve any issues related to bias and to build transparent systems. Some of these ethical concerns are interlinked. For instance, transparent systems will help recognise and address questions of bias and accountability. The dignity of humans is at the heart of the question of AI ethics. All human beings are born free and equal in dignity and rights, so no form of technology should be assigned a level of human identity or dignity. AI systems must be conceived, designed and implemented to serve and protect human beings and the environment in which they live. Human dignity needs to include security, human rights, transparency, justice, accountability, opportunity, innovation, and inclusiveness. AI can help in achieving these needs, but at times even threaten them.

AI and robotics have raised fundamental questions about what we should do with these systems, what the systems should do, and what the long term risks are. We are still unaware of all the possible issues that AI could lead to. We will have to watch the developments closely to identify the issues early on and take necessary precautions.

AI is Imperfect— What if It Makes a Mistake?

AI systems are not immune to making mistakes and machine learning takes time to become useful. If trained well using good data, then AI systems can perform well. However, if we feed AI systems bad data or make errors with internal programming, the AI systems can be harmful. Take Microsoft's AI chatbot, Tay (Thinking About You), which was released on Twitter in 2016. In less than one day, due to the information it was receiving and learning from other Twitter users, the robot learned to spew racist slurs and Nazi propaganda. Microsoft shut the chatbot down immediately since allowing it to live would have obviously damaged the company's reputation.

Yes, AI systems make mistakes. But do they make greater or fewer mistakes than humans? How many lives have humans taken with mistaken decisions? Is it better or worse when an AI makes the same mistake?

COMPUTER ETIQUETTES

Always cover the computer when it is not in use.

Let's Implement

A Tick (✓) the correct option.

1. __________ is a set of moral principles which help us discern between right and wrong.

   a. AI ◯ b. Ethics ◯ c. Bias ◯ d. None of these ◯

2. __________ have the ability to think, to retain knowledge, to learn from past experience, use reasoning to solve problems, and to adapt to new situations.

   a. AI ◯ b. Humans ◯ c. Both a and b ◯ d. None of these ◯

3. AI relies on __________ volumes of data to produce the output.

   a. small ◯ b. large ◯ c. medium ◯ d. None of these ◯

4. One of the primary concerns people have with AI is future __________ of jobs.

   a. gain ◯ b. loss ◯ c. Both a and b ◯ d. None of these ◯

5. AI systems __________ make mistakes.

   a. can ◯ b. cannot ◯ c. may ◯ d. None of these ◯

B Fill in the blanks.

1. ______________ are not immune to make mistakes and machine learning takes time to become useful.
2. Your data can be used to change your behaviour through __________________ and ______________.
3. AI is a technology designed by humans to replicate, enhance, or even replace ______________.
4. ______________ are moral principles that govern a person's behaviour or the conduct of an activity.
5. AI is also behind self-driving cars, predictive policing, and autonomous weapons that can kill without ______________.
6. The data fed into the algorithm, could cause bias because of ______________ reasons.

C Write **True** or **False**.

1. The AI ethics framework is important because it helps in considering the risks and benefits of AI. [ ]
2. As a job shift occurs from humans to AI, an employer could increase the costs by using AI. [ ]
3. Controlling who collects which data and who has access, is difficult in the digital world. [ ]
4. The big five companies use AI to collect this data in order to gain, maintain, and direct your attention. [ ]
5. In September 2019, Twitter admitted letting advertisers access its users' personal data to improve their marketing campaigns. [ ]
6. The implementation of ethics is crucial for AI systems to provide safety guidelines that can prevent risks for humans. [ ]

D Answer the following questions.

1. What are ethics?
2. What do you mean by AI Ethics?
3. Why are AI ethics important?
4. Give one example of the following AI concerns:

   a. AI bias   b. Manipulation of behaviour
5. What is the idea of singularity?
6. How can a developer make AI separate right from wrong?

Refreshment Corner

- **Explore the internet to find one more example of the following.**
  1. Example of AI bias: ______________________
  2. Algorithmic AI bias: ______________________
  3. Societal AI bias: ______________________
  4. Example of Privacy and Security ethical concerns: ______________________
  5. Example of Singularity ethical concerns: ______________________
  6. Example of Manipulation of Behaviour ethical concerns: ______________________

Competency Based Questions

Critical Thinking

1. Should AI not replace laborious jobs? Will the lives of people improve if they keep on being unskilled?
2. While browsing the internet, most websites collect and store your personal data and use it to advertise products in which you might be interested in exchange for using their free services. Is this behaviour justified?

RACKING YOUR BRAIN

Life Skills and Values

As a school captain, you are on the judging panel to allot scores to the science exhibition presentations. Your best friend is also taking part in the competition and has asked you to give her the highest score. You visit the exhibition and find that there are other students with better presentations than her. What will you do?

Just Have a Discussion

Communication

Divide the class into two groups and discuss what defines Ethics according to you in context of Artificial intelligence machines.

Teacher's Note

The teacher should demonstrate some more examples of ethical concerns to the students in detail.

8

INTRODUCTION TO PYTHON

Outlines

- Introduction
- Getting Started with Python
- Working in Interactive Mode
- Working in Script Mode
- Installing Python
- Components of IDLE Python
- Variables in Python

Introduction

In this chapter, we will learn basics of the python programming language, which one of the most powerful high-level languages. Python is a simple programming language. It is used to write codes for the computer programs. Python was created by Guido van Rossum when he was working at Centrum Wiskunde and Informatica (CWI), which is a National Research Institute for Mathematics and Computer Science in Netherlands. The language was released in 1991. Some of the features that make Python so popular are as follows:

- It is an easy to learn, general-purpose and high-level programming language.
- It is a platform-independent programming language, which means it can be used on any machine and in any operating system.
- It has a simple syntax.
- Python is a case-sensitive language.
- It is an interpreted language.

- It is free to use and even free for commercial redistribution.

Python is a popular and easy-to-learn language. It can be used to do the following:

- Build a website
- Develop games
- Program robots

Installing Python

Python can be easily installed on your computer by following the given steps:

- To download Python, visit the site: www.python.org/downloads and click on the Download Python button under 'Download the latest version for Windows'. You will get an .exe file.

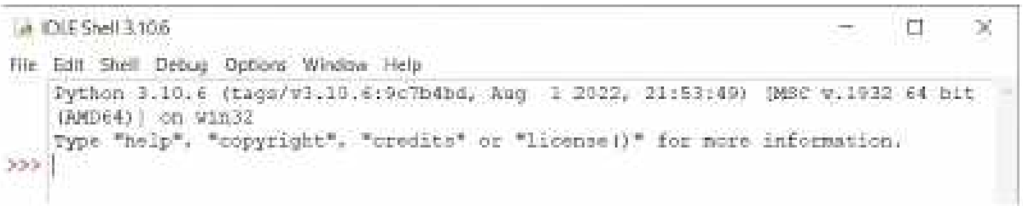
- Double-click on the .exe file, click on Run or Install Now in the dialog box, to start the installation. Keep following the instructions to install Python.

Getting Started with Python

After installing Python, you can start with any of the following two modes:

- Interactive mode
- Script mode

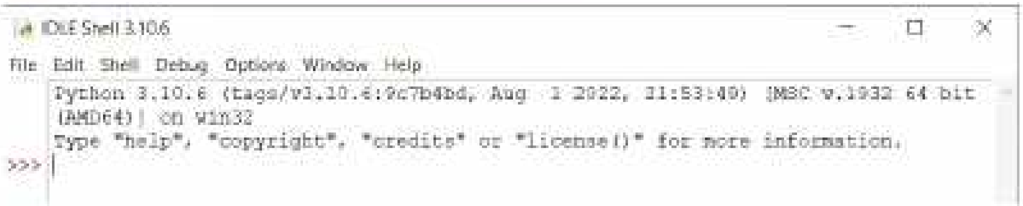
The Interactive mode is the default mode. To launch the interactive mode,



click on Start > Scroll down to Python 3.7 > IDLE (Python 3.7). The IDLE screen appears as shown in the figure.

Components of IDLE Python

Just like any other application window, the IDLE Python screen has the following components:



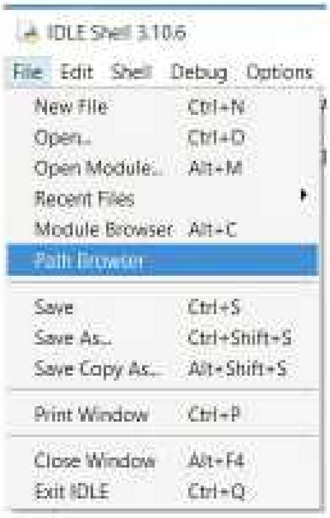
Title Bar

This is the area that displays the name of the application and the document. It also contains command buttons, like Minimize, Maximize/Restore and Close.

Menu Bar

The menu bar consists of various menus with different options. The menus of the Python window are File, Edit, Shell, Debug, Options, Windows and Help.

- **The File Menu :** This menu is used to work with the options related to the currently used file. It includes sub-options, like opening a new file, opening an existing file, viewing recently used files, saving the files, closing the files and exiting the application.



- **The Edit Menu :** This menu is generally used to edit the file in use. You can cut, copy or paste the selected text. You can also search for a particular text or phase, and even replace it with some other text.

Command Prompt

The command prompt is denoted by '>>>'. This is the area where you type a code.

Status Bar

The Status bar tells you the current status of your control or cursor on the window.

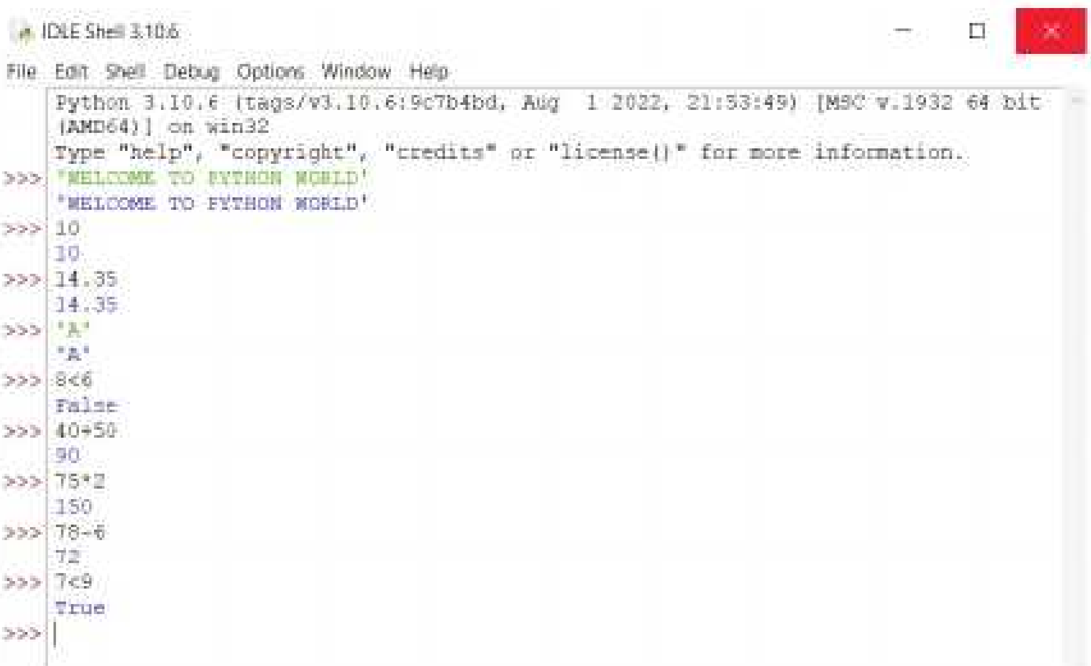
Working in Interactive Mode

In the Interactive mode of Python, the instructions are executed, line-by-line, giving the output.

In this mode, the commands are typed next to the Python Command prompt(>>>). For example, if you type 3 + 6 next to the Python prompt, it will give you the result as 9.

To work in the interactive mode, follow the process given below:

- Click on the Start> Python 3.7> IDLE (Python 3.7). It will open the Python Shell window (interactive mode) where you will see the Python prompt (>>>).
- Type the commands next to Python prompt one-by-one as shown in the figure given below, and press Enter. Python will immediately give the output for each of them.



The interactive mode is useful for testing the code where you type the commands, one at a time, and get the result or error immediately.

Printing a Message

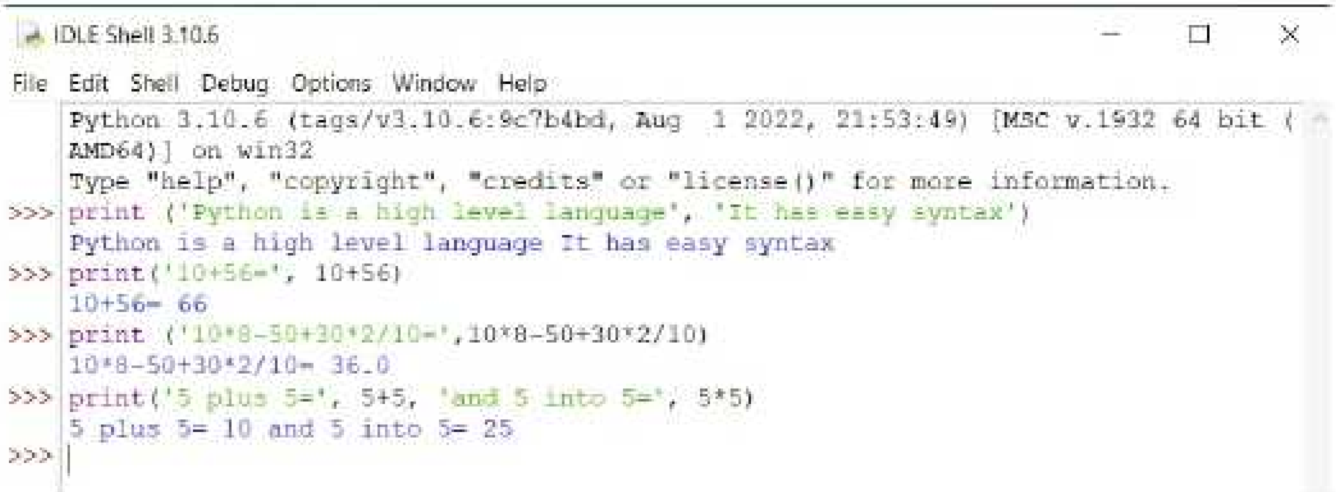
The print( ) function is used to display the output of any command on the screen. It can be used to print the specified messages.

For example:

>>>print ("Hello World") will display Hello World.

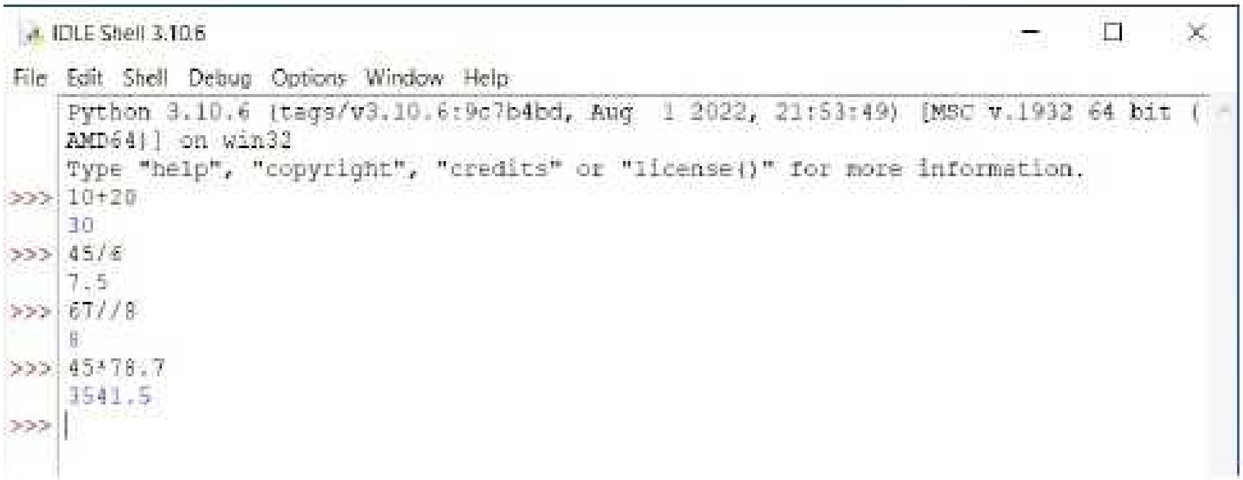
Python is a case-sensitive programming language. It means that Python differentiates between capital and small letters. For example, Print (P capital) and Print (p small) are two different things for Python, where print is a valid command in Python, while Print is just a word and not a command.

You can also pass more than one argument to the print( ) function. In such a case, the arguments are separated by commas as shown in the figure.



Using IDLE as a Calculator

To evaluate an arithmetic expression, it is not necessary to use the print( ) function. If you type an arithmetic expression at the Python command prompt and press the Enter key, the interpreter automatically evaluates it and shows the result. In this case, the interpreter acts as a simple calculator. The expressions are evaluated using operators, like +, -, *, /, etc. The basic BODMAS rules are followed for performing calculations.



Info Zone In IDLE, you can press Alt + P to repeat the previous command.

Variables in Python

If you need to store some food for future use, what will you require? Well, you will need a container to store the food. Similarly, when you are working with the values in Python, you require some storage location to hold the values for later use. Named storage locations in the computer memory, which are used to store data are called variables. A variable can store only one data value at a time. When a new value is stored in a variable, its previous value gets overwritten.

Assigning Values to Variables

Values are assigned to variables using the assignment operator (=). For example, the statement x=25 assigns the value 25 to the variable x.

Observe the following codes to understand the use of the assignment operators:

Variable	Description
a=10	# value 10 is assigned to a.
b=20	# value 20 is assigned to b.
c=a+b	# The numbers are added and the value is assigned to the variable c, i.e., 30 is assigned to the variable c.

a=c-10	# Value 20 (30-10) is assigned to the variable a. So, now the value of the variable a is 20, and not 10.

Rules to Write the Variables Names

There are certain rules in Python that have to be followed to form valid variable names. The rules for valid identifier (variable name) are :

- A variable name must start with an alphabet (capital or small) or an underscore (_).
- A variable name can consist of letters, digits and underscore. No other character is allowed.
- A keyword cannot be used as a variable name.
- A variable name can be of any length.
- Variable names are case-sensitive (e.g., age and age are different variable names).

Examples of some valid variable names are:

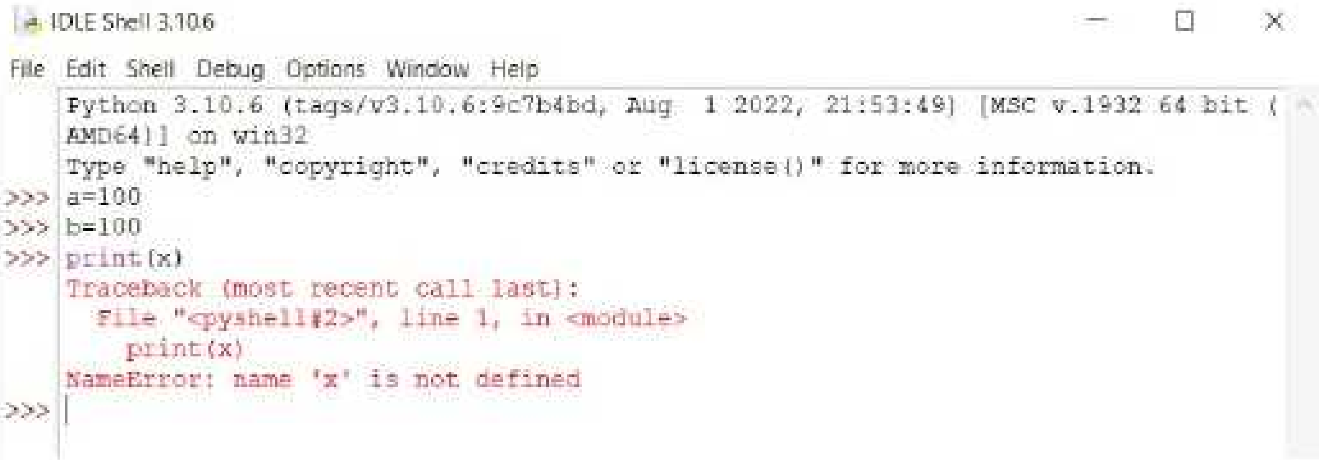
Class, emp_code, totsalary, minvalue, bal_fee, age1980, a45r

Examples of some invalid variable names are:

Invalid variable name	Reason
Employee code	Space is not allowed in a variable name.
10code	Variable name cannot start with a digit.
A.B.C. Ltd	(.) is not allowed in a variable name.
ABC@ltd	Special characters are not allowed in a variable name.
else	'else' is a keyword in Python, so it cannot be used as a variable name.

Note: If a variable is being used without assigning a value, i.e., if the variable is not defined, it gives an error.

The following code gives an error because the variable x is used in the print() function without being defined.



Working in Script Mode

The interactive mode is preferred for small programs, where only a few commands are to be executed. This is because the output in the Interactive mode is compressed between the statements and may create confusion while writing long programs.

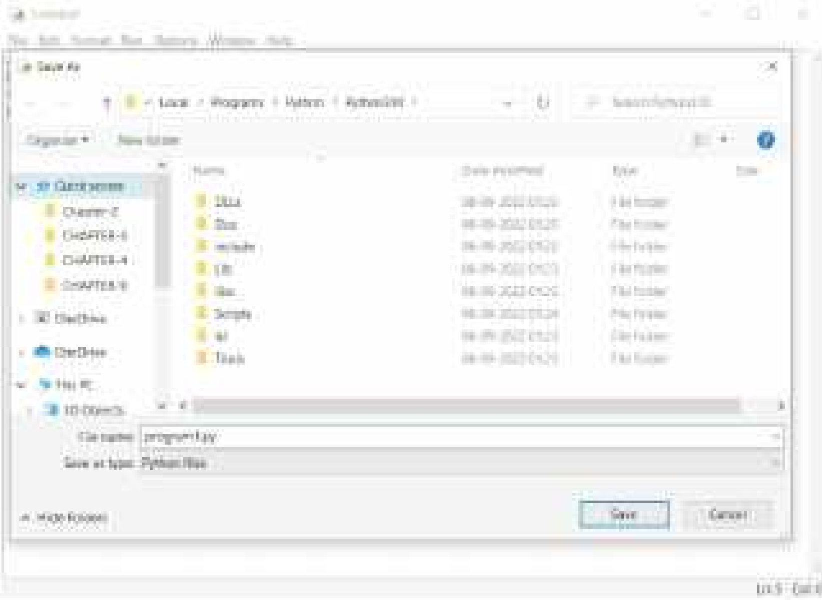
For writing lengthy programs in Python, the Script mode is used. Using this mode, you can create and edit Python programs. In this mode, you can also save your files so that they can be used later. The complete script is written in an editor.

Creating Module/Script/Program File

To start with the Script mode for creating a module or program, follow the given steps:

- Open IDLE Python Shell, i.e., the Interactive mode.
- Click on File > New File in IDLE Python Shell as shown in the figure.
- In the new window that opens, type the commands you want to save as a program as shown in the figure.

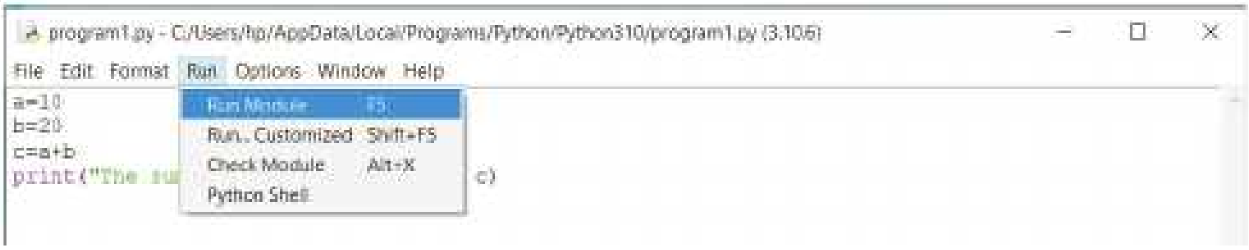
- Click on File > Save, and save the file with the name 'prog.py' where .py is the extension for Python files.



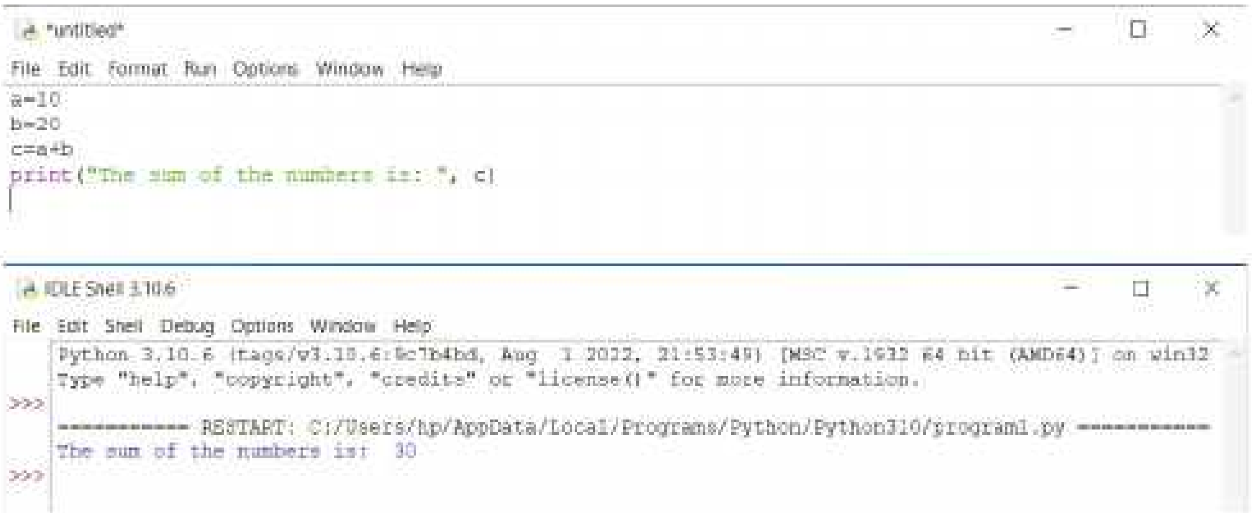
Running Module/Script/Program

After the Python program file is created, you can run the program by following the given steps:

- Open IDLE Python Shell.
- Click on File > Open. (If the file is already opened, you can directly follow the next step).
- Click on Run > Run Module or press F5.
- It will execute all the commands.



Observe the difference between the Script mode and the Interactive mode as shown in the figure given below.



COMPUTER ETIQUETTES

Never clean the computer with the wet cloth.

Let's Implement

A **Tick (✓) the correct option.**

1. Python language was released in ____________.
   a. 1990 ◯ b. 1991 ◯ c. 1993 ◯ d. None of these ◯
2. When a new value is stored in a variable, its previous value gets ____________.
   a. accepted ◯ b. overwritten ◯
   c. overlapped ◯ d. None of these ◯
3. Values are assigned to variables using the ____________ operator.
   a. string ◯ b. print() ◯ c. assignment ◯ d. None of these ◯
4. ____________ is a valid variable name.
   a. A.B.C Ltd ◯ b. 10 code ◯ c. Minvalue ◯ d. None of these ◯
5. ____________ is an invalid variable name.
   a. 10 code ◯ b. class ◯ c. Totsalary ◯ d. None of these ◯

B **Fill in the blanks.**

1. Python was created by ____________.
2. The ____________ key is pressed to execute a Python program.
3. In the interactive mode of Python, the instructions are executed ____________.
4. The ____________ mode is the default mode.
5. Python ____________ cannot be used as a variable name.

C **Write True or False.**

1. A variable name can consists of letters, digits, and underscore. ☐
2. Python language has a difficult syntax. ☐
3. We cannot build a website with Python. ☐
4. A keyword cannot be used as a variable name. ☐
5. A variable can store only one data value at a time. ☐

D **Answer the following questions.**

1. Write any three features of the Python language.
2. How can you install Python on your computer?
3. What is variable?

4. Explain the different working modes of Python.
5. What is the use of print() function? Give one example.

Refreshment Corner

- **Draw and colour the logo of Python in the space given below.**

Practical Session

Experiential Learning

- **Write the code for the following program in the Script mode.**
  1. Write a program to find the product of two numbers a and y, where x=150 and y=200.
  2. It takes 3 hours to drive a distance of 192km. Create a program to calculate the average speed in km/h?

Competency Based Question

Critical Thinking

- Anshika's computer teacher has given her an assignment of creating a lengthy program in Python. She also has to edit and save this program. Which mode should she use for this purpose?

RACKING YOUR BRAIN

Life Skills and Values

What will you do if you are unable to work with Python and your friend knows it well? Will you ask your friend for help or will you prefer taking help from your teacher? Justify your answer.

Just Have a Discussion

Communication

Discuss on Interactive mode vs Script mode in the class in detail.

Teacher's Note

Students should be made to practice a lot of programs on the computer to hone their programming skills.

QUICK REVIEW 2

Tick (✓) the correct option.

1. The cell reference __________ can be included in relative reference.

   a. \$B8 ◯ b. B\$8 ◯ c. B8\$ ◯ d. B8 ◯

2. The files stored on Google Drive __________ opened from any device.

   a. can be ◯ b. Cannot be ◯
   c. Both a and b ◯ d. None of these ◯

3. AI relies on __________ volumes of data to produce the output.

   a. small ◯ b. large ◯
   c. medium ◯ d. None of these ◯

4. __________ is a valid variable name.

   a. A.B.C. Ltd ◯ b. 10 code ◯
   c. Minvalue ◯ d. None of these ◯

5. __________ is not in the AutoSum menu.

   a. Add ◯ b. Sum ◯
   c. Total ◯ d. Average ◯

6. Google Drive was launched on:

   a. April 12, 2012 ◯ b. April 24, 2012 ◯
   c. April 12, 2009 ◯ d. April 24, 2009 ◯

7. __________ have the ability to think, to retain knowledge, to learn from past experience, use reasoning to solve problems, and to adapt to new situations.

   a. AI ◯ b. Humans ◯
   c. Both a and b ◯ d. None of these ◯

8. Values are assigned to variables using the __________ operator.

   a. string ◯ b. print() ◯
   c. assignment ◯ d. None of these ◯

9. __________ calculates the greatest common factor or highest common factor of two or more integers.

   a. LCM ◯ b. GCD ◯ c. EXP ◯ d. INT ◯

10. Which authorisation option restricts people to only comment on or view the file?

    a. Editor ◯ b. Viewer ◯ c. Commentor ◯ d. None of these ◯

TEST PAPER 1

A Fill in the blanks.

1. ____________ is data that has been processed into a meaningful and useful form.
2. ____________ is easy-to-use and provides a user-friendly environment.
3. The Merge Cells option is present in the ____________ group on the Layout tab.
4. To insert a new column, click on the Insert ____________ option.
5. Types of ______________ computers include hand-held notebook, laptop, portable and super microcomputer.

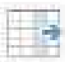
B Write **True** or **False**.

1. Windows 10 automatically changes to the Tablet mode if it detects that there is no keyboard attached. [ ]
2. The Borders option is available on the Design tab. [ ]
3. Merging refers to the rotation of the text in different angles inside the cell. [ ]
4. Microcomputers are also called Personal Computers. [ ]
5. You can drag and drop an app from one desktop to another in the Task View pane. [ ]

C Answer the following questions.

1. How can we convert text to a table?
2. List out the steps to copy data from one location to another.
3. What is the difference between input and auxiliary storage?
4. How is the 'Edge' feature useful?
5. How can we delete rows or columns?

D Match the following.

1.  a. Insert Above
2.  b. Insert Left
3.  c. Delete
4.  d. Insert Below
5.  e. Insert Right

TEST PAPER 2

A Fill in the blanks.

1. ____________ reference is a combination of relative and absolute reference.
2. Google Drive gives you an instant access to ____________.
3. AI is a technology designed by humans to replicate, enhance, or even replace ____________.
4. The ____________ mode is the default mode.
5. The cell address in a formula that does not change on copying is considered as ____________.

B Write **True** or **False**.

1. The storage capacity of Google Drive cannot be increased beyond 15 GB. ☐
2. As a job shift occurs from humans to AI, an employer could increase the costs by using AI. ☐
3. We cannot build a website with Python. ☐
4. There are five types of cell references. ☐
5. You cannot create a folder in Google Drive. ☐

C Answer the following questions.

1. What do you mean by function? Name some of the commonly used functions.
2. List any three features of Google Drive.
3. Why are AI ethics important?
4. Write any three features of the Python language.
5. What is cell reference? Mention its types.

D Match the following.

1.	LCM	a.	returns a square root
2.	ROUND (number, num-digit)	b.	returns the result of a number raised to some number
3.	INT (number)	c.	rounds a number to the specified digits
4.	SQRT (number)	d.	return the least common multiple of integers
5.	POWER (number)	e.	rounds a number to the nearest integer